# ivdtools handbook

hiox-tech

# Contents

# Welcome

This book is an executable, auditable usage guide for `ivdtools`, intended for IVD practitioners, statisticians, and readers new to R. It covers everything from data entering the project to result export, experimental scenarios, and the statistical algorithms actually implemented in the package source code. Pedagogical recommendations are not equivalent to regulatory or standard requirements.

**!** Scope of Use

This book is for teaching statistical workflows and verifying implementations. It is not a substitute for product intended use, risk analysis, current regulations, or the original text of CLSI standards. Any performance claims should be supported by approved protocols, real samples, and independent review.

## Reader's Guide

- Beginners read Part 1 in sequence, then move on to the function chapters by task.
- Experienced analysts can directly look up function signatures, object fields, and boundary behavior.

## Version and Citation

```
packageVersion("ivdtools")
```

```
[1] '0.1.1'
```

In papers or reports, at least record: package name and version, R version, functions and key parameters, data snapshot checksums, and analysis date. The software may be cited as: hiox-tech (2026), *ivdtools: Statistical Tools for in Vitro Diagnostic Method Evaluation*, version 0.1.1. Please submit issues to the project's issue page.

# 1. ivdtools Package Overview

## 1.1 Package Positioning

`ivdtools` is an R statistical toolkit for in vitro diagnostic (IVD) reagent evaluation, providing a set of reproducible workflows spanning data inspection, statistical analysis, and result summarization and visualization. The current version is 0.1.1, requires R >= 4.1.0, and is released under the MIT license.

The package covers method comparison, ROC, qualitative agreement, precision, reference intervals, stability, quality control, curve fitting, analytical sensitivity, outlier and normality assessment, and sample-size calculation. Statistical results should be interpreted in the context of a pre-specified study protocol, applicable standards, laboratory requirements, and clinical or analytical allowable limits, and cannot substitute for professional judgment.

## 1.2 Design Features

### 1.2.1 Standalone Workflows

Standalone interfaces complete one task at a time, for example:

```
normal_test(dat, col = "candidate")
reference_interval(dat, col = "candidate")
```

### 1.2.2 Progressive Workflows

Modules such as method comparison, ROC, qualitative analysis, and precision organize the analysis process using S3 objects:

1. A constructor function creates an analysis object and stores the raw data and metadata;
2. An analysis method receives the object, computes results, and returns the updated object;
3. `summary()` summarizes the analyses that have been completed;
4. `plot()` produces plots from results already present in the object;
5. `predict()` performs prediction in supported modules.

```
data -> constructor -> analysis method -> updated object -> summary/plot/predict
```

The progressive interface requires reassignment:

```
obj <- analysis_method(obj)
```

If you call only `analysis_method(obj)` without the `obj <-`, the new results will not accumulate into the original object.

## 1.3 Functional Modules

| Module | Main entry | Key features |
|---|---|---|
| Method comparison | `mcr()` | Descriptive statistics, correlation, OLS/WLS, Deming, weighted Deming, Passing–Bablok, Bland–Altman, bias and outlier analysis |

| Module | Main entry | Key features |
|---|---|---|
| ROC | `roc()` | ROC curve, AUC, optimal cut-off, multivariate logistic regression and prediction |
| Qualitative analysis | `raw_to_table()`, `counts_to_table()` | Four-fold table, diagnostic accuracy metrics, Kappa agreement and McNemar test |
| Precision | `precision()` | Variance components, confidence intervals, outliers, normality and Sadler precision profile |
| Reference interval | `reference_interval()` | Percentile, parametric and robust reference intervals |
| Bottle/lot ANOVA | `bottle_anova()` | One-way or nested ANOVA, Tukey HSD and compact letter display |
| Stability | `stability_bias()`, `stability_regression()` | Bias, stability regression, time to exceed limit, stability plan, MKT and Arrhenius models |
| Quality control | `qc_chart()`, `youden_plot()` | Levey–Jennings chart, Westgard rules and Youden plot |
| Curve fitting | `fit_equation()` | Linear, exponential, 4PLC, 5PLC and other equation fits, weighted/constrained fits and model comparison |
| Analytical sensitivity | `lob_lod_loq()` | Limit of blank, limit of detection, limit of quantitation determination |
| Outliers and distribution | `outliers_test()`, `normal_test()` | Grubbs, ESD, Dixon, IQR and multiple normality tests |
| Sample size | `sample_size_*()` | Bland–Altman agreement, single-proportion confidence interval and one-arm target-value test |

## 1.4 Quick Start

The following code uses the deterministic example data bundled with the package. See《Installing and Using the Analysis Environment》for installation and loading instructions.

### 1.4.1 Method Comparison

```
library(ivdtools)

mc <- mcr(
  ivd_mcr_example,
  id = "sid",
  candidate = "test",
  reference = "ref"
)
```

```
mc <- describe(mc)
mc <- correlation(mc, method = "pearson")
mc <- regression(mc, method = "deming")
mc <- bland_altman(mc, type = "difference")

summary(mc)
plot(mc, type = "regression")
plot(mc, type = "bland_altman")
```

Regression methods include OLS, WLS, Deming, weighted Deming, and Passing-Bablok. The specific method, weights, confidence level, and outlier-handling strategy should be decided before analysis.

### 1.4.2 ROC

```
ro <- roc(
  ivd_roc_example,
  cols = c("x1", "x2"),
  reference = "ref",
  positive = 1
)
ro <- describe(ro)
ro <- auc(ro)
ro <- cutoff(ro)
ro <- mlr(ro, cols = c("x1", "x2"), name = "combined")

summary(ro)
plot(ro, mlr = "all", cutpoint = "Youden")
```

The reference variable must be binary. If the original reference result is continuous, first use `continuous_to_binary()`, and explicitly record the threshold and the positive direction.

### 1.4.3 Qualitative Agreement

```
qa <- raw_to_table(
  ivd_qualitative_example,
  candidate = "new",
  reference = "gold",
  id = "id",
  positive = "positive"
)
qa <- describe(qa)
qa <- diagnostics(qa)
qa <- kappa(qa)
qa <- mcnemar(qa)

summary(qa)
```

When TP, FP, TN, FN counts are already available, you can also create an object with `counts_to_table()`.

### 1.4.4 Reference Interval

```
set.seed(20260806)
ri_data <- data.frame(value = rnorm(160, mean = 100, sd = 15))

ri <- reference_interval(
  ri_data,
  col = "value",
  method = "all"
)
```

```
print(ri)
plot(ri)
```

The percentile, parametric, or robust method should be chosen based on sample source, sample size, distribution characteristics, and the study protocol.

### 1.4.5 Bottle/Lot ANOVA

```
ba <- bottle_anova(ivd_bottle_example, value ~ bottle)
print(ba)

posthoc <- tukey(ba)
print(posthoc)
```

### 1.4.6 Standalone Tools

```
# Outliers and normality
outliers_test(ri_data, "value", method = "grubbs")
normal_test(ri_data, "value", method = "shapiro")

# View available fit equations and Westgard rules
list_equation()
list_westgard()

# Sample size / power
sample_size_bland_altman(
  power = 0.80,
  mu = 0.2,
  sd = 1,
  delta = 2.5
)
```

### 1.4.7 Getting Help

View the function documentation:

```
help(mcr,"ivdtools")
```

View the help documentation and browse topic-specific files.

```
vignette("ivdtools")
```

Visit the project homepage to view/download the complete function manual and help documentation.

## 1.5 AI Automated Analysis

The `ivdtools` R package provides statistical functions, while the `ivdtools-analysis` Skill provides a constrained automated analysis workflow for AI. Once the user supplies the data file, study design, and analysis requirements, the Skill uses `ivdtools` as its statistical engine to organize data pre-check, parameter confirmation, statistical analysis, result verification, and Chinese report generation.

The value of using the Skill in a conversation is not to "replace statistical judgment" but to standardize repetitive technical steps, and to save the data source, analysis parameters, excluded records, warnings, result tables, figures, and software environment together, reducing the risk of omissions and manual copy errors.

`ivdtools-analysis` is not installed with `ivdtools`; it can be obtained from the project homepage.

## 1.6 Dependencies

`ivdtools` mainly depends on `ggplot2`, `ggrepel`, `VCA`, `VFP`, `minpack.lm`, `nloptr`, `nls2`, `nortest`, and `rlang`.

## 1.7 Scope of Use

- Software output does not automatically constitute a pass/fail determination; acceptance criteria should be defined before analysis.
- Before use in regulated scenarios, applicable software validation, process verification, and result review should be completed.

## 1.8 Project Information

- Project homepage: https://github.com/hiox-tech/ivdtools
- Issue feedback: https://github.com/hiox-tech/ivdtools/issues
- License: MIT

# 2. Installing and Using the Analysis Environment

## 2.1 Purpose of This Document

This document explains how to set up the `ivdtools` analysis environment, create an R Markdown project, import and validate data, run a progressive analysis, and save reproducible results.

## 2.2 Installing R

`ivdtools` 0.1.1 requires R >= 4.1.0. It is recommended to use a still-supported newer R version and to keep the same major version and dependency versions across all people in the project.

### 2.2.1 Windows and macOS

Download the installer for your system from CRAN: https://cran.r-project.org/.

- Windows: install R; Rtools of a matching version is needed only when building packages containing native code from source.
- macOS: choose the installer according to your processor architecture; Xcode Command Line Tools may be needed when building from source.

### 2.2.2 Debian/Ubuntu

The base R can be installed from the system repository:

```
sudo apt update
sudo apt install r-base r-base-dev
```

### 2.2.3 Fedora

```
sudo dnf install R R-devel
```

The R version in Linux distribution repositories may be older; if the project needs a newer R, configure the software source according to the CRAN instructions for the corresponding distribution.

## 2.3 Choosing an Editor

It is recommended to use `.Rmd` or `.qmd` files to record code, text, tables, and figures.

| Software | Windows | Linux | macOS | Suitable for |
|---|---|---|---|---|
| RStudio | ✓ | ✓ | ✓ | Works out of the box; suitable for beginners and routine analysis |
| Positron | ✓ | ✓ | ✓ | Suitable for users who want the new-generation Posit IDE |
| Visual Studio Code | ✓ | ✓ | ✓ | Requires configuring R, Quarto and other extensions |
| Jupyter | ✓ | ✓ | ✓ | Requires Python, Jupyter and an R kernel |

| Software | Windows | Linux | macOS | Suitable for |
|---|---|---|---|---|
| PyCharm | ✓ | ✓ | ✓ | Requires corresponding R support and extra configuration |

RStudio and Positron can be obtained from https://posit.co/downloads/; Visual Studio Code can be obtained from https://code.visualstudio.com/Download.

## 2.4 Installing R Packages

Once `ivdtools` has been released on the CRAN mirror you use:

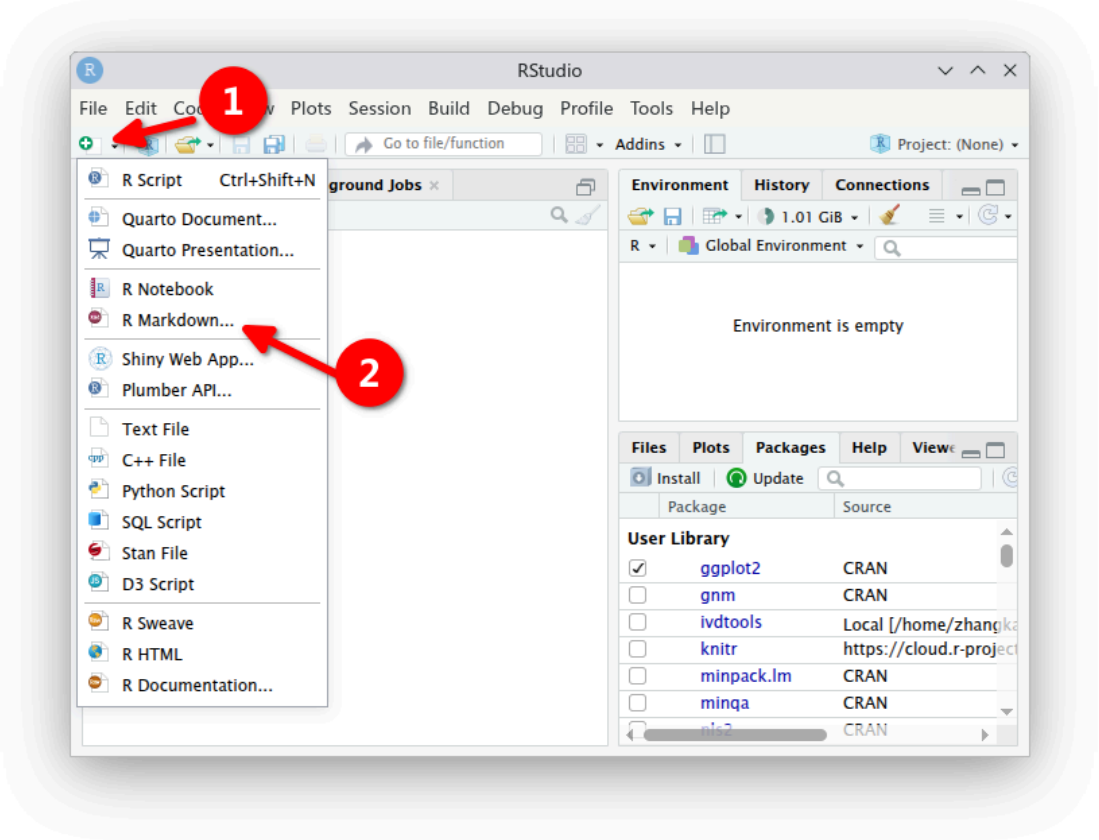
```
install.packages(c("ivdtools", "rmarkdown", "knitr"))
```

Additional packages can be installed as needed for data import:

```
install.packages(c("readr", "readxl"))
```

## 2.5 Creating a New Project

In RStudio, choose **File > New File > R Markdown**.




Minimal document template:

```
---
title: "IVD Analysis Report"
author: "Analyst"
date: "Enter the report date"
output:
  html_document:
```

```
    toc: true
    number_sections: true
---
```

Then insert an R code chunk named `setup` with `include=FALSE` set, containing the following:

```
knitr::opts_chunk$set(collapse = TRUE, comment = "#>")
library(ivdtools)
```

## 2.6 Importing and Validating Data

### 2.6.1 CSV

Base R:

```
dat <- read.csv(
  "data-raw/method-comparison.csv",
  stringsAsFactors = FALSE,
  check.names = FALSE
)
```

Using `readr`:

```
dat <- readr::read_csv(
  "data-raw/method-comparison.csv",
  show_col_types = FALSE
)
```

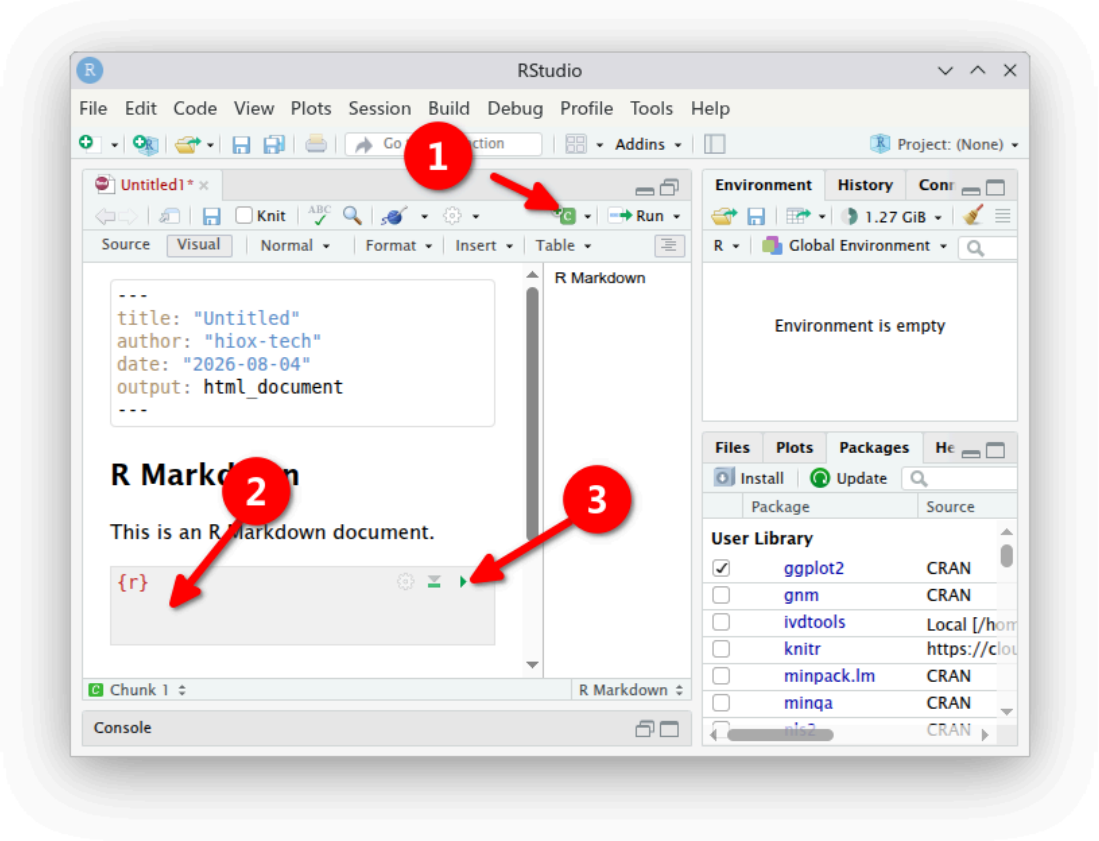
### 2.6.2 Excel

```
dat <- readxl::read_excel(
  "data-raw/method-comparison.xlsx",
  sheet = "data"
)
```

If the Excel file contains formulas, merged cells, or manual formatting, first confirm that the imported values, column names, and data types match the original sheet.

## 2.7 Running Code

In a `.Rmd`, insert an R code chunk and use the button at the top-right of the chunk to run the current chunk, or run all code from top to bottom.




To ensure reproducibility, the final report should be rendered from scratch in a fresh R session, rather than relying on objects already present in the Console.

## 2.8 Exporting Results

### 2.8.1 Tables

After reading the result fields described on the help page, you can write out a CSV:

```
# Example: the actual fields should be confirmed against the help page of the corresponding object
result_table <- mc$regression
saveRDS(result_table, "tables/regression-result.rds")
```

Complex objects are best saved as RDS to preserve their class and attributes; when exchanging with other software, tidy them into flat data frames and export as CSV.

### 2.8.2 Figures

```
p <- plot(mc, type = "bland_altman")
ggplot2::ggsave(
  "figures/bland-altman.png",
  plot = p,
  width = 7,
  height = 5,
  dpi = 300
)
```

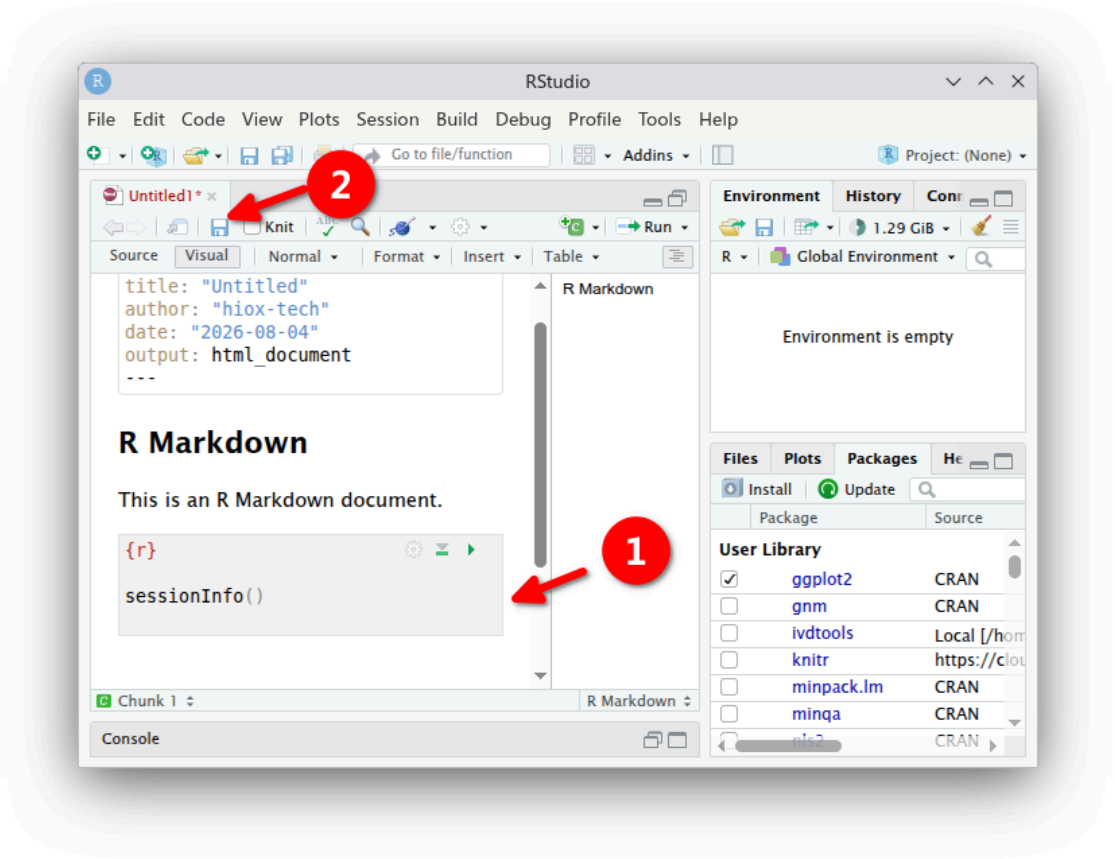
## 2.9 Rendering and Saving

In RStudio, click **Knit**, or run:

```
rmarkdown::render(
  "report/analysis.Rmd",
  output_format = "html_document",
  clean = TRUE
)
```

After the analysis is complete, record the environment information:

```
sessionInfo()
```




Recommended for archiving:

1. The unmodified raw data;
2. The `.Rmd` source file;
3. The rendered HTML/PDF;
4. Readable results and figures;
5. The `sessionInfo()` environment information;

# 3. 4PLC Equation Calibration Curve Fitting

## 3.1 Analysis Objectives

This document demonstrates how to fit a four-parameter logistic (4PLC) calibration curve using `fit_equation()` from the `ivdtools` package, and how to examine convergence status, residuals, weight effects, and forward and inverse prediction.

4PLC is a nonlinear model. A successfully returned fit object does not imply the model is adequate; formal analysis should also incorporate the calibration point design, lower and upper asymptote coverage, parameter stability, residual structure, repeat-measurement precision, and pre-specified acceptance criteria.

```
library(ivdtools)
library(readr)
```

## 3.2 4PLC Model and Main Functions

In `ivdtools`, 4PLC corresponds to `E07` and can also be invoked by the name `"4PLC"`:

$$y = A + \frac{B - A}{1 + (x/C)^D} \tag{3.1}$$

Here, $A$ and $B$ denote the two asymptotic plateaus, $C$ denotes the concentration scale at the curve midpoint, and $D$ controls the curve direction and steepness. The meaning of the parameters should be interpreted in the context of the actual signal direction.

| Function | Purpose |
|---|---|
| `list_equation()` | View the built-in equation registry |
| `replicate_to_mean()` | Summarize replicate measurements and optionally compute weights |
| `fit_equation()` | Fit built-in or custom equations |
| `coef()` | Extract model parameters |
| `residuals()` | Extract residuals (observed minus fitted) |
| `predict()` | Forward prediction of response or inverse estimation of concentration |
| `plot()` | Plot observed points, fitted curve, and intervals |
| `compare_equation()` | Compare candidate equations |

View the response-curve models:

```
list_equation(category = "Response")
```

```
  ID       Name                          Formula        Params nP Engine
-------------------------------------------------------------------------
 E01     Linear                           a + b*x          a, b  2     lm
 E02  Quadratic                  a + b*x + c*x^2       a, b, c  3     lm
 E03 ExpoGrowth                   a * exp(b * x)          a, b  2    nls
 E04  ExpoDecay                  a * exp(-b * x)          a, b  2    nls
 E05      Power                           a * x^b          a, b  2    nls
```

```
 E06        Log                  a + b * log(x)          a, b  2     lm
 E07       4PLC   A + (B - A) / (1 + (x / C)^D)    A, B, C, D  4    nls
 E08       5PLC A + (B - A) / (1 + (x / C)^D)^E A, B, C, D, E  5    nls
 E09   Gaussian a * exp(-(x - b)^2 / (2 * c^2))       a, b, c  3    nls
 E10  Michaelis             Vmax * x / (Km + x)      Vmax, Km  2    nls
 E11 Sinusoidal          a * sin(b * x + c) + d    a, b, c, d  4    nls
 E12      Cubic         a + b*x + c*x^2 + d*x^3    a, b, c, d  4     lm
```

`fit_equation()` can use automatic starting values or take them via `start`; `lower`, `upper`, and `constraints` set parameter bounds or constraints. `weights` accepts a numeric vector or the built-in schemes `"equal"`, `"1/y"`, `"1/y^2"`, `"1/x"`, `"1/x^2"`.

> **Note:** Weights should reflect the measurement error or variance structure; they should not be chosen only because the curve is positively or negatively correlated. It is best to pre-specify weights based on repeat-measurement precision, residual diagnostics, or the study protocol.

## 3.3 Example 1: Basic Fitting Workflow

### 3.3.1 Reading and Validating Data

```
u1 <- read_csv("./data/4PLC-U1.csv", show_col_types = FALSE)
u1
```

```
# A tibble: 6 × 3
  Sample     Con    RLU
  <chr>    <dbl>  <dbl>
1 s1         0      1581
2 s2         4.08   3857
3 s3        24.9   12931
4 s4        92.8   36153
5 s5       339.   106083
6 s6      1018.   303726
```

```
str(u1)
```

```
spc_tbl_ [6 × 3] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ Sample: chr [1:6] "s1" "s2" "s3" "s4" ...
 $ Con   : num [1:6] 0 4.08 24.93 92.75 338.54 ...
 $ RLU   : num [1:6] 1581 3857 12931 36153 106083 ...
 - attr(*, "spec")=
  .. cols(
  ..   Sample = col_character(),
  ..   Con = col_double(),
  ..   RLU = col_double()
  .. )
 - attr(*, "problems")=<externalptr>
```

```
summary(u1[c("Con", "RLU")])
```

```
      Con                RLU
 Min.   :   0.000   Min.   :  1581
 1st Qu.:   9.293   1st Qu.:  6126
 Median :  58.840   Median : 24542
 Mean   : 246.308   Mean   : 77388
 3rd Qu.: 277.092   3rd Qu.: 88600
 Max.   :1017.550   Max.   :303726
```

```
anyDuplicated(u1$Sample)
```

```
[1] 0
```

```
sum(!is.finite(u1$Con) | !is.finite(u1$RLU))
```

```
[1] 0
```

This data contains 6 concentration levels. Because 4PLC requires estimating 4 parameters, the residual degrees of freedom are very few; in addition, the highest concentration does not yet clearly cover the upper plateau, so the model parameters may be unstable. This example is mainly intended to illustrate the operating workflow and should not be used directly as a formal calibration design example.

If each concentration contains replicates, you can fit all observations directly, or first summarize them with `replicate_to_mean()`. Whether to summarize and how to weight should be stated in the analysis plan.

### 3.3.2 Fitting and Checking Convergence

```
fit_u1 <- fit_equation(
  "4PLC",
  data = u1,
  x = "Con",
  y = "RLU"
)
print(fit_u1)
```

```
E07 (4PLC)
Formula: A + (B - A) / (1 + (x / C)^D)

Call:
  fit_equation(eq = "4PLC", data = u1, x = "Con", y = "RLU")

Residuals:
      Min       1Q   Median       3Q      Max
    -1991.214  -1335.541  -252.849  960.620  2652.956

Coefficients:
      Estimate  Std. Error  t value Pr(>|t|)
A  3.10445e+03 2.09875e+03  1.47919 0.277197
B  6.13447e+07 2.60964e+09  0.02351 0.983380
C  2.60778e+05 1.18118e+07  0.02208 0.984391
D -9.58182e-01 1.20691e-01 -7.93917 0.015498 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2775.94 on 2 degrees of freedom
AIC: 115.58  Iterations: 50
```

Do not ignore the optimizer's convergence information:

```
fit_u1$fit$convInfo[c("isConv", "finIter", "stopCode", "stopMessage")]
```

```
$isConv
[1] FALSE

$finIter
[1] 50
```

```
$stopCode
[1] -1

$stopMessage
[1] "Number of iterations has reached `maxiter' == 50."
```

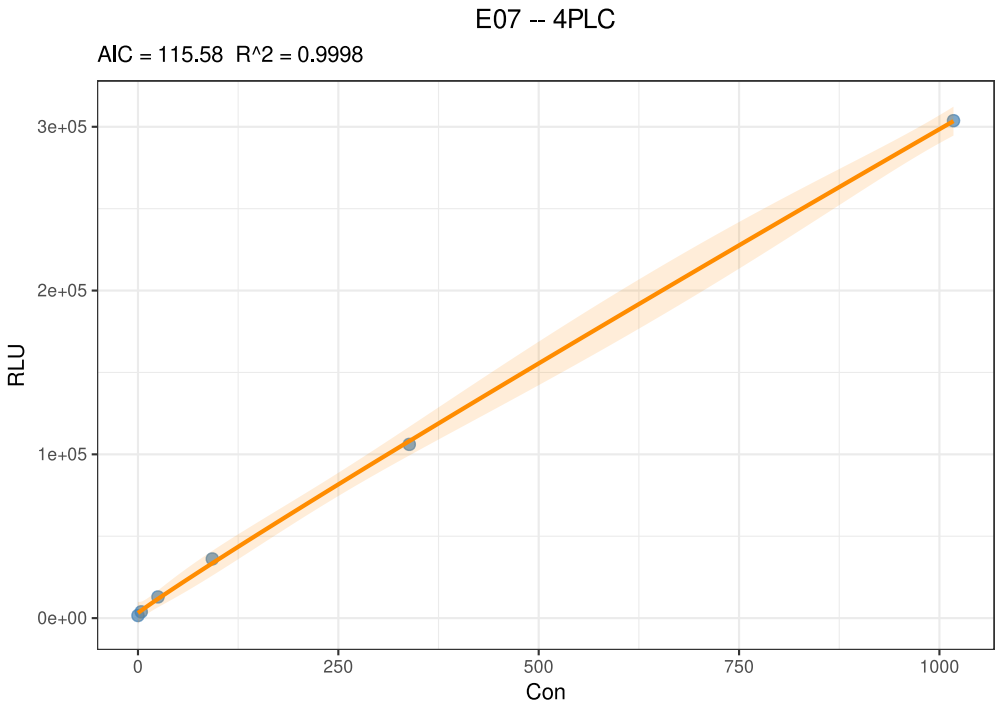
The default fit for this data reaches the iteration limit without declaring convergence. Increasing the maximum number of iterations sometimes helps, but cannot resolve a calibration range that does not cover a plateau or parameters that are not identifiable. When convergence is not achieved, inspect the data, model direction, starting values, parameter bounds, and calibration point design in turn, rather than accepting the result based only on the curve appearance.

### 3.3.3 Residuals and Graphics

```
u1_diagnostics <- data.frame(
  Sample = u1$Sample,
  Con = u1$Con,
  RLU = u1$RLU,
  Residual = residuals(fit_u1),
  Relative_residual_pct = residuals(fit_u1) / u1$RLU * 100
)
knitr::kable(u1_diagnostics, digits = 2)
```

| Sample | Con | RLU | Residual | Relative_residual_pct |
|---|---|---|---|---|
| s1 | 0.00 | 1581 | −1523.45 | −96.36 |
| s2 | 4.08 | 3857 | −771.83 | −20.01 |
| s3 | 24.93 | 12931 | 1192.12 | 9.22 |
| s4 | 92.75 | 36153 | 2652.96 | 7.34 |
| s5 | 338.54 | 106083 | −1991.21 | −1.88 |
| s6 | 1017.55 | 303726 | 266.13 | 0.09 |

```
plot(fit_u1, interval = "confidence", level = 0.95)
```




Residuals should be examined on both the original response scale and the relative scale, and interpreted together with the repeat-measurement variance. For responses spanning several orders of magnitude, looking at original residuals alone may let high-signal points dominate the judgment.

### 3.3.4 Predicting Response from Concentration

```
predict(
  fit_u1,
  newdata = data.frame(Con = c(20, 200, 1000)),
  interval = "confidence",
  level = 0.95
)
```

```
    x         y   y_ci_lwr  y_ci_upr
   20  10095.70   5194.229  14997.17
  200  66540.68  59494.141  73587.22
 1000 298518.29 289967.527 307069.05
```

The mean confidence interval describes the uncertainty of the fitted mean; the prediction interval also includes single-observation error and is usually wider:

```
predict(
  fit_u1,
  newdata = data.frame(Con = c(20, 200, 1000)),
  interval = "prediction",
  level = 0.95
)
```

```
    x         y      y_pi_lwr  y_pi_upr
   20  10095.70     -7.215948  20198.61
  200  66540.68  55240.318293  77841.05
 1000 298518.29 286223.575001 310813.01
```

### 3.3.5 Estimating Concentration from Response (Inverse Prediction)

```
predict(
  fit_u1,
  newdata = data.frame(RLU = c(20000, 100000, 250000)),
  inverse = TRUE
)
```

```
        x      y
  50.2395  20000
 311.3672 100000
 828.5559 250000
```

**Prediction boundaries:** Avoid extrapolating beyond the calibration concentration range. Inverse prediction is very sensitive to response error in the plateau region, and non-monotonic models can also have multiple solutions. The current example fit did not converge reliably, so the predictions are used only to demonstrate the interface and cannot be used for quantitative conclusions.

## 3.4 Example 2: Sandwich Assay and Weights

### 3.4.1 Equal-Weight Fit

Sandwich assay data typically show increasing signal with increasing concentration. This example first performs an equal-weight fit:

```
u2 <- read_csv("./data/4PLC-U2.csv", show_col_types = FALSE)

fit_u2_equal <- fit_equation(
  "4PLC",
  data = u2,
  x = "Con",
  y = "RLU"
)
print(fit_u2_equal)
```

```
E07 (4PLC)
Formula: A + (B - A) / (1 + (x / C)^D)

Call:
  fit_equation(eq = "4PLC", data = u2, x = "Con", y = "RLU")

Residuals:
      Min       1Q   Median       3Q      Max
    -1034493.852  -58331.509  17644.023  45830.410  785037.098

Coefficients:
      Estimate  Std. Error  t value   Pr(>|t|)
A -4.19069e+04 2.74298e+05 -0.15278 0.88288329
B  4.01980e+08 9.51404e+08  0.42251 0.68532695
C  1.12895e+03 3.16059e+03  0.35720 0.73147078
D -1.02539e+00 1.32241e-01 -7.75390 0.00011121 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 565331 on 7 degrees of freedom
AIC: 327.64  Iterations: 24
```

### 3.4.2 Inverse Response-Squared Weight Fit

When the response variance clearly increases with signal, an inverse-response-squared weight can be evaluated:

```
fit_u2_weighted <- fit_equation(
  "4PLC",
  data = u2,
  x = "Con",
  y = "RLU",
  weights = "1/y^2"
)
print(fit_u2_weighted)
```

```
E07 (4PLC)
Formula: A + (B - A) / (1 + (x / C)^D)

Call:
  fit_equation(eq = "4PLC", data = u2, x = "Con", y = "RLU", weights = "1/y^2")

Residuals:
      Min       1Q   Median       3Q      Max
    -1038926.254  -2031.562  491.283  35707.649  809345.965

Coefficients:
      Estimate  Std. Error    t value   Pr(>|t|)
A  5.66828e+03 2.30614e+02   24.57907 4.7016e-08 ***
B  2.00185e+08 6.72603e+07    2.97627   0.020624 *
C  4.83741e+02 1.74222e+02    2.77658   0.027433 *
D -1.08082e+00 1.02400e-02 -105.54858 1.8062e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.042025 on 7 degrees of freedom
AIC: 265.29  Iterations: 7
```

Compare the relative residuals of the two models at each calibration point:

```
u2_comparison <- data.frame(
  Sample = u2$Sample,
  Con = u2$Con,
  RLU = u2$RLU,
  Equal_weight_pct = residuals(fit_u2_equal) / u2$RLU * 100,
  Inverse_y2_weight_pct = residuals(fit_u2_weighted) / u2$RLU * 100
)
knitr::kable(u2_comparison, digits = 2)
```

| Sample | Con | RLU | Equal_weight_pct | Inverse_y2_weight_pct |
|---|---|---|---|---|
| c1 | 0.00 | 5637 | 843.43 | −0.55 |
| c2 | 0.05 | 16014 | 275.49 | 3.07 |
| c3 | 0.20 | 47520 | 67.84 | −4.70 |
| c4 | 0.51 | 125038 | 14.11 | −1.46 |
| c5 | 1.03 | 277621 | 4.57 | 4.70 |
| c6 | 5.19 | 1498023 | −4.39 | 0.98 |
| c7 | 10.92 | 3334387 | −1.53 | 1.70 |
| c8 | 20.83 | 6148959 | −6.53 | −5.30 |

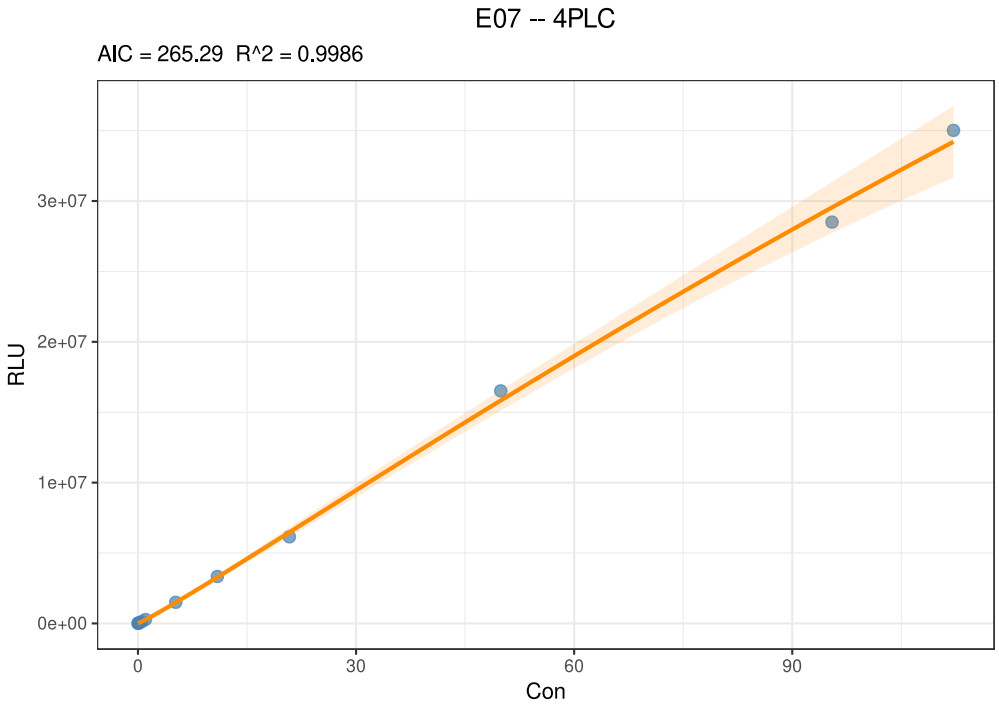
| Sample | Con | RLU | Equal_weight_pct | Inverse_y2_weight_pct |
|---|---|---|---|---|
| c9 | 49.90 | 16515788 | 4.75 | 4.13 |
| c10 | 95.47 | 28504677 | −3.63 | −3.64 |
| c11 | 112.18 | 35017203 | 1.75 | 2.31 |

```
data.frame(
  Model = c("Equal weight", "1/y^2 weight"),
  Median_absolute_relative_residual_pct = c(
    median(abs(u2_comparison$Equal_weight_pct)),
    median(abs(u2_comparison$Inverse_y2_weight_pct))
  ),
  Maximum_absolute_relative_residual_pct = c(
    max(abs(u2_comparison$Equal_weight_pct)),
    max(abs(u2_comparison$Inverse_y2_weight_pct))
  )
) |>
  knitr::kable(digits = 2)
```

| Model | Median_absolute_relative_residual_pct | Maximum_absolute_relative_residual_pct |
|---|---|---|
| Equal weight | 4.75 | 843.43 |
| 1/y^2 weight | 3.07 | 5.30 |

In this example data, the `1/y^2` weight clearly reduces the relative residuals at low-signal points; this only indicates that this weight better matches the relative-error evaluation objective adopted for this data. The formal choice still needs to combine the variance of the replicates at each concentration, the back-calculated concentration bias, model parameter stability, and pre-specified acceptance criteria. The residual standard errors and AIC under different weights have different scales and should not be compared directly without an error model.

```
plot(fit_u2_weighted, interval = "confidence", level = 0.95)
```



## 3.5 Example 3: Competitive Assay

Competitive assay data typically show decreasing signal with increasing concentration:

```
u3 <- read_csv("./data/4PLC-U3.csv", show_col_types = FALSE)

fit_u3 <- fit_equation(
  "4PLC",
  data = u3,
  x = "Con",
  y = "RLU"
)
print(fit_u3)
```

```
E07 (4PLC)
Formula: A + (B - A) / (1 + (x / C)^D)

Call:
  fit_equation(eq = "4PLC", data = u3, x = "Con", y = "RLU")

Residuals:
      Min       1Q   Median       3Q      Max
    -1937.581  -454.210  -28.126  625.582  1807.765

Coefficients:
      Estimate  Std. Error   t value   Pr(>|t|)
A -2.81576e+04 2.17005e+03  -12.9755 1.1793e-06 ***
B  1.40035e+06 1.08896e+03 1285.9562 < 2.22e-16 ***
C  1.29783e+02 6.49733e-01  199.7479 4.4162e-16 ***
D  7.82733e-01 2.79342e-03  280.2058 < 2.22e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1093.1 on 8 degrees of freedom
AIC: 207.11  Iterations: 5
```

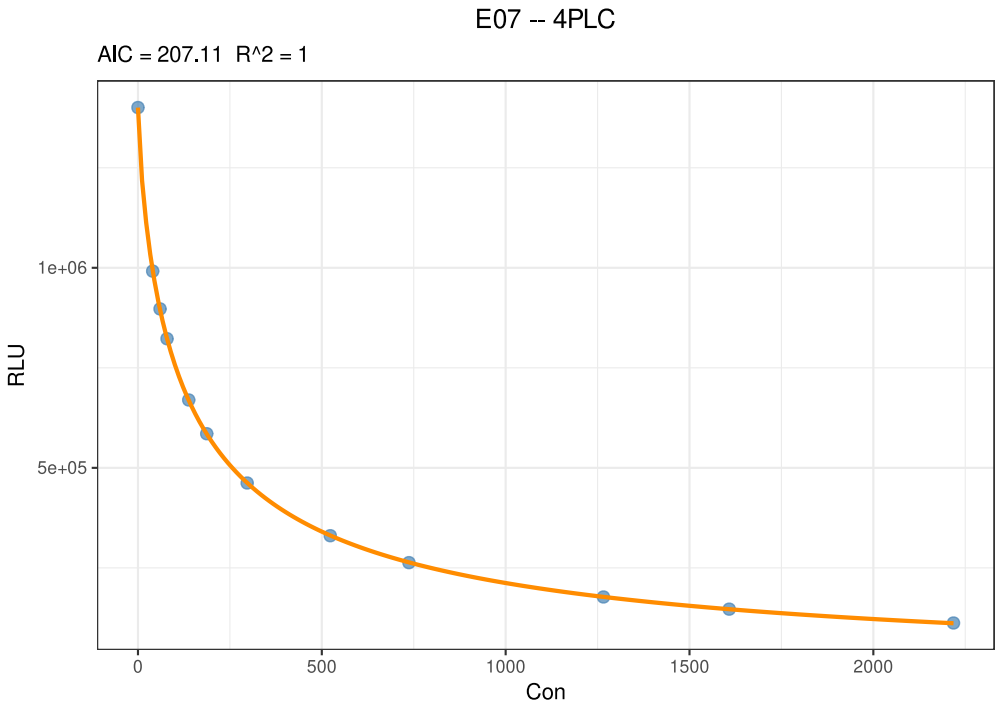
```
fit_u3$fit$convInfo[c("isConv", "finIter", "stopCode", "stopMessage")]
```

```
$isConv
[1] TRUE

$finIter
[1] 5

$stopCode
[1] 1

$stopMessage
[1] "Relative error in the sum of squares is at most `ftol'."
```

```
u3_diagnostics <- data.frame(
  Sample = u3$Sample,
  Con = u3$Con,
  RLU = u3$RLU,
  Relative_residual_pct = residuals(fit_u3) / u3$RLU * 100
)
knitr::kable(u3_diagnostics, digits = 2)
```

| Sample | Con | RLU | Relative_residual_pct |
|---|---|---|---|
| m1 | 0 | 1400548 | 0.01 |
| m2 | 40 | 991716 | −0.20 |
| m3 | 60 | 897247 | 0.20 |
| m4 | 79 | 822749 | −0.05 |
| m5 | 138 | 669646 | 0.11 |
| m6 | 187 | 585293 | 0.10 |
| m7 | 297 | 462119 | −0.07 |
| m8 | 523 | 330399 | −0.19 |
| m9 | 737 | 263095 | −0.25 |
| m10 | 1266 | 177260 | −0.12 |
| m11 | 1608 | 146827 | 0.11 |
| m12 | 2218 | 112252 | 0.61 |

```
plot(fit_u3, interval = "confidence", level = 0.95)
```




In this example, the equal-weight fit has already converged and the in-sample relative residuals are small, but you should still decide whether weights are needed based on the replicate measurement variance. The signal direction itself is not a basis for choosing between equal-weight and weighted fitting.

## 3.6 Key Points for Interpreting Results

1. **Calibration design**: Concentration points should cover the intended working range and, where possible, provide information about the upper and lower plateaus.
2. **Convergence status**: Check `convInfo`; do not treat reaching the maximum iterations as successful convergence.
3. **Parameter plausibility**: Judge stability in light of the plateaus, curve midpoint, direction, standard errors, and parameter correlations.

4. **Residual structure**: Examine the original residuals, relative residuals, and their variation with concentration or fitted value.
5. **Weight rationale**: Prefer variance structures estimated from replicate measurements, and specify the selection rule before analysis.
6. **Back-calculation performance**: Evaluate the back-calculated concentration bias and precision at each calibration point as required by the project.
7. **Prediction range**: Avoid extrapolation, especially unstable inverse prediction in the plateau region.
8. **Reproducibility**: Save the raw data, code, model parameters, package version, figures, and complete session information.

# 4. Stability Studies

## 4.1 Analysis Objectives

Stability studies evaluate the extent to which measurement results change over time for reagents, calibrators, or quality-control materials under specified storage, transport, or use conditions. This document demonstrates how to use the `ivdtools` package to establish a stability analysis plan, complete node bias, regression of trends within and between conditions relative to the first node, prediction of the time to reach an allowable limit, Arrhenius accelerated stability analysis, and mean kinetic temperature (MKT) computation.

```
library(ivdtools)
library(readr)
```

All data in this document are deterministic teaching examples used only to validate the analysis workflow. The 5% allowable drift, 10% accelerated degradation limit, target temperature, and activation energy in the examples are not universal acceptance criteria; formal studies must use parameters pre-specified in the protocol, risk assessment, or applicable standards. Whether to choose a concentration value or a signal value as the response should be decided according to the protocol.

## 4.2 Overview of Functions

| Function | Main purpose | Key input or output |
|---|---|---|
| `stability_plan()` | Plan the minimum number of time points for a regression series | Allowable drift, variability, replicates, expected drift and power |
| `stability_bias()` | Compute absolute/relative bias of each condition relative to the first node | Node summary, within-condition bias, between-condition bias and limit status |
| `stability_regression()` | Regression of a single-condition trend or test/reference paired bias | Slope, one-sided confidence limit, node results and diagnostics |
| `stability_time()` | Predict the time when the point estimate or one-sided confidence limit reaches the allowable bias | Predicted time, whether extrapolated, and search range |
| `arrhenius()` | Extrapolate the shelf life at target temperature from degradation rates at multiple elevated temperatures | Kinetic order, rate, activation energy and predicted shelf life |
| `mkt()` | Compute equally spaced or time-weighted mean kinetic temperature | Per-probe and combined MKT |

`stability_regression()` supports two modes:

- `mode = "self"`: regress the measurement results of a single condition over time;
- `mode = "compare"`: compute the bias of the test condition relative to the reference condition at matched time points, then regress the bias over time.

Before analysis, clarify the condition mapping, time unit, bias scale, expected direction of change, and allowable limits. Conclusions about acceptability can be drawn only when the protocol provides acceptance criteria.

## 4.3 Example 1: Establishing an Analysis Plan

Suppose the study protocol specifies an allowable relative drift of 5%, an expected drift of 2.5%, repeatability variability of 1.5%, and a target power of 90%. Compare the minimum number of time points needed when measuring 1, 2, or 3 times at each time point:

```
plan <- ivdtools::stability_plan(
  allowable_drift = 5,
  variability = 1.5,
  replicates = c(1, 2, 3),
  expected_drift = 2.5,
  power = 0.9,
  mode = "simple",
  bias_type = "relative"
)
print(plan)
```

```
Stability Study Time-Point Plan
  Allowable drift: 5.0000; expected drift: 2.5000
  Success target:  90%; method: CLSI EP25 Appendix A Table A2

  Replicates     Residual  Allow/Resid  Min nodes
  -------------------------------------------------
           1       1.5000       3.3333        n/a
           2       1.0607       4.5455         28
           3       0.8660       5.7471         18

  n/a -- No feasible entry in EP25 table.
```

## 4.4 Example 2: Bias Analysis

### 4.4.1 Reading and Validating Raw Data

```
bias_data <- read_csv(
  "./data/stability-bias-example.csv",
  show_col_types = FALSE
)
bias_data
```

```
# A tibble: 30 × 4
   Condition   Day Replicate Result
   <chr>     <dbl>     <dbl>  <dbl>
 1 2-8C          0         1  100.
 2 2-8C          0         2   99.8
 3 2-8C          0         3  100
 4 2-8C          7         1   99.9
 5 2-8C          7         2  100.
 6 2-8C          7         3   99.8
 7 2-8C         14         1   99.7
 8 2-8C         14         2   99.9
 9 2-8C         14         3   99.6
10 2-8C         21         1   99.5
# ℹ 20 more rows
```

```
dim(bias_data)
```

```
[1] 30  4
```

```
str(bias_data)
```

```
spc_tbl_ [30 × 4] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ Condition: chr [1:30] "2-8C" "2-8C" "2-8C" "2-8C" ...
 $ Day      : num [1:30] 0 0 0 7 7 7 14 14 14 21 ...
 $ Replicate: num [1:30] 1 2 3 1 2 3 1 2 3 1 ...
 $ Result   : num [1:30] 100.2 99.8 100 99.9 100.1 ...
 - attr(*, "spec")=
  .. cols(
  ..   Condition = col_character(),
  ..   Day = col_double(),
  ..   Replicate = col_double(),
  ..   Result = col_double()
  .. )
 - attr(*, "problems")=<externalptr>
```

### 4.4.2 Node Bias Computation

Using `2-8C` as the reference condition, normalize to each condition's own first node, and evaluate whether the relative bias is within the illustrative ±5% limit:

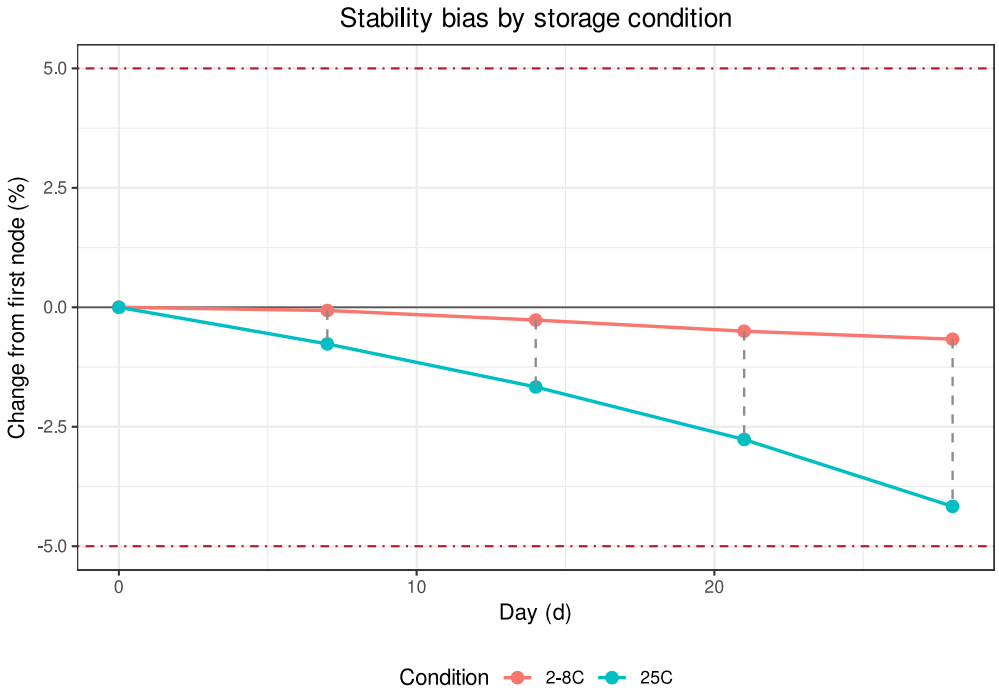
```
bias_result <- ivdtools::stability_bias(
  data = bias_data,
  condition = "Condition",
  time = "Day",
  value = "Result",
  reference_condition = "2-8C",
  limit = 5,
  bias_type = "relative",
  time_unit = "d"
)
print(bias_result)
```

```

Stability Bias Analysis
  Conditions:      2 (2-8C, 25C)
  Reference:       2-8C
  Time unit:       d
  Node comparisons:20
  Acceptance:      +/- 5.0000 (relative); 10/10 nodes passed

  Within-condition bias (vs first node)
  Cond         Time      Value    Bias(%) Status
  ---------------------------------------------
  2-8C       0.0000   100.0000     0.0000 Pass
  2-8C       7.0000    99.9333    -0.0667 Pass
  2-8C      14.0000    99.7333    -0.2667 Pass
  2-8C      21.0000    99.5000    -0.5000 Pass
  2-8C      28.0000    99.3333    -0.6667 Pass
  25C        0.0000   100.0000     0.0000 Pass
  25C        7.0000    99.2333    -0.7667 Pass
  25C       14.0000    98.3333    -1.6667 Pass
  25C       21.0000    97.2333    -2.7667 Pass
  25C       28.0000    95.8333    -4.1667 Pass
```

```
Between-condition bias (reference: 2-8C)
    Time Condition    TestVal  Bias(%)
-------------------------------------
  0.0000 25C        100.0000     0.0000
  7.0000 25C         99.2333    -0.7005
 14.0000 25C         98.3333    -1.4037
 21.0000 25C         97.2333    -2.2781
 28.0000 25C         95.8333    -3.5235
```

Within-condition bias uses each condition's own first node as the baseline; between-condition bias compares the test and reference conditions at the same time points.

```
plot(bias_result)
```




## 4.5 Example 3: Regression Analysis Comparing Storage Conditions

### 4.5.1 Reading and Validating Raw Data

```
regression_data <- read_csv(
  "./data/stability-regression-example.csv",
  show_col_types = FALSE
)
regression_data
```

```
# A tibble: 36 × 4
   Condition   Day Replicate Result
   <chr>     <dbl>     <dbl>  <dbl>
 1 2-8C          0         1  100.
 2 2-8C          0         2   99.9
 3 2-8C          0         3  100
 4 2-8C          7         1  100
 5 2-8C          7         2   99.8
 6 2-8C          7         3  100.
```

```
 7 2-8C         14         1   99.9
 8 2-8C         14         2  100
 9 2-8C         14         3   99.8
10 2-8C         21         1   99.8
# ℹ 26 more rows
```

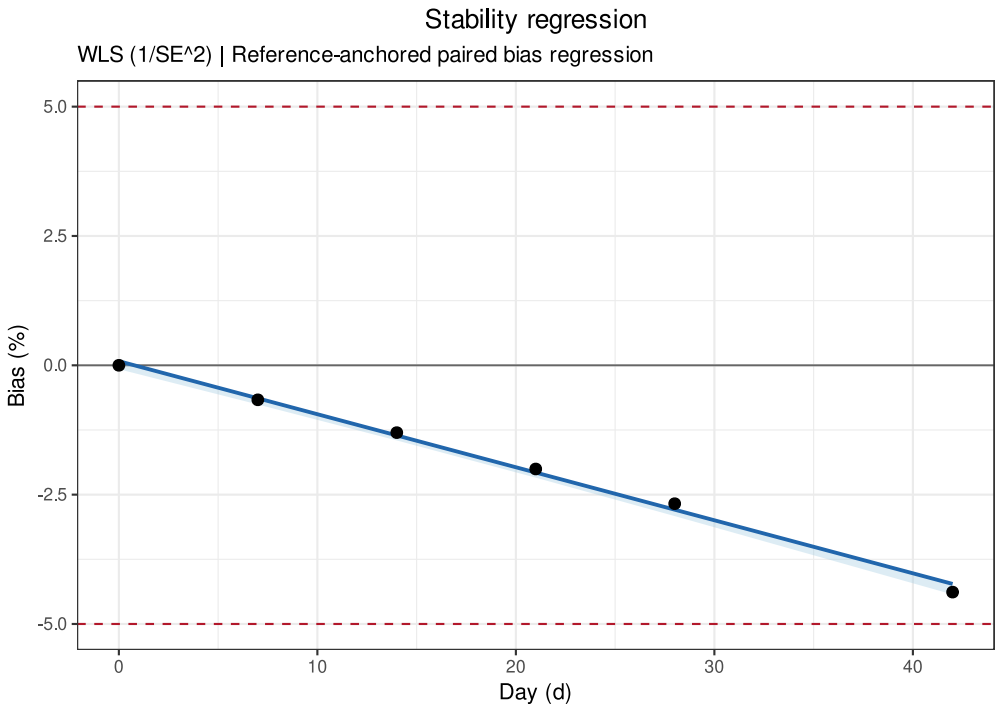
### 4.5.2 Comparing Test and Reference Conditions

Compute the relative bias of `25C` relative to `2-8C` at each matched time point, and regress the bias over time. The example pre-specifies that results decrease over time, so `direction = "decrease"` is used:

```
regression_result <- ivdtools::stability_regression(
  data = regression_data,
  time = "Day",
  value = "Result",
  condition = "Condition",
  mode = "compare",
  test_condition = "25C",
  reference_condition = "2-8C",
  bias_type = "relative",
  direction = "decrease",
  conf.level = 0.95,
  limit = 5,
  time_unit = "d"
)
print(regression_result)
```

```
Stability Regression
  Mode:            compare
  Test condition:  25C
  Reference:       2-8C
  Fit:             WLS (1/SE^2); intercept = 0.0799, slope = -0.1025
  Direction:       decrease
  One-sided CI:    95.0%
  R-squared:       0.9956
  Cook distance:   row(s) 6 > 1; not removed

     Time      ObsBias   Estimate   CI_limit Status
  ------------------------------------------------
    0.0000       0.0000     0.0799    -0.0750
    7.0000      -0.6669    -0.6376    -0.7566
   14.0000      -1.3013    -1.3552    -1.4522
   21.0000      -2.0040    -2.0728    -2.1717
   28.0000      -2.6747    -2.7904    -2.9140
   42.0000      -4.3842    -4.2256    -4.4296
```

```
plot(regression_result)
```




The model uses a one-sided confidence limit to evaluate the least favorable direction. This example identifies a node with a large Cook's distance, but the function does not automatically delete it; you should go back to the original records, experimental procedure, and protocol requirements, and perform an include/exclude sensitivity analysis when there is sufficient justification.

## 4.6 Example 4: Regression Analysis Relative to the First Node

### 4.6.1 Single-Condition Analysis Relative to the First Node

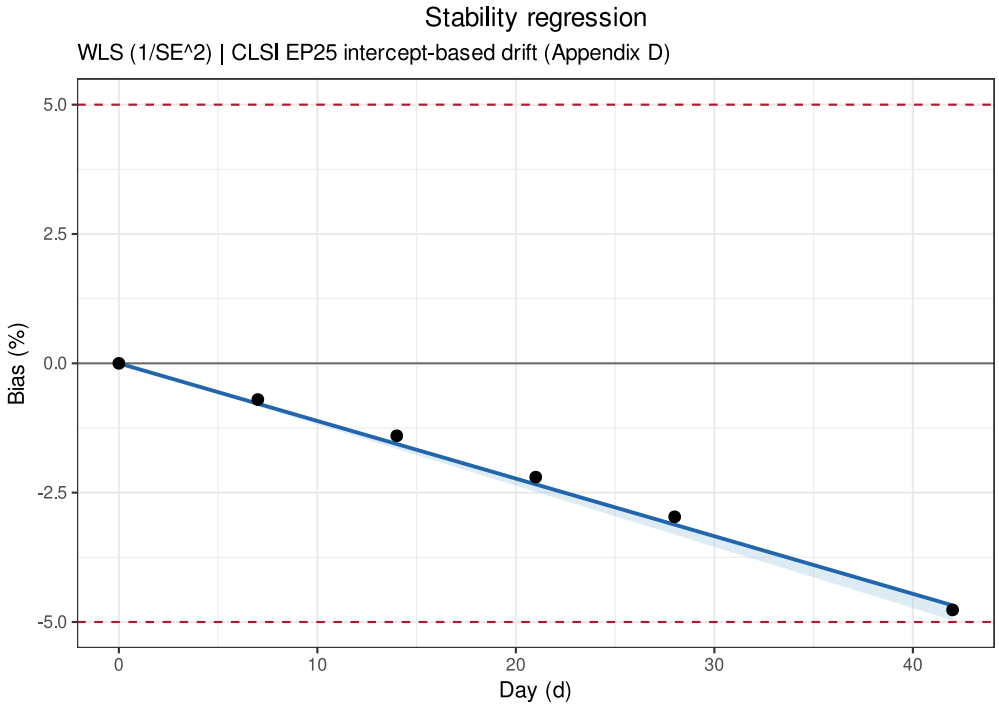
In addition to the test/reference paired regression of Example 3, `mode = "self"` can analyze how a single condition changes over time on its own. The regression example data is still used below, but only the `25C` condition is selected; the `observed_bias` at each node is computed relative to that condition's first-node mean:

```
self_regression_result <- ivdtools::stability_regression(
  data = regression_data,
  time = "Day",
  value = "Result",
  condition = "Condition",
  mode = "self",
  test_condition = "25C",
  bias_type = "relative",
  direction = "decrease",
  conf.level = 0.95,
  limit = 5,
  time_unit = "d"
)
print(self_regression_result)
```

```

Stability Regression
  Mode:            self
  Test condition:  25C
  Fit:             WLS (1/SE^2); intercept = 100.0849, slope = -0.1115
  Direction:       decrease
  One-sided CI:    95.0%
```

```
  R-squared:       0.9966
  Cook distance:   row(s) 1, 6 > 1; not removed

     Time      ObsBias   Estimate   CI_limit Status
-------------------------------------------------
   0.0000       0.0000    -0.0000    -0.0000
   7.0000      -0.7000    -0.7797    -0.8278
  14.0000      -1.4000    -1.5594    -1.6555
  21.0000      -2.2000    -2.3391    -2.4833
  28.0000      -2.9667    -3.1188    -3.3110
  42.0000      -4.7667    -4.6782    -4.9665
```

```
plot(self_regression_result)
```




The observed bias at the first node is fixed at 0; the model estimates, however, are fitted jointly from all time points and do not require the regression intercept to pass exactly through the first node. A one-sided confidence limit is used to evaluate the pre-specified least favorable direction; the illustrative ±5% limit still cannot replace the study protocol requirements.

## 4.7 Example 5: Predicting the Time to Reach a Limit

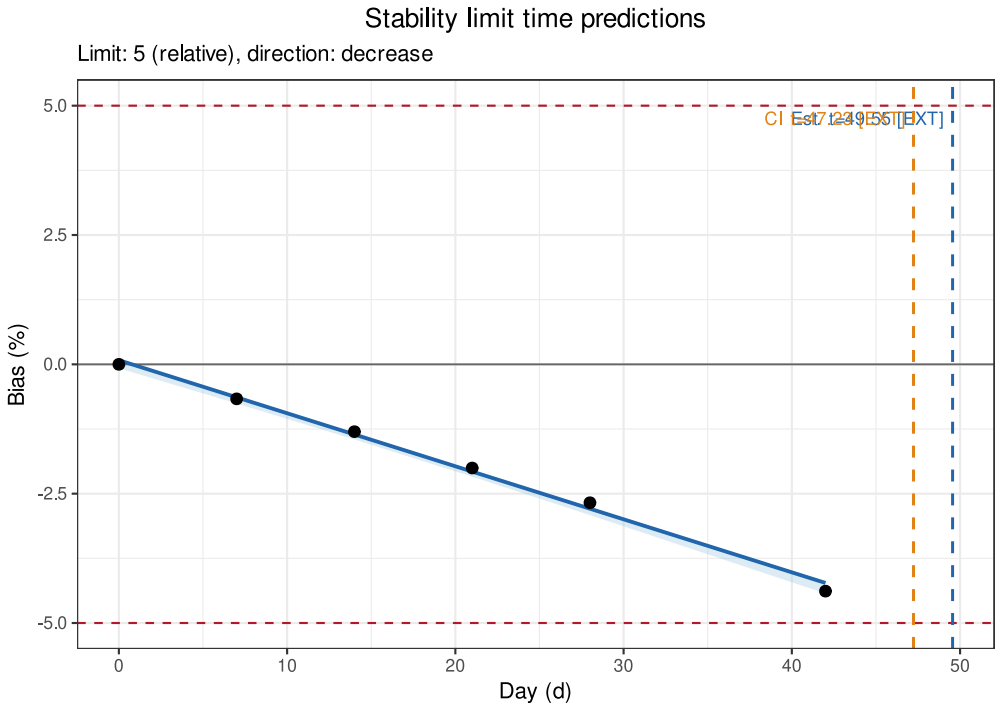
`stability_time()` uses the regression object from Example 3 to compute the times at which the point estimate and the one-sided confidence limit first reach a −5% bias:

```
time_result <- ivdtools::stability_time(
  regression_result,
  limit = 5,
  max_time = 90
)
print(time_result)
```

```

Stability Limit Time
  Limit: 5.0000 (relative); direction: decrease
```

```
  estimate       49.5548 d [EXTRAPOLATED]
  one_sided_ci   47.2316 d [EXTRAPOLATED]
```

```
plot(time_result)
```




In this example, the longest actual observation time is 42 days, while the predicted time to reach the limit exceeds the observation range, so the output is marked `EXTRAPOLATED`. Extrapolated results rely on the assumption that the linear trend continues beyond the unobserved interval and cannot be treated as measured stability evidence; report the extrapolation distance and uncertainty prominently, and prefer confirmation with longer actual observation data.

## 4.8 Example 6: Arrhenius Accelerated Stability Analysis

### 4.8.1 Reading and Validating Raw Data

```
arrhenius_data <- read_csv(
  "./data/stability-arrhenius-example.csv",
  show_col_types = FALSE
)
arrhenius_data
```

```
# A tibble: 24 × 4
   Temperature   Day Replicate Result
         <dbl> <dbl>     <dbl>  <dbl>
 1          30     0         1  100.
 2          30     0         2   99.8
 3          30     7         1   99.6
 4          30     7         2   99.4
 5          30    14         1   99.1
 6          30    14         2   98.9
 7          30    21         1   98.5
 8          30    21         2   98.3
 9          37     0         1  100.
```

```
10          37     0         2   99.9
# ℹ 14 more rows
```

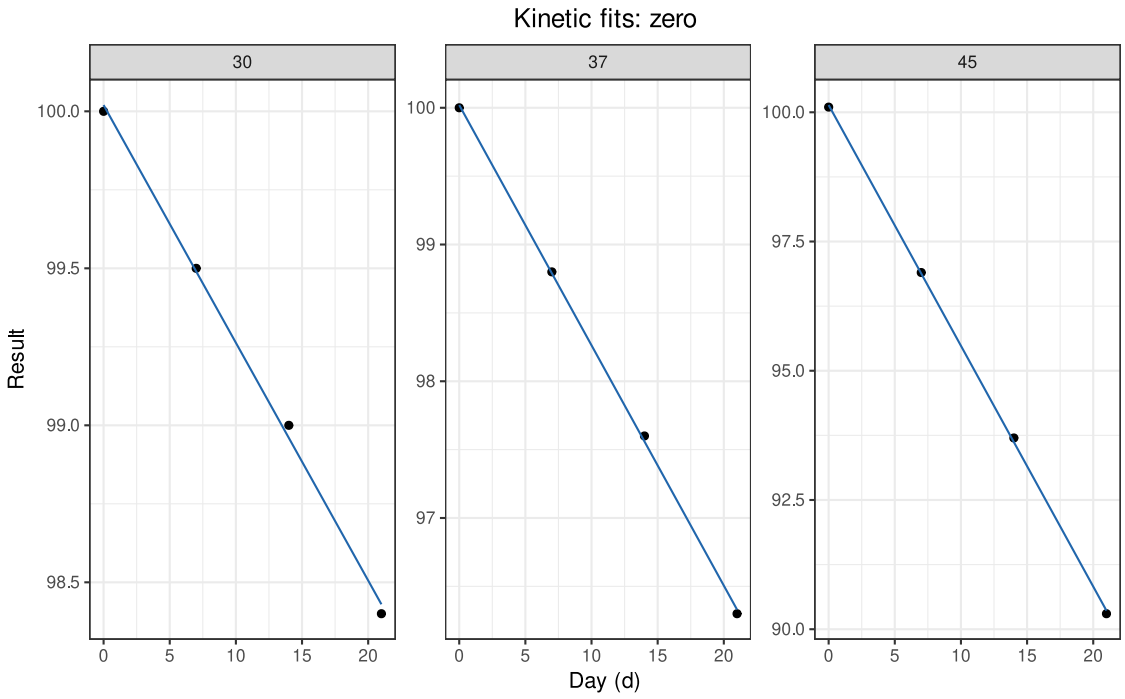
### 4.8.2 Kinetic and Arrhenius Fitting

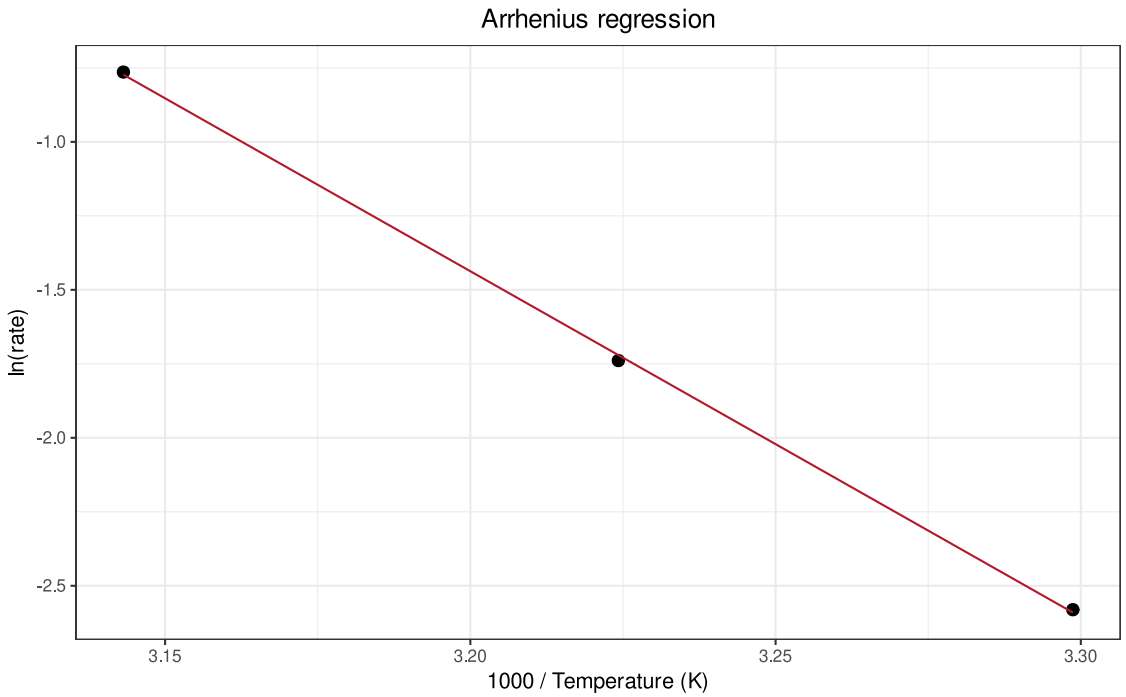
The example uses accelerated data at 30, 37, and 45 °C, extrapolated to 5 °C; allows a relative change of 10%, and pre-specifies the degradation direction as decreasing:

```
arrhenius_result <- ivdtools::arrhenius(
  data = arrhenius_data,
  temperature = "Temperature",
  time = "Day",
  value = "Result",
  target_temp = 5,
  order = "auto",
  direction = "decrease",
  limit = 10,
  bias_type = "relative",
  temp_unit = "C",
  time_unit = "d",
  conf.level = 0.95
)
print(arrhenius_result)
```

```

Arrhenius Stability Analysis
  Kinetic order:   zero (auto-selected)
  Direction:       decrease
  Activation Ea:   97.159 kJ/mol
  Target rate:     0.002346 /d
  Limit time:      4263.7878 d [1585.3589, 11467.3633]
```

```
plot(arrhenius_result)
```






`order = "auto"` selects a model from the zero-order and first-order kinetic candidates according to the implementation rules. Formal analysis should determine the candidate orders under experimental design and mechanistic knowledge, and verify the degradation direction, rate significance, extent of degradation, and model residuals at each temperature. Extrapolating from elevated temperatures to the target storage temperature may span a large range; the predicted shelf life and its interval must be interpreted together with the model assumptions, the actual time data, and validation studies.

## 4.9 Example 7: MKT Computation

### 4.9.1 Reading and Validating Raw Temperature Data

```
mkt_data <- read_csv(
  "./data/stability-mkt-example.csv",
  show_col_types = FALSE
)
mkt_data$Timestamp <- as.POSIXct(
  mkt_data$Timestamp,
  format = "%Y-%m-%d %H:%M:%S",
  tz = "UTC"
)
mkt_data
```

```
# A tibble: 8 × 3
  Timestamp           Probe_A Probe_B
  <dttm>                <dbl>   <dbl>
1 2026-07-01 00:00:00      25    24.5
2 2026-07-01 06:00:00      26    25.5
3 2026-07-01 12:00:00      29    28.5
4 2026-07-01 18:00:00      28    27.5
5 2026-07-02 00:00:00      25    24.8
6 2026-07-02 08:00:00      24    24.2
7 2026-07-02 16:00:00      27    26.5
8 2026-07-03 00:00:00      26    25.8
```

### 4.9.2 Time-Weighted MKT

Using two temperature probes and the actual time intervals, with temperature in degrees Celsius and an example activation energy of 83.144 kJ/mol:

```
mkt_result <- ivdtools::mkt(
  data = mkt_data,
  temp_cols = c("Probe_A", "Probe_B"),
  time = "Timestamp",
  temp_unit = "C",
  ea = 83.144
)
print(mkt_result)
```

```
Mean Kinetic Temperature
  Ea: 83.144 kJ/mol
  Method: Time-weighted trapezoidal MKT

  Probe_A              299.538 K     26.388 C
  Probe_B              299.194 K     26.044 C
  Combined             299.367 K     26.217 C
```

MKT is not a simple arithmetic mean of temperature; high-temperature exposure receives greater weight through the Arrhenius relationship. Formal computations should confirm probe calibration status, time ordering, missing intervals, temperature units, exposure duration, and the activation energy assumption. MKT can only summarize temperature exposure; it cannot replace product-specific stability studies.

## 4.10 Splitting Multi-Sample Data

When different concentration levels, sample types, or lots need to be evaluated separately, you should retain the sample identifier, split the data first, and then analyze each sample with the same pre-specified parameters. Read and validate the multi-sample raw data:

```
multiple_samples_data <- read_csv(
  "./data/stability-multiple-samples-example.csv",
  show_col_types = FALSE
)
multiple_samples_data
```

```
# A tibble: 36 × 4
   Sample     Day Replicate Result
   <chr>    <dbl>     <dbl>  <dbl>
 1 Sample_A     0         1  100.
 2 Sample_A     0         2   99.8
 3 Sample_A     0         3  100
 4 Sample_A     7         1   99.7
 5 Sample_A     7         2   99.5
 6 Sample_A     7         3   99.8
 7 Sample_A    14         1   99.2
 8 Sample_A    14         2   99.4
 9 Sample_A    14         3   99.1
10 Sample_A    21         1   98.7
# ℹ 26 more rows
```

Use `split()` to build a list of data named by sample, and first look at one of the samples:

```
sample_data_list <- split(
  multiple_samples_data,
  multiple_samples_data$Sample
```

```
)
names(sample_data_list)
```

```
[1] "Sample_A" "Sample_B"
```

```
sample_a_data <- sample_data_list[["Sample_A"]]
sample_a_data
```

```
# A tibble: 18 × 4
   Sample     Day Replicate Result
   <chr>    <dbl>     <dbl>  <dbl>
 1 Sample_A     0         1  100.
 2 Sample_A     0         2   99.8
 3 Sample_A     0         3  100
 4 Sample_A     7         1   99.7
 5 Sample_A     7         2   99.5
 6 Sample_A     7         3   99.8
 7 Sample_A    14         1   99.2
 8 Sample_A    14         2   99.4
 9 Sample_A    14         3   99.1
10 Sample_A    21         1   98.7
11 Sample_A    21         2   98.5
12 Sample_A    21         3   98.8
13 Sample_A    28         1   98
14 Sample_A    28         2   98.3
15 Sample_A    28         3   98.1
16 Sample_A    42         1   96.1
17 Sample_A    42         2   96.4
18 Sample_A    42         3   96.2
```

Perform the analysis on each sample in turn.

## 4.11 Key Points for Interpreting Results

1. **Protocol first**: Before analysis, specify conditions, nodes, replicates, time units, direction, bias scale, and acceptance limits.
2. **Raw data read-only**: Report missing, duplicate, non-numeric records, and every excluded row at each step; do not automatically delete anomalous records.
3. **Correct pairing**: Condition comparisons must be made at matched time points, with the test and reference conditions explicitly identified.
4. **Model diagnostics**: Report slope, confidence limits, fitting method, residuals, influential points, and whether a WLS fallback occurred.
5. **Prominent extrapolation**: Predictions of stability time beyond the observed range should be clearly labeled and not treated as measured evidence.
6. **Explicit temperature assumptions**: Arrhenius analysis and MKT must record Celsius/Kelvin units, time units, and activation energy.
7. **Separate statistics from acceptability**: Statistical significance is not the same as product acceptability; judgments can only be based on pre-specified allowable limits.
8. **Independent review**: Clinical, regulatory, or release decisions should be independently reviewed by appropriate professionals.

# 5. Precision Analysis

## 5.1 Analysis Objectives

Precision evaluates the degree of agreement among results measured on the same sample by the same measurement system under repeatable conditions, usually expressed as the standard deviation (SD) or coefficient of variation (CV%). CLSI EP05 subdivides precision into variance components by condition, such as within-run, between-run, and between-day, and provides estimates, confidence intervals, and acceptance limits for judging each component.

This document demonstrates how to use the `precision()` function in the `ivdtools` package to establish a precision variance-component analysis, complete outlier checking, normality assessment, estimation of variance components and confidence intervals, Sadler precision profile fitting, and rearrangement of the results into an EP05-style variance component table. The example data are deterministic teaching data used only to demonstrate the analysis workflow; formal studies must use the experimental design, acceptance criteria, and analysis parameters pre-specified in the protocol or standard.

```
library(ivdtools)
library(readr)
```

**Data format requirement:** `precision()` ultimately calls `VCA::anovaVCA()`, which requires a plain `data.frame`. Since `readr::read_csv()` reads in a tibble, this document consistently converts to a `data.frame` first, then converts the experimental factor columns (day, run, replicate, etc.) to factors.

## 5.2 Overview of Functions

| Function | Main purpose | Key input or output |
|---|---|---|
| `precision()` | Create a precision variance-component analysis object | Nested formula, `by` grouping, balance check |
| `outlier()` | Per-sample outlier detection | Grubbs or IQR method |
| `normal()` | Per-sample normality test | Shapiro-Wilk etc., `level` controls the confidence level |
| `variance()` / `vc()` | Estimate variance components | VC, %Total, SD, CV%, `NegVC` passed here |
| `ci()` | Confidence intervals for SD/CV of each component | Satterthwaite approximation |
| `profile()` | Sadler precision profile fitting | 10 candidate models, AIC selection |
| `list_sadler()` | View the 10 Sadler models | Model formula and type |
| `summary()` / `plot()` | Summarize and plot | `plot(type=...)` selects the graphic |

After `precision()` creates the object, analysis results accumulate into the object step by step through `outlier()`, `normal()`, `variance()`, `ci()`, and `profile()`, and **must be reassigned**:

```
precision() -> outlier() -> normal() -> variance() -> ci() -> profile()
```

Only analyses that have been executed appear in `summary()`, and some `plot()` graphics depend on prior analyses.

## 5.3 Example 1: EP05 20×2×2 Single-Sample Precision

### 5.3.1 Reading and Validating Data

20×2×2 means 20 days, 2 runs per day, and 2 replicates per run, for a total of 80 results; it is one of the classic EP05 designs:

```
ep05 <- read_csv("./data/precision-ep05.csv", show_col_types = FALSE)
ep05 <- as.data.frame(ep05)
str(ep05)
```

```
'data.frame':   80 obs. of  4 variables:
 $ day  : num  1 1 1 1 2 2 2 2 3 3 ...
 $ run  : num  1 1 2 2 1 1 2 2 1 1 ...
 $ rep  : num  1 2 1 2 1 2 1 2 1 2 ...
 $ value: num  96.9 93.5 99.3 101.5 103 ...
```

The experimental factor columns are integers in the CSV and are converted to factors after reading:

```
ep05$day <- factor(ep05$day)
ep05$run <- factor(ep05$run)
ep05$rep <- factor(ep05$rep)
```

```
dim(ep05)
```

```
[1] 80  4
```

```
summary(ep05$value)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  86.06   94.83   99.33   99.11  103.53  115.01
```

`value ~ day/run` means the variance components are nested layer by layer as between-day (day) -> within-day run (run). `/` is the nesting notation, equivalent to treating each deeper factor as a factor nested within the factor above it.

### 5.3.2 Creating the Analysis Object and Pre-Check

```
p <- precision(ep05, value ~ day/run)
p
```

```
Precision analysis object
  Data class          : data.frame
  Total rows          : 80
  Formula             : value ~ day/run
  Samples             : 1

Factor levels
  day         : 1 (4), 2 (4), 3 (4), 4 (4), 5 (4), 6 (4), 7 (4), 8 (4), 9 (4), 10 (4),
11 (4), 12 (4), 13 (4), 14 (4), 15 (4), 16 (4), 17 (4), 18 (4), 19 (4), 20 (4)
  run         : 1 (40), 2 (40)
  rep         : 1 (40), 2 (40)

Analysis status

  [ ] outlier
  [ ] normal
  [ ] variance
```

```
  [ ] ci
  [ ] profile
```

The balance check output reports the ratio of the largest to smallest cell size; when the data are highly unbalanced, `VCA` will suggest switching to REML.

Perform per-sample outlier detection and normality testing:

```
p <- outlier(p, method = "grubbs")
```

```
Outlier detection -- Grubbs (alpha = 0.05)
  No outliers detected.
```

```
p <- normal(p, method = "shapiro")
```

```
Normality test -- Shapiro-Wilk
  Sample               n           W  p-value
  ------------------------------------------
  all                 80      0.9868  0.5858
```

**Parameter notes:** The `method` of `outlier()` can be `"grubbs"` (default) or `"iqr"`, and `alpha` is the Grubbs significance level. For `normal()`, `method="auto"` selects the test based on sample size (Shapiro-Wilk for n≤50, Anderson-Darling for larger samples); when `method="shapiro"` is specified, the reported `W` column is the true Shapiro-Wilk W statistic. For single-sample data with n=80, either approach is acceptable; this example specifies it explicitly for illustration.

### 5.3.3 Variance Components and Confidence Intervals

```
p <- variance(p)
```

```
Variance components
 sample component      VC %Total     SD  CV[%]
-----------------------------------------------
    all       day 18.8358   42.6 4.3400 4.3788
    all   day:run 17.4343   39.4 4.1754 4.2128
    all     error  7.9482   18.0 2.8193 2.8445
```

The `NegVC` parameter needs to be passed at `variance()`:

```
p <- variance(p, NegVC = TRUE)   # allow negative variance components (default FALSE)
```

```
p <- ci(p)
```

```
Confidence intervals (SD)
 sample component estimate  lower  upper
----------------------------------------
    all     total   6.6497 5.4288 8.5842
    all       day   4.3400 0.0000 6.2255
    all   day:run   4.1754 2.0129 5.5513
    all     error   2.8193 2.3147 3.6073

Confidence intervals (%CV)
 sample component estimate  lower  upper
----------------------------------------
    all     total   6.7091 5.4773 8.6609
    all       day   4.3788 0.0000 6.2812
```

```
    all   day:run   4.2128 2.0309 5.6009
    all     error   2.8445 2.3353 3.6395
```

`ci()` uses the Satterthwaite effective-degrees-of-freedom approximation to compute confidence intervals for the SD and CV of each component.

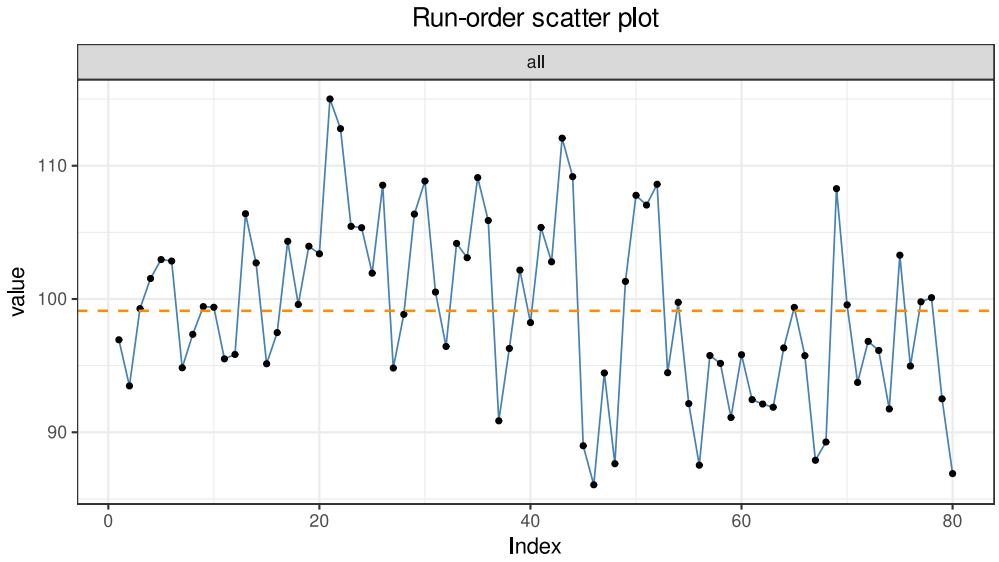
### 5.3.4 Converting to an EP05 Terminology Table

Classify the variance components into within-run, between-run, and between-day by the nesting depth of the component name, compute the total SD/CV, and further merge the component confidence intervals to form a complete EP05 report table with intervals. **Whether the statistical results are acceptable must be judged against the precision limits pre-specified in the protocol.**

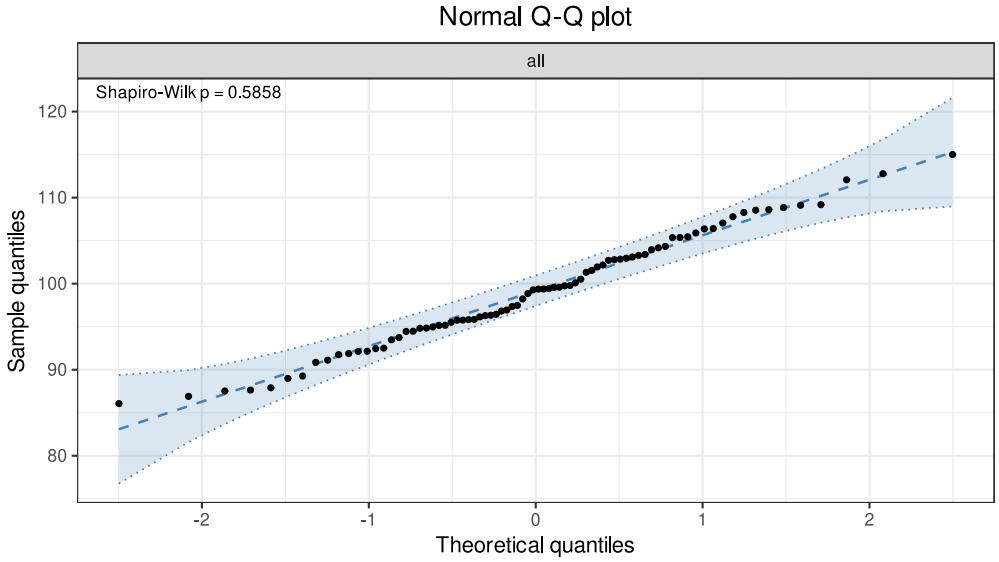
### 5.3.5 Graphics

A scatter plot can be used to quickly identify test anomalies and the distribution of measurements:

```
plot(p, type = "dot")
```




```
plot(p, type = "qq")
```



**Graphic dependencies:** `plot(type="dot")` is a scatter plot by run order (no prior analysis needed), `type="his"` is a standard histogram; `type="var"` requires `variance()` to have been run, `type="qq"` requires `normal()`, and `type="profile"` requires `profile()`. Confirm that the corresponding analysis has been run before plotting.

```
summary(p)
```

```
Precision analysis -- summary
---------------------------------------

Precision analysis object
  Data class          : data.frame
  Total rows          : 80
  Formula             : value ~ day/run
  Samples             : 1

Factor levels
  day         : 1 (4), 2 (4), 3 (4), 4 (4), 5 (4), 6 (4), 7 (4), 8 (4), 9 (4), 10 (4),
11 (4), 12 (4), 13 (4), 14 (4), 15 (4), 16 (4), 17 (4), 18 (4), 19 (4), 20 (4)
  run         : 1 (40), 2 (40)
  rep         : 1 (40), 2 (40)

Analysis status

  [x] outlier
  [x] normal
  [x] variance
  [x] ci
  [ ] profile

Outlier detection -- Grubbs (alpha = 0.05)
  No outliers detected.

Normality test -- Shapiro-Wilk
  Sample               n           W  p-value
  ------------------------------------------------
  all                 80      0.9868  0.5858

Variance components
 sample component      VC %Total     SD  CV[%]
-----------------------------------------------
    all       day 18.8358   42.6 4.3400 4.3788
    all   day:run 17.4343   39.4 4.1754 4.2128
    all     error  7.9482   18.0 2.8193 2.8445

Confidence intervals (SD)
 sample component estimate  lower  upper
----------------------------------------
    all     total   6.6497 5.4288 8.5842
    all       day   4.3400 0.0000 6.2255
    all   day:run   4.1754 2.0129 5.5513
    all     error   2.8193 2.3147 3.6073

Confidence intervals (%CV)
 sample component estimate  lower  upper
----------------------------------------
    all     total   6.7091 5.4773 8.6609
    all       day   4.3788 0.0000 6.2812
```

```
    all   day:run   4.2128 2.0309 5.6009
    all     error   2.8445 2.3353 3.6395
```

## 5.4 Example 2: Multi-Sample 3×5 and Sadler Profile

### 5.4.1 Reading Data

When multiple concentration levels need to be evaluated, use `by` to specify the sample grouping column, and estimate the variance components separately for each sample:

```
profile_data <- as.data.frame(read_csv(
  "./data/precision-profile.csv",
  show_col_types = FALSE
))
profile_data$sample <- factor(profile_data$sample)
profile_data$day <- factor(profile_data$day)
profile_data$rep <- factor(profile_data$rep)
```

### 5.4.2 Per-Sample Variance Components

```
p2 <- precision(profile_data, value ~ day, by = "sample")
p2 <- variance(p2)
```

```
Variance components
 sample component       VC %Total      SD  CV[%]
-------------------------------------------------
     L1       day   0.6376   63.0  0.7985 3.9446
     L1     error   0.3750   37.0  0.6124 3.0250
     L2       day   5.7139   75.1  2.3904 4.7584
     L2     error   1.8937   24.9  1.3761 2.7393
     L3       day   3.3913   15.7  1.8415 1.8088
     L3     error  18.1466   84.3  4.2599 4.1840
     L4       day  30.5549   54.6  5.5276 2.7529
     L4     error  25.4525   45.4  5.0451 2.5126
     L5       day   0.0000    0.0  0.0000 0.0000
     L5     error 171.7986  100.0 13.1072 3.3003
```

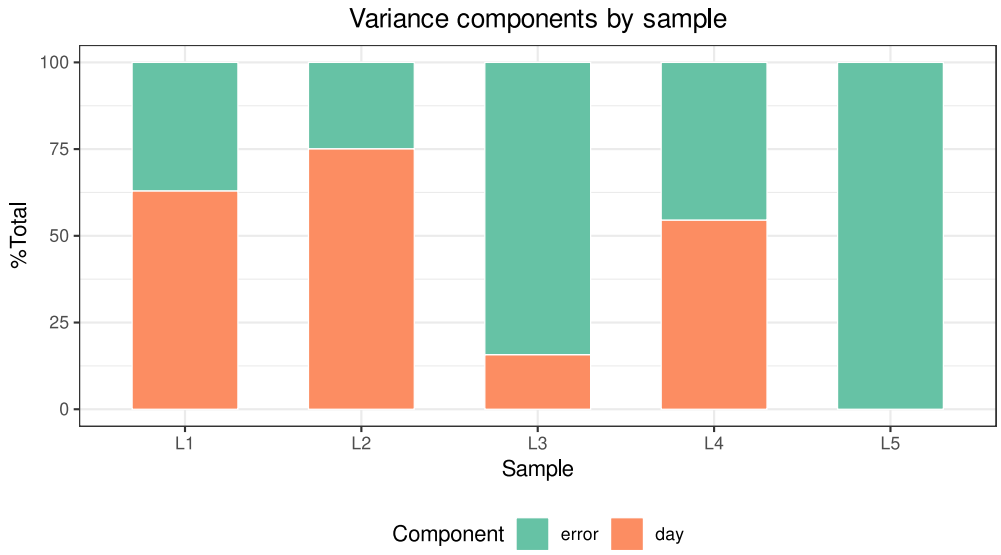
```
p2 <- ci(p2)
```

```
Confidence intervals (SD)
 sample component estimate  lower   upper
----------------------------------------
     L1     total   1.0063 0.6007  2.9368
     L1       day   0.7985 0.0000  1.4267
     L1     error   0.6124 0.4391  1.0108
     L2     total   2.7582 1.5714  9.9889
     L2       day   2.3904 0.0000  4.2023
     L2     error   1.3761 0.9868  2.2716
     L3     total   4.6409 3.2872  7.8819
     L3       day   1.8415 0.0000  4.1779
     L3     error   4.2599 3.0547  7.0319
     L4     total   7.4838 4.6166 19.1652
     L4       day   5.5276 0.0000 10.0268
     L4     error   5.0451 3.6177  8.3280
     L5     total  13.1072 9.5629 20.8255
     L5       day   0.0000     NA      NA
     L5     error  13.1072 9.3990 21.6365
```

```
Confidence intervals (%CV)
 sample component estimate  lower   upper
----------------------------------------
     L1     total   4.9710 2.9673 14.5079
     L1       day   3.9446 0.0000  7.0480
     L1     error   3.0250 2.1692  4.9936
     L2     total   5.4905 3.1281 19.8843
     L2       day   4.7584 0.0000  8.3652
     L2     error   2.7393 1.9643  4.5219
     L3     total   4.5583 3.2287  7.7416
     L3       day   1.8088 0.0000  4.1035
     L3     error   4.1840 3.0003  6.9067
     L4     total   3.7271 2.2992  9.5447
     L4       day   2.7529 0.0000  4.9936
     L4     error   2.5126 1.8017  4.1476
     L5     total   3.3003 2.4078  5.2436
     L5       day   0.0000     NA      NA
     L5     error   3.3003 2.3666  5.4478
```

Comparing the proportion of variability across samples can be used for product optimization and testing scheme design:

```
plot(p2, type = "var")
```




### 5.4.3 Precision Profile

`profile()` fits the Sadler model family to each variance component and selects the best model by AIC:

```
list_sadler()
```

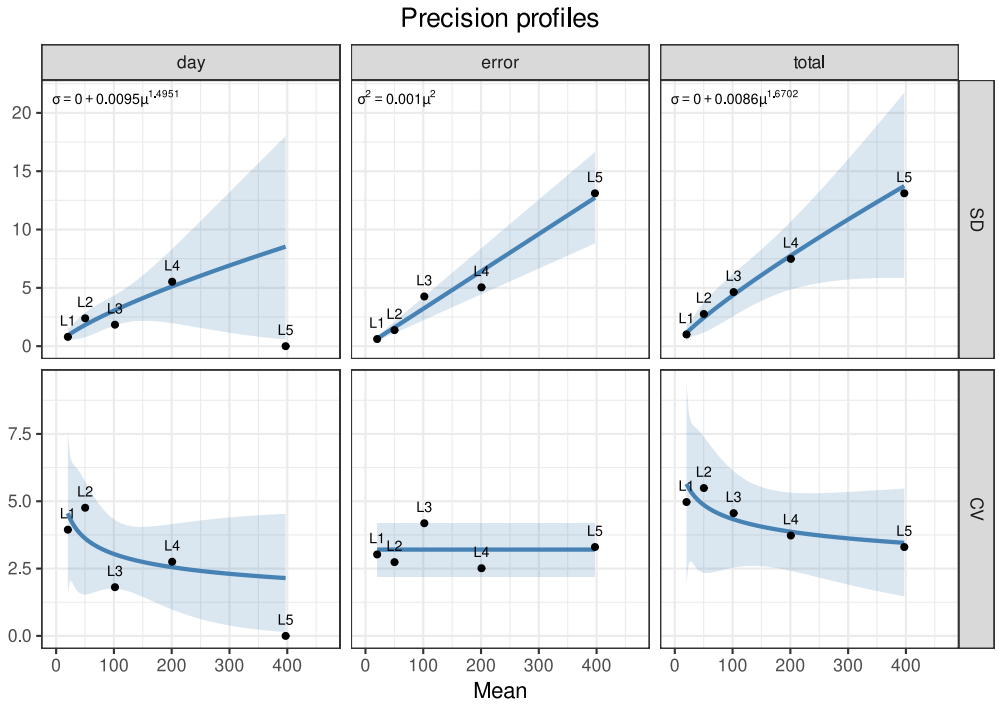
```
Sadler Precision Profile Models

   1  Constant SD           sigma^2 = b1
   2  Constant CV           sigma^2 = b1 * mu^2
   3  Linear (variance)     sigma^2 = b1 + b2 * mu
   4  Power (fixed K)       sigma^2 = b1 * mu^K
   5  Power (fixed K, +intercept)  sigma^2 = b1 + b2 * mu^K
   6  Linear (SD)           sigma = b1 + b2 * mu
   7  Power (SD)            sigma = b1 + b2 * mu^b3
```

```
   8  Power (variance)      sigma^2 = b1 + b2 * mu^b3
   9  Exponential (log-log)  log(sigma^2) = b1 + b2 * log(mu)
  10  Power (full)          sigma^2 = b1 * mu^b2
```

```
p2 <- profile(p2, model.no = 1:10)
```

```
Precision profiles -- Sadler
  total: sigma = 0.0000 + 0.0086 * mu^1.6702  (AIC = 56.45, R^2 = 0.9847)
  day: sigma = 0.0000 + 0.0095 * mu^1.4951  (AIC = 43.46, R^2 = 0.8919)
  error: sigma^2 = 0.0010 * mu^2  (AIC = 55.96, R^2 = 0.9806)
```

```
plot(p2, type = "profile")
```




**Parameter notes:** `profile(model.no = 1:10)` specifies the candidate model numbers; `...` can be passed to the internal Sadler fit (for example `K`). Each variance component requires at least 3 samples with a finite SD to be fitted, so a multi-sample design should include at least 3 concentration levels. The profile can be used, for example, in functional-sensitivity analyses to predict the SD/CV at any concentration (see 《Functional Sensitivity》).

In this example, the best models for the total and day components are power-function type (model 7), and the day:rep component is a constant-CV type (model 2), consistent with the common setting where "SD increases with concentration while CV is relatively stable".

## 5.5 Example 3: Multi-Sample Multi-Site (Site) Precision

### 5.5.1 Reading Data

The data contain 2 sites, 3 samples, and per site per sample 3 days × 3 runs × 5 replicates:

```
sites_data <- as.data.frame(read_csv(
  "./data/precision-sites.csv",
  show_col_types = FALSE
```

```
))
sites_data[] <- lapply(sites_data, function(x)
  if (is.numeric(x) && all(x == round(x))) factor(x) else x)
str(sites_data)
```

```
'data.frame':   270 obs. of  6 variables:
 $ site  : chr  "A" "A" "A" "A" ...
 $ sample: chr  "S1" "S1" "S1" "S1" ...
 $ day   : Factor w/ 3 levels "1","2","3": 1 1 1 1 1 1 1 1 1 1 ...
 $ run   : Factor w/ 3 levels "1","2","3": 1 1 1 1 1 2 2 2 2 2 ...
 $ rep   : Factor w/ 5 levels "1","2","3","4",..: 1 2 3 4 5 1 2 3 4 5 ...
 $ value : num  49.2 52.3 48.8 49.4 51.5 ...
```

### 5.5.2 Multi-Column Grouping

`by` can accept multiple column names and groups internally by the interaction, obtaining all site × sample combinations at once:

```
p3 <- precision(sites_data, value ~ day/run, by = c("sample", "site"))
p3 <- variance(p3)
```

```
Variance components
 sample component      VC %Total     SD  CV[%]
-----------------------------------------------
   S1.A       day  0.0000    0.0 0.0000 0.0000
   S1.A   day:run  0.1646    6.8 0.4057 0.8231
   S1.A     error  2.2677   93.2 1.5059 3.0553
   S2.A       day  0.3241    4.3 0.5693 0.5602
   S2.A   day:run  0.1523    2.0 0.3903 0.3840
   S2.A     error  7.0483   93.7 2.6549 2.6124
   S3.A       day  4.4871   15.6 2.1183 1.0420
   S3.A   day:run  1.7006    5.9 1.3041 0.6415
   S3.A     error 22.5797   78.5 4.7518 2.3375
   S1.B       day  0.0000    0.0 0.0000 0.0000
   S1.B   day:run  0.5459   23.5 0.7389 1.4758
   S1.B     error  1.7784   76.5 1.3335 2.6637
   S2.B       day  0.0000    0.0 0.0000 0.0000
   S2.B   day:run  1.6690   21.5 1.2919 1.2891
   S2.B     error  6.0856   78.5 2.4669 2.4616
   S3.B       day  0.7633    2.3 0.8737 0.4448
   S3.B   day:run  0.0000    0.0 0.0000 0.0000
   S3.B     error 31.8033   97.7 5.6394 2.8709
```

```
names(p3$results)
```

```
[1] "S1.A" "S2.A" "S3.A" "S1.B" "S2.B" "S3.B"
```

### 5.5.3 Splitting by Site for Separate Analysis

An equivalent approach is to first `split()` by site and then run the same analysis parameters on each site subset. When the variance-component structure is expected to differ between sites, or separate reports are needed, splitting and analyzing separately is clearer:

```
for (s in c("A", "B")) {
  sub <- sites_data[sites_data$site == s, ]
  ps <- precision(sub, value ~ day/run, by = "sample")
  ps <- variance(ps)
```

```
  ps <- ci(ps)
}
```

**Note:** Multi-center studies should unify the analysis plan, variance-component model, and reporting convention; when "reporting per center first and then combining", be sure to record whether each center's raw data were included, and whether the exclusion criteria and factor definitions are consistent.

## 5.6 Example 4: Custom More Complex Models

### 5.6.1 Reading Data

Besides the standard day/run/replicate nesting, the `precision()` formula can be any VCA nested structure, for example adding an operator level: `value ~ operator/day/run`, i.e., nested layer by layer as operator -> day -> run:

```
custom_data <- as.data.frame(read_csv(
  "./data/precision-custom.csv",
  show_col_types = FALSE
))
custom_data[] <- lapply(custom_data, function(x)
  if (is.numeric(x) && all(x == round(x))) factor(x) else x)
```

### 5.6.2 Fitting and Interpretation

```
p4 <- precision(custom_data, value ~ operator/day/run)
p4 <- variance(p4)
```

```
Variance components
 sample        component     VC %Total     SD  CV[%]
-------------------------------------------------------
    all         operator 2.2914   11.8 1.5137 1.4898
    all     operator:day 6.3159   32.4 2.5132 2.4734
    all operator:day:run 3.2463   16.7 1.8017 1.7733
    all            error 7.6322   39.2 2.7627 2.7190
```

## 5.7 Key Points for Interpreting Results

1. **Design match**: First clarify the nesting structure of the study design; the `precision()` formula must faithfully reflect the experimental hierarchy.
2. **Data format**: After `read_csv`, convert to `data.frame` and explicitly convert factor columns to factor to avoid VCA errors or ambiguity.
3. **Call order**: `outlier → normal → variance → ci → profile`, reassign after each analysis; plotting depends on prior analyses.
4. **Negative components**: When a factor has too few levels or the data happen to be nearly identical, a variance component may be set to zero; interpret carefully and specify the handling approach in the protocol.
5. **CI applicability**: The Satterthwaite approximation becomes less accurate with very few variance components or unbalanced data; note the method in the report.
6. **Terminology table**: Rearrange the numeric quantities in `$results` (`sd_comp`/`cv_comp`/`vc`) into an EP05 terminology table; do not directly use the display-oriented string `$vc`.
7. **Acceptability judgment**: Statistical estimates do not automatically constitute a pass/fail determination; they should be compared with pre-specified precision limits and reviewed by professionals.

# 6. Linearity Analysis

## 6.1 Analysis Objectives

Linearity evaluates whether there is an acceptable linear relationship between the measured result and the analyte concentration (or target value) throughout the entire working range of a measurement system, usually using dilution data and analyzing bias per CLSI EP06. This document demonstrates how to use the `fit_equation()` function in the `ivdtools` package to complete linear fitting, weighted fitting, evaluation of bias and residuals at each level, and EP06-A2-style polynomial analysis to determine whether higher-order terms are needed.

The example data are deterministic teaching data used only to demonstrate the workflow; formal studies should use the dilution scheme, concentration levels, replicate counts, and acceptance criteria pre-specified in the protocol.

```
library(ivdtools)
library(readr)
```

## 6.2 Overview of Functions

| Function | Main purpose | Key input or output |
|---|---|---|
| `list_equation()` | View the built-in equation registry | `category`, `engine` filters |
| `replicate_to_mean()` | Summarize replicates and optionally compute weights | `"n"`, `"1/sd"`, `"1/sd^2"` |
| `fit_equation()` | Fit built-in or custom equations | `eq`, `weights`, `lower/upper`, `constraints` |
| `compare_equation()` | Fit multiple candidate equations and rank by AIC | `eqs`, deltaAIC |
| `coef()` / `residuals()` | Extract parameters and residuals | |
| `predict()` | Forward prediction or inverse back-calculation of concentration | `inverse=TRUE`, `interval` |
| `plot()` | Plot fitted curve and intervals | `interval`, `frame` |

## 6.3 Example 1: Linearity Bias of a Serial Dilution

### 6.3.1 Reading and Validating Data

The dilution series contains 8 concentration levels with 3 replicates each, for a total of 24 rows:

```
dilution <- read_csv("./data/linearity-dilution.csv", show_col_types = FALSE)
dilution <- as.data.frame(dilution)
head(dilution, 6)
```

```
  conc replicate signal
1   50         1   48.6
```

```
2   50         2   49.2
3   50         3   50.0
4  100         1  101.3
5  100         2   99.7
6  100         3  101.4
```

```
dim(dilution)
```

```
[1] 24  3
```

```
summary(dilution)
```

```
      conc        replicate     signal
 Min.   :  50   Min.   :1   Min.   :  48.6
 1st Qu.: 175   1st Qu.:1   1st Qu.: 173.6
 Median : 600   Median :2   Median : 595.6
 Mean   :1594   Mean   :2   Mean   :1593.0
 3rd Qu.:2000   3rd Qu.:3   3rd Qu.:2004.5
 Max.   :6400   Max.   :3   Max.   :6406.9
```

### 6.3.2 Viewing Available Response Equations

```
list_equation(category = "Response")
```

```
  ID       Name                          Formula        Params nP Engine
-------------------------------------------------------------------------
 E01     Linear                             a + b*x          a, b  2     lm
 E02  Quadratic                   a + b*x + c*x^2       a, b, c  3     lm
 E03 ExpoGrowth                    a * exp(b * x)          a, b  2    nls
 E04  ExpoDecay                   a * exp(-b * x)          a, b  2    nls
 E05      Power                           a * x^b          a, b  2    nls
 E06        Log                    a + b * log(x)          a, b  2     lm
 E07       4PLC   A + (B - A) / (1 + (x / C)^D)    A, B, C, D  4    nls
 E08       5PLC A + (B - A) / (1 + (x / C)^D)^E A, B, C, D, E  5    nls
 E09   Gaussian a * exp(-(x - b)^2 / (2 * c^2))       a, b, c  3    nls
 E10  Michaelis             Vmax * x / (Km + x)      Vmax, Km  2    nls
 E11 Sinusoidal          a * sin(b * x + c) + d    a, b, c, d  4    nls
 E12      Cubic         a + b*x + c*x^2 + d*x^3    a, b, c, d  4     lm
```

The linear equation corresponds to registry ID `E01` (the name `"Linear"` is also accepted).

### 6.3.3 Summarizing Replicate Measurements

First use `replicate_to_mean()` to obtain the mean, SD, and replicate count at each level.

```
rep_sum <- replicate_to_mean(dilution, x = "conc", y = "signal")
rep_sum
```

```
     x     y_mean       y_sd y_n
1   50   49.26667  0.7023769   3
2  100  100.80000  0.9539392   3
3  200  202.66667  4.4523402   3
4  400  397.83333  7.3961702   3
5  800  798.86667 10.2617412   3
6 1600 1596.90000 13.7535450   3
7 3200 3207.80000 20.9408691   3
8 6400 6389.56667 19.9234368   3
```

### 6.3.4 Linear Fitting and Bias

```
fit_lin <- fit_equation("E01", data = rep_sum, x = "x", y = "y_mean")
print(fit_lin)
```

```
E01 (Linear)
Formula: a + b*x

Call:
  stats::lm(formula = frm, data = d, weights = weights)

Residuals:
      Min       1Q   Median       3Q      Max
    -4.330   -2.404   -1.439    0.430   10.364

Coefficients:
  Estimate  Std. Error    t value Pr(>|t|)
a 0.975290 2.205878722    0.44213  0.67388
b 0.998894 0.000844269 1183.14720  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.944 on 6 degrees of freedom
Multiple R-squared: 1.0000,  Adjusted R-squared: 1.0000
F-statistic: 1.39984e+06 on 1 and 6 DF,  p-value: <2e-16
```

Compute the original residual (the linearity bias, i.e., observed mean minus predicted) and the recovery bias (observed mean minus theoretical value):

```
predict <- predict(fit_lin)
```

```
    x          y
   50   50.91999
  100  100.86468
  200  200.75408
  400  400.53286
  800  800.09044
 1600 1599.20559
 3200 3197.43588
 6400 6393.89648
```

```
lin_diag <- data.frame(
  conc = rep_sum$x,
  signal = rep_sum$y_mean,
  predict = predict$y,
  Residual = residuals(fit_lin),
  Recover_bias = rep_sum$y_mean - rep_sum$x
)
print(lin_diag)
```

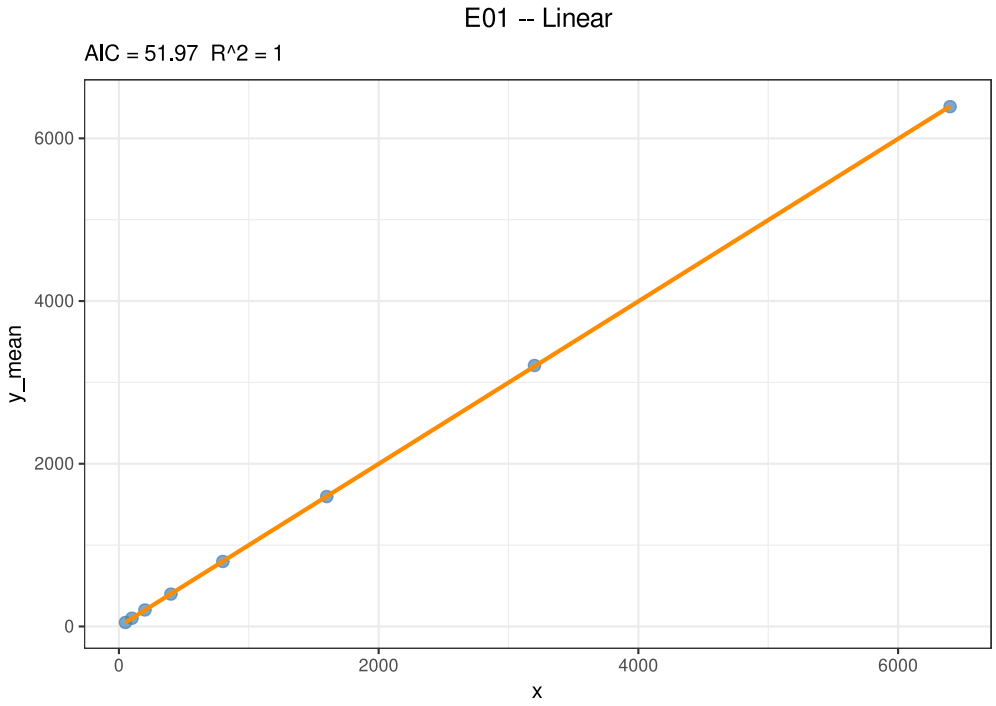
```
  conc     signal    predict    Residual Recover_bias
1   50   49.26667   50.91999 -1.65332058   -0.7333333
2  100  100.80000  100.86468 -0.06468401    0.8000000
3  200  202.66667  200.75408  1.91258912    2.6666667
4  400  397.83333  400.53286 -2.69953130   -2.1666667
5  800  798.86667  800.09044 -1.22377212   -1.1333333
6 1600 1596.90000 1599.20559 -2.30558710   -3.1000000
7 3200 3207.80000 3197.43588 10.36411628    7.8000000
8 6400 6389.56667 6393.89648 -4.32981030  -10.4333333
```

**Whether the bias is acceptable must be compared with the linearity acceptance limits pre-specified in the protocol.**

**Parameter notes:** The `weights` of `fit_equation()` accepts a numeric vector or the built-in schemes `"equal"`, `"1/y"`, `"1/y^2"`, `"1/x"`, `"1/x^2"`, `"inverse"`, `"inverse2"`. Weighted and equal-weight fits should both be compared comprehensively in terms of residual structure, back-calculation bias, and pre-specified acceptance criteria; do not rely on a single fit metric.

### 6.3.5 Graphics and Prediction

```
plot(fit_lin, interval = "confidence", level = 0.95)
```




```
predict(
  fit_lin,
  newdata = data.frame(x = c(0.5, 2, 8)),
  interval = "confidence"
)
```

```
   x        y  y_ci_lwr  y_ci_upr
0.5 1.474737 -3.922223  6.871698
2.0 2.973078 -2.421993  8.368150
8.0 8.966442  3.578916 14.353968
```

Back-calculate concentration from response:

```
predict(
  fit_lin,
  newdata = data.frame(y_mean = c(650, 3300, 6500)),
  inverse = TRUE
)
```

```
        x    y
649.7434  650
```

```
3302.6777 3300
6506.2210 6500
```

**Prediction boundaries:** Inverse back-calculation should avoid extrapolating beyond the calibration concentration range. Linear back-calculation within range is stable, but in plateau or nonlinear regions, inverse prediction is very sensitive to response error.

## 6.4 Example 2: Polynomial Analysis of a Twofold Dilution

### 6.4.1 Reading and Validating Data

The data are 5 concentration levels × 3 replicates:

```
poly_data <- read_csv("./data/linearity-polynomial.csv", show_col_types = FALSE)
poly_data <- as.data.frame(poly_data)
head(poly_data, 6)
```

```
    conc replicate signal
1  75.00         1   75.0
2  75.00         2   75.0
3  75.00         3   74.4
4 483.75         1  454.1
5 483.75         2  455.3
6 483.75         3  456.0
```

### 6.4.2 Compute replicate means and introduce weights:

```
rep_sum_w <- replicate_to_mean(
  poly_data,
  x = "conc",
  y = "signal",
  weights = "1/sd^2"
)
rep_sum_w
```

```
        x    y_mean       y_sd y_n     weight
1   75.00   74.8000  0.3464102   3 8.33333333
2  483.75  455.1333  0.9609024   3 1.08303249
3  892.50  865.9667  8.0257918   3 0.01552474
4 1301.25 1328.4667 24.4933324   3 0.00166688
5 1710.00 1710.4333  8.1193185   3 0.01516914
```

**Parameter notes:** The `weights` of `replicate_to_mean()` can be `NULL` (no weight column generated), `"n"`, `"1/sd"`, or `"1/sd^2"` (inverse-variance weighting). Weights should be determined based on the actual variance structure of the replicates or the protocol.

### 6.4.3 Candidate Equation Comparison

EP06-A2 typically compares linear, quadratic, and cubic models to determine whether higher-order terms are needed and to compare the residual standard errors:

```
fit_poly_lin <- fit_equation("E01", data = rep_sum_w, x = "x", y = "y_mean", weights
= rep_sum_w$weights)
print(fit_poly_lin)
```

```
E01 (Linear)
Formula: a + b*x

Call:
  stats::lm(formula = frm, data = d, weights = weights)

Residuals:
      Min       1Q   Median       3Q      Max
    -20.993  -17.367   -5.447   16.760   27.047

Coefficients:
   Estimate Std. Error  t value   Pr(>|t|)
a -18.00771 20.0633790 -0.89754    0.43557
b   1.01397  0.0188681 53.73983 1.4192e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24.3886 on 3 degrees of freedom
Multiple R-squared: 0.9990,  Adjusted R-squared: 0.9986
F-statistic: 2887.97 on 1 and 3 DF,  p-value: 1.42e-05
```

```
fit_poly_qua <- fit_equation("E02", data = rep_sum_w, x = "x", y = "y_mean", weights
= rep_sum_w$weights)
print(fit_poly_qua)
```

```
E02 (Quadratic)
Formula: a + b*x + c*x^2

Call:
  stats::lm(formula = frm, data = d, weights = weights)

Residuals:
      Min       1Q   Median       3Q      Max
    -13.443  -13.294  -13.146    8.912   30.970

Coefficients:
      Estimate  Std. Error t value  Pr(>|t|)
a -7.14812e+00 3.09846e+01 -0.2307 0.8389994
b  9.72049e-01 8.28550e-02 11.7319 0.0071872 **
c  2.34851e-05 4.48020e-05  0.5242 0.6524439
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 28.0077 on 2 degrees of freedom
Multiple R-squared: 0.9991,  Adjusted R-squared: 0.9982
F-statistic: 1095.05 on 2 and 2 DF,  p-value: 0.000912
```

```
fit_poly_cub <- fit_equation("E12", data = rep_sum_w, x = "x", y = "y_mean", weights
= rep_sum_w$weights)
print(fit_poly_cub)
```

```
E12 (Cubic)
Formula: a + b*x + c*x^2 + d*x^3

Call:
  stats::lm(formula = frm, data = d, weights = weights)

Residuals:
      Min       1Q   Median       3Q      Max
```

```
    -13.146   -2.191   -2.191    8.764    8.764

Coefficients:
      Estimate  Std. Error  t value Pr(>|t|)
a  2.04822e+01 2.48863e+01  0.82303  0.56161
b  7.25244e-01 1.39796e-01  5.18788  0.12123
c  3.86253e-04 1.91646e-04  2.01544  0.29321
d -1.35487e-07 7.07339e-08 -1.91545  0.30631

Residual standard error: 18.3308 on 1 degrees of freedom
Multiple R-squared: 0.9998,  Adjusted R-squared: 0.9992
F-statistic: 1705.48 on 3 and 1 DF,  p-value: 0.0178
```

## 6.5 Key Points for Interpreting Results

1. **Separate fit from model judgment**: A model being successfully fitted does not mean it is appropriate; judge in light of residuals, bias, acceptance criteria, and the protocol.
2. **Residuals on two scales**: Examine both original and relative residuals; for data spanning orders of magnitude, focus on relative residuals.
3. **Weight rationale**: Prefer the variance structure of the replicates or protocol requirements.
4. **Range and extrapolation**: Conclusions are valid only within the validated concentration range; avoid inverse prediction in plateau regions or beyond range.
5. **Reproducibility**: Save the raw data, code, model parameters, convergence information, package version, and complete session information.

# 7. Analytical Sensitivity

## 7.1 Analysis Objectives

Analytical sensitivity describes the ability of a measurement system to distinguish "signal from noise" at the low-concentration end, and consists of three progressively increasing limits:

- **LoB (Limit of Blank)**: The highest measurement that blank samples may produce (95th percentile).
- **LoD (Limit of Detection)**: The **lowest concentration** that can be distinguished from blank at a 95% confidence level.
- **LoQ (Limit of Quantitation)**: The **lowest concentration** at which the measurement result reaches a pre-specified precision target (for example CV = 20%).

This document demonstrates how to compute analytical sensitivity with the `lob_lod_loq()` function of the `ivdtools` package according to **CLSI EP17-A2**:

- **LoB** by the 5.3.3.1 parametric method: `LoB = M_B + cp·SD_B`, pooling all blank results into one pool, with `cp = 1.645/√(1 − 1/(4(B−K)))` the small-sample-adjusted 95% one-sided quantile (B is the total number of blank results, K the number of blank samples).
- **LoD** by 5.4.3: first fit a Sadler precision profile model (variance as a function of concentration) to the non-blank low-concentration samples, then perform **fixed-point iteration** from LoB, `X = LoB + cp·SD_WL(X)`, with `cp = 1.645/√(1 − 1/(4(N_TOT−K)))`.
- **LoQ** by Appendix D1 Example 1 (functional sensitivity): find the intersection of the precision profile with the target CV level line, `√Var(X)/X = target_cv`. This LoQ **reflects only precision and does not include bias**.

The example data are deterministic teaching data used only to demonstrate the workflow; formal studies must use the experimental design, target concentration, and acceptance criteria pre-specified in the protocol or standard.

```
library(ivdtools)
library(readr)
```

## 7.2 Overview of Functions

| Function | Main purpose | Key input or output |
|---|---|---|
| `lob_lod_loq()` | Compute LoB / LoD / LoQ in one call | Summarized data, blank sample names, target CV |
| `replicate_to_mean()` | Summarize replicate measurements | Mean, SD, replicate count n |
| `fit_equation()` | Fit the Sadler precision profile (called internally) | `eq="sadler"`, `df_col` |
| `print()` / `plot()` | Result display | Tabular summary, SD/CV dual panels |

The input to `lob_lod_loq()` is an **already summarized** data.frame: one row per sample, containing a sample-name column, a mean column (i.e., concentration), an SD column, and a replicate-count n column; the column names can be freely specified. It internally and automatically completes the optimal Sadler model fitting, so no manual fitting is required.

## 7.3 Example: LoB / LoD / LoQ Analysis

### 7.3.1 Reading and Validating Data

The data contain 2 blank samples (blank_A, blank_B) and 6 low-concentration samples (L1–L6), each with replicate measurements (40 for each blank, 30 for each low concentration), for a total of 260 rows. The SD increases with concentration, consistent with the common low-end situation of "variance increasing with the mean":

```
sens <- read_csv("./data/sensitivity.csv", show_col_types = FALSE)
sens <- as.data.frame(sens)
str(sens)
```

```
'data.frame':   260 obs. of  2 variables:
 $ sample: chr  "blank_A" "blank_A" "blank_A" "blank_A" ...
 $ value : num  0.024 -0.006 0.055 -0.072 0.057 -0.023 -0.05 0.003 0.051 0.029 ...
```

```
dim(sens)
```

```
[1] 260   2
```

```
table(sens$sample)
```

```

blank_A blank_B      L1      L2      L3      L4      L5      L6
     40      40      30      30      30      30      30      30
```

### 7.3.2 Per-Sample Summarization

Use `replicate_to_mean()` to group by sample name and compute the mean, SD, and replicate count (when the grouping variable is character, the original values are retained, so each blank occupies one row):

```
rep <- replicate_to_mean(data.frame(x = sens$sample, y = sens$value),
                         x = "x", y = "y")
rep$sample <- rep$x
rep$mean   <- rep$y_mean
rep$sd     <- rep$y_sd
rep$n      <- rep$y_n
rep[c("sample", "mean", "sd", "n")]
```

```
   sample       mean         sd  n
1 blank_A  0.0080750 0.04561623 40
2 blank_B -0.0036500 0.05287360 40
3      L1  0.5327667 0.11511694 30
4      L2  0.9601000 0.14135803 30
5      L3  1.9882667 0.21709110 30
6      L4  5.0710667 0.56385703 30
7      L5 10.1809667 0.96523481 30
8      L6 19.8890000 2.08779088 30
```

**Note:** `lob_lod_loq()` requires the **variance model response to be the variance** and weighted by degrees of freedom, so after summarizing you also need to add the two columns `var = sd^2` and `df = n - 1` to each row before passing them to `lob_lod_loq()`. If your data already contain these two columns, this step is unnecessary.

### 7.3.3 Computing LoB / LoD / LoQ

The `blank` parameter specifies the blank sample names (multiple allowed), and `target_cv` specifies the precision target for LoQ:

```
summ <- rep[c("sample", "mean", "sd", "n")]
res <- lob_lod_loq(summ, sample_col = "sample", mean_col = "mean",
                   sd_col = "sd", n_col = "n",
                   blank = c("blank_A", "blank_B"), target_cv = 0.20)
res
```

```
Limit of Blank / Detection / Quantitation (EP17-A2)
Precision profile: Sadler; best model = Model_3
  Equation            : sigma^2 = b1 + b2 * mu^2
  Parameters          : b1 = 0.010196, b2 = 0.010388
  Fit                 : AIC = -479.4; RSS = 0.08315; GoF p = 0.9437
  LoB  (parametric, B = 80, K = 2) : 0.08357
  LoD  (fixed-point)                : 0.2552
  LoQ  (CV = 20%)                   : 0.5868
Note: LoQ is based on precision ONLY; bias is not included.
```

Interpreting the results:

- **LoB = 0.0836**: the 95th percentile of blank measurements (parametric method), pooling B = 80 blank results and K = 2 blank samples.
- **LoD = 0.26**: the fixed point of `X = LoB + cp·SD(X)` on the precision profile, i.e., measurements above this concentration can be judged "non-blank".
- **LoQ = 0.59** (CV = 20%): the lowest concentration needed to reach the 20% precision target, **based only on precision**.

### 7.3.4 Returned Object Structure

`lob_lod_loq()` returns a `sensitivity` object, with each limit and parameter stored as a named component:

```
str(res, max.level = 1)
```

```
List of 13
 $ call      : language lob_lod_loq(data = summ, sample_col = "sample", mean_col =
"mean", sd_col = "sd",      n_col = "n", blank = c("bl| __truncated__
 $ lob       : num 0.0836
 $ lod       : num 0.255
 $ loq       : num 0.587
 $ n_blank   : int 80
 $ B         : int 80
 $ K         : int 2
 $ target_cv: num 0.2
 $ cp_lob    : num 1.65
 $ cp_lod    : num 1.65
 $ fit       :List of 12
  ..- attr(*, "class")= chr "fit_equation"
 $ model     : chr "Model_3"
 $ notes     : chr(0)
 - attr(*, "class")= chr "sensitivity"
```

```
res$lob
```

```
[1] 0.0835705
```

```
res$lod
```

```
[1] 0.2552205
```

```
res$loq
```

```
[1] 0.5867594
```

```
res$B
```

```
[1] 80
```

```
res$K
```

```
[1] 2
```

```
res$cp_lob
```

```
[1] 1.647643
```

```
res$cp_lod
```

```
[1] 1.646183
```

```
res$model
```

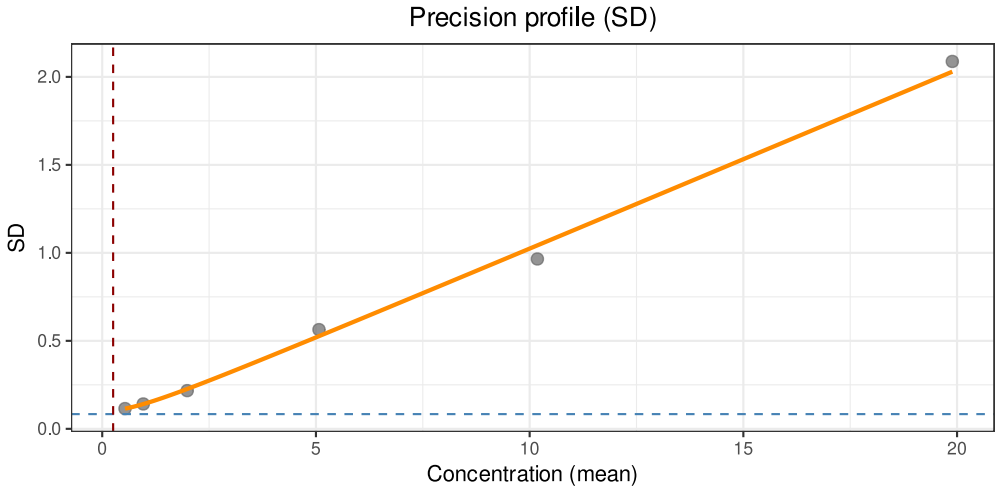
```
[1] "Model_3"
```

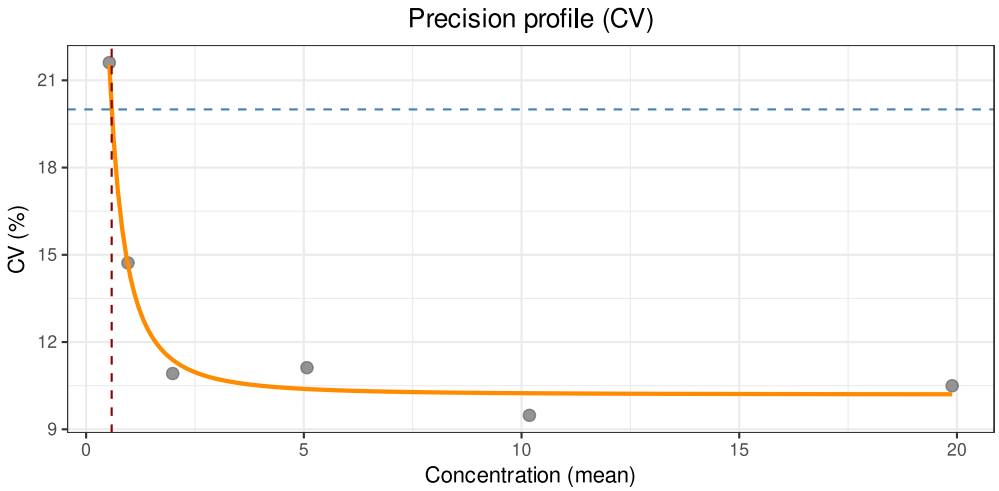
`cp_lob` and `cp_lod` are the two small-sample correction factors of EP17; `model` is the optimal Sadler model selected by AIC (in this example, model 3, the mixed-variance type).

### 7.3.5 Graphics

`plot()` draws the two precision-profile panels for SD and CV, overlays the LoB / LoD / LoQ reference lines, and visually shows the geometric meaning of each limit:

```
plot(res)
```




## 7.4 Other Input Scenarios

### 7.4.1 Scenario 1: Blank Only (LoB Only)

If there are no low-concentration levels, only LoB is computed, and LoD / LoQ are `NA`:

```
res_b <- lob_lod_loq(summ[summ$sample %in% c("blank_A", "blank_B"), ],
                     sample_col = "sample", mean_col = "mean",
                     sd_col = "sd", n_col = "n",
                     blank = c("blank_A", "blank_B"))
res_b
```

```
Limit of Blank / Detection / Quantitation (EP17-A2)
  LoB  (parametric, B = 80, K = 2) : 0.08357
  LoD  (fixed-point)               : --
  LoQ  (CV = 20%)                  : --
Note: LoQ is based on precision ONLY; bias is not included.
```

### 7.4.2 Scenario 2: No Blank (LoQ Only)

If there are no blank samples (`blank = NULL`), only LoQ is computed, and LoB / LoD are `NA`:

```
res_q <- lob_lod_loq(summ[!summ$sample %in% c("blank_A", "blank_B"), ],
                     sample_col = "sample", mean_col = "mean",
```

```
                    sd_col = "sd", n_col = "n")
res_q
```

```
Limit of Blank / Detection / Quantitation (EP17-A2)
Precision profile: Sadler; best model = Model_3
  Equation            : sigma^2 = b1 + b2 * mu^2
  Parameters          : b1 = 0.010196, b2 = 0.010388
  Fit                 : AIC = -479.4; RSS = 0.08315; GoF p = 0.9437
  LoB  (parametric, B = 0, K = 0) : --
  LoD  (fixed-point)                : --
  LoQ  (CV = 20%)                   : 0.5868
Note: LoQ is based on precision ONLY; bias is not included.
```

### 7.4.3 Scenario 3: LoQ < LoD

When the LoQ computed from the target CV is below the LoD (for example, large blank noise but good precision at low concentrations), LoQ takes `max(LoQ, LoD)` and issues a warning, ensuring **LoQ ≥ LoD**:

```
# Construct data with "large blank noise but good low-concentration precision"
d_noisy <- data.frame(
  sample = c("blank", paste0("L", 1:6)),
  mean   = c(0, 0.5, 1, 2, 5, 10, 20),
  sd     = c(5, sqrt(0.01 + 0.02 * c(0.5, 1, 2, 5, 10, 20)^2)),
  n      = c(30L, rep(5L, 6))
)
res_n <- lob_lod_loq(d_noisy, sample_col = "sample", mean_col = "mean",
                     sd_col = "sd", n_col = "n", blank = "blank")
res_n
```

```
Limit of Blank / Detection / Quantitation (EP17-A2)
Precision profile: Sadler; best model = Model_3
  Equation            : sigma^2 = b1 + b2 * mu^2
  Parameters          : b1 =   0.01, b2 =   0.02
  Fit                 : AIC = -469.2; RSS = 2.924e-29; GoF p =     1
  LoB  (parametric, B = 30, K = 1) : 8.261
  LoD  (fixed-point)                : 10.79
  LoQ  (CV = 20%)                   : 10.79
Note: LoQ is based on precision ONLY; bias is not included.
  Note: LoQ (precision) < LoD; LoQ reported as LoD = 10.79.
```

## 7.5 Key Points for Interpreting Results

1. **Target first**: Before analysis, determine the target CV (20% in this document is only a teaching example), confidence level, and acceptance criteria.
2. **Blank pooled pool**: LoB pools all blank results for computation; `cp` includes a `B–K` degrees-of-freedom correction that cannot be ignored when the sample size is small.
3. **LoQ contains only precision**: With this workflow, LoQ does not include bias and cannot replace a total-error (TE)-based limit-of-quantitation claim.
4. **Model selection**: Different Sadler models extrapolate very differently at the low-concentration end; the report should note the AIC-optimal model and its ranking.
5. **LoQ ≥ LoD**: The program guarantees LoQ is not below LoD; if the CV-based LoQ is below the LoD, it warns and takes `max`.
6. **Data format**: The input to `lob_lod_loq()` is an already-summarized data.frame; the original replicate data must first be processed by `replicate_to_mean()`.

# 8. Bottle-to-Bottle Variability Analysis

## 8.1 Analysis Objectives

Bottle-to-bottle variability evaluates the differences in measurement results among different bottles (or different lots) of the same product. The analysis is usually done in two steps: first, use `bottle_anova()` to test whether the means of the bottles (or lots) differ statistically, and then use variance-component analysis to separate the between-bottle variability from the within-bottle variability, judging whether the magnitude of the difference meets the pre-specified precision limits. This document demonstrates both scenarios: the bottle-to-bottle difference of 15 bottles × 5 measurements, and the between-lot difference when different reagent lots measure the same quality-control material.

The example data are deterministic teaching data; statistical significance is not equivalent to product unacceptability; judgments can only be made against the acceptance limits pre-specified in the protocol.

```
library(ivdtools)
library(readr)
```

## 8.2 Overview of Functions

| Function | Main purpose | Key input or output |
|---|---|---|
| `bottle_anova()` | One-way or nested ANOVA | `value ~ batch` or `value ~ batch/vial` |
| `tukey()` | Tukey HSD pairwise comparison and compact letters | `term` specifies the effect |
| `precision()` / `variance()` | Split between-bottle/within-bottle variance components | `value ~ bottle` one-way |

## 8.3 Example 1: Bottle-to-Bottle Difference of 15 Bottles × 5 Measurements

### 8.3.1 Reading and Validating Data

```
bottle <- read_csv("./data/bottle-difference.csv", show_col_types = FALSE)
bottle <- as.data.frame(bottle)
bottle$bottle <- factor(bottle$bottle)
str(bottle)
```

```
'data.frame':   75 obs. of  3 variables:
 $ bottle   : Factor w/ 15 levels "B01","B02","B03",..: 1 1 1 1 1 2 2 2 2 2 ...
 $ replicate: num  1 2 3 4 5 1 2 3 4 5 ...
 $ value    : num  97.5 100.5 98.8 97.5 97.8 ...
```

```
dim(bottle)
```

```
[1] 75  3
```

### 8.3.2 Step 1: Test for Significant Difference

The one-way model `value ~ bottle` tests whether the means of the 15 bottles are equal:

```
ba <- bottle_anova(bottle, value ~ bottle)
print(ba)
```

```
Bottle ANOVA Analysis
  Response:        value
  Group variable:  bottle (15 groups)
  Complete cases:  75 / 75
  Confidence level: 0.95

  Descriptive Statistics:
 ------------------------------------------------------------
  bottle              n      Mean        SD
 ------------------------------------------------------------
  B01                 5   98.4520    1.2739
  B02                 5  101.7140    0.9360
  B03                 5   99.8060    0.8153
  B04                 5  102.7980    0.4992
  B05                 5   98.7520    0.8662
  B06                 5  100.0960    1.4971
  B07                 5  100.7180    0.8466
  B08                 5  100.5000    1.3179
  B09                 5   99.8160    1.0246
  B10                 5  101.6400    0.8034
  B11                 5   99.3300    0.9360
  B12                 5   98.4700    1.0884
  B13                 5   99.7440    1.1004
  B14                 5  100.9040    0.4803
  B15                 5   99.7920    1.4477

  ANOVA Table:
 ------------------------------------------------------------
  Term              Df    Sum Sq    Mean Sq   F-value    p-value  Sig
 ------------------------------------------------------------
  bottle            14  107.7233     7.6945    7.1460     0.0000  ***
  Residuals         60   64.6057     1.0768
 ------------------------------------------------------------
  Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

  Assumption Checks:
    Bartlett test: K-squared = 9.9325, df = 14, p-value = 0.7671 (OK)
    Shapiro-Wilk:  W = 0.9828, p-value = 0.4011 (OK)

  Post-hoc: use tukey() for pairwise comparisons
```

The ANOVA table gives the between-bottle F test; the `assumptions` section reports the Bartlett variance-homogeneity and Shapiro-Wilk normality tests.

### 8.3.3 Pairwise Comparison

```
posthoc <- tukey(ba)
posthoc
```

```
Tukey HSD Post-hoc Test
  Response:        value
```

```
  Complete cases:  75 / 75
  Confidence level: 0.95

  Term: bottle
-------------------------------------------------------------
  Comparison                Diff       lwr       upr   p-value  Sig
-------------------------------------------------------------
  B02-B01                 3.2620    0.9412    5.5828    0.0005  ***
  B03-B01                 1.3540   -0.9668    3.6748    0.7526
  B04-B01                 4.3460    2.0252    6.6668    0.0000  ***
  B05-B01                 0.3000   -2.0208    2.6208    1.0000
  B06-B01                 1.6440   -0.6768    3.9648    0.4526
  B07-B01                 2.2660   -0.0548    4.5868    0.0624
  B08-B01                 2.0480   -0.2728    4.3688    0.1408
  B09-B01                 1.3640   -0.9568    3.6848    0.7431
  B10-B01                 3.1880    0.8672    5.5088    0.0008  ***
  B11-B01                 0.8780   -1.4428    3.1988    0.9896
  B12-B01                 0.0180   -2.3028    2.3388    1.0000
  B13-B01                 1.2920   -1.0288    3.6128    0.8078
  B14-B01                 2.4520    0.1312    4.7728    0.0287  *
  B15-B01                 1.3400   -0.9808    3.6608    0.7656
  B03-B02                -1.9080   -4.2288    0.4128    0.2229
  B04-B02                 1.0840   -1.2368    3.4048    0.9391
  B05-B02                -2.9620   -5.2828   -0.6412    0.0026  **
  B06-B02                -1.6180   -3.9388    0.7028    0.4794
  B07-B02                -0.9960   -3.3168    1.3248    0.9686
  B08-B02                -1.2140   -3.5348    1.1068    0.8674
  B09-B02                -1.8980   -4.2188    0.4228    0.2298
  B10-B02                -0.0740   -2.3948    2.2468    1.0000
  B11-B02                -2.3840   -4.7048   -0.0632    0.0384  *
  B12-B02                -3.2440   -5.5648   -0.9232    0.0006  ***
  B13-B02                -1.9700   -4.2908    0.3508    0.1830
  B14-B02                -0.8100   -3.1308    1.5108    0.9952
  B15-B02                -1.9220   -4.2428    0.3988    0.2134
  B04-B03                 2.9920    0.6712    5.3128    0.0022  **
  B05-B03                -1.0540   -3.3748    1.2668    0.9507
  B06-B03                 0.2900   -2.0308    2.6108    1.0000
  B07-B03                 0.9120   -1.4088    3.2328    0.9852
  B08-B03                 0.6940   -1.6268    3.0148    0.9990
  B09-B03                 0.0100   -2.3108    2.3308    1.0000
  B10-B03                 1.8340   -0.4868    4.1548    0.2778
  B11-B03                -0.4760   -2.7968    1.8448    1.0000
  B12-B03                -1.3360   -3.6568    0.9848    0.7693
  B13-B03                -0.0620   -2.3828    2.2588    1.0000
  B14-B03                 1.0980   -1.2228    3.4188    0.9330
  B15-B03                -0.0140   -2.3348    2.3068    1.0000
  B05-B04                -4.0460   -6.3668   -1.7252    0.0000  ***
  B06-B04                -2.7020   -5.0228   -0.3812    0.0092  **
  B07-B04                -2.0800   -4.4008    0.2408    0.1258
  B08-B04                -2.2980   -4.6188    0.0228    0.0549
  B09-B04                -2.9820   -5.3028   -0.6612    0.0023  **
  B10-B04                -1.1580   -3.4788    1.1628    0.9026
  B11-B04                -3.4680   -5.7888   -1.1472    0.0002  ***
  B12-B04                -4.3280   -6.6488   -2.0072    0.0000  ***
  B13-B04                -3.0540   -5.3748   -0.7332    0.0016  **
  B14-B04                -1.8940   -4.2148    0.4268    0.2326
  B15-B04                -3.0060   -5.3268   -0.6852    0.0020  **
  B06-B05                 1.3440   -0.9768    3.6648    0.7619
  B07-B05                 1.9660   -0.3548    4.2868    0.1854
  B08-B05                 1.7480   -0.5728    4.0688    0.3515
```

```
  B09-B05                1.0640   -1.2568    3.3848    0.9470
  B10-B05                2.8880    0.5672    5.2088    0.0037  **
  B11-B05                0.5780   -1.7428    2.8988    0.9999
  B12-B05               -0.2820   -2.6028    2.0388    1.0000
  B13-B05                0.9920   -1.3288    3.3128    0.9696
  B14-B05                2.1520   -0.1688    4.4728    0.0968
  B15-B05                1.0400   -1.2808    3.3608    0.9556
  B07-B06                0.6220   -1.6988    2.9428    0.9997
  B08-B06                0.4040   -1.9168    2.7248    1.0000
  B09-B06               -0.2800   -2.6008    2.0408    1.0000
  B10-B06                1.5440   -0.7768    3.8648    0.5576
  B11-B06               -0.7660   -3.0868    1.5548    0.9972
  B12-B06               -1.6260   -3.9468    0.6948    0.4711
  B13-B06               -0.3520   -2.6728    1.9688    1.0000
  B14-B06                0.8080   -1.5128    3.1288    0.9953
  B15-B06               -0.3040   -2.6248    2.0168    1.0000
  B08-B07               -0.2180   -2.5388    2.1028    1.0000
  B09-B07               -0.9020   -3.2228    1.4188    0.9866
  B10-B07                0.9220   -1.3988    3.2428    0.9837
  B11-B07               -1.3880   -3.7088    0.9328    0.7199
  B12-B07               -2.2480   -4.5688    0.0728    0.0670
  B13-B07               -0.9740   -3.2948    1.3468    0.9739
  B14-B07                0.1860   -2.1348    2.5068    1.0000
  B15-B07               -0.9260   -3.2468    1.3948    0.9831
  B09-B08               -0.6840   -3.0048    1.6368    0.9992
  B10-B08                1.1400   -1.1808    3.4608    0.9125
  B11-B08               -1.1700   -3.4908    1.1508    0.8956
  B12-B08               -2.0300   -4.3508    0.2908    0.1498
  B13-B08               -0.7560   -3.0768    1.5648    0.9976
  B14-B08                0.4040   -1.9168    2.7248    1.0000
  B15-B08               -0.7080   -3.0288    1.6128    0.9988
  B10-B09                1.8240   -0.4968    4.1448    0.2859
  B11-B09               -0.4860   -2.8068    1.8348    1.0000
  B12-B09               -1.3460   -3.6668    0.9748    0.7601
  B13-B09               -0.0720   -2.3928    2.2488    1.0000
  B14-B09                1.0880   -1.2328    3.4088    0.9374
  B15-B09               -0.0240   -2.3448    2.2968    1.0000
  B11-B10               -2.3100   -4.6308    0.0108    0.0523
  B12-B10               -3.1700   -5.4908   -0.8492    0.0009  ***
  B13-B10               -1.8960   -4.2168    0.4248    0.2312
  B14-B10               -0.7360   -3.0568    1.5848    0.9982
  B15-B10               -1.8480   -4.1688    0.4728    0.2668
  B12-B11               -0.8600   -3.1808    1.4608    0.9914
  B13-B11                0.4140   -1.9068    2.7348    1.0000
  B14-B11                1.5740   -0.7468    3.8948    0.5257
  B15-B11                0.4620   -1.8588    2.7828    1.0000
  B13-B12                1.2740   -1.0468    3.5948    0.8226
  B14-B12                2.4340    0.1132    4.7548    0.0311  *
  B15-B12                1.3220   -0.9988    3.6428    0.7819
  B14-B13                1.1600   -1.1608    3.4808    0.9015
  B15-B13                0.0480   -2.2728    2.3688    1.0000
  B15-B14               -1.1120   -3.4328    1.2088    0.9266
 ---------------------------------------------------------------
 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 Compact Letter Display:
 ---------------------------------------------------------------
   B04: a
   B02: a
   B10: a
```

```
    B14: a
    B07: ab
    B08: ab
    B06: b
    B09: b
    B03: b
    B15: b
    B13: b
    B11: b
    B05: b
    B12: b
    B01: b
```

Tukey HSD gives all pairwise comparisons and compact letter displays; bottles with different letters differ significantly in mean. In this example, there are many bottles, so some significant pairwise differences are normal; the key question is whether the **magnitude of the difference** is acceptable.

### 8.3.4 Step 2: Whether the Difference Magnitude Meets Expectations

Use `precision(value ~ bottle)` to split the total variability into the between-bottle component (bottle) and the within-bottle component (error):

```
bp <- precision(bottle, value ~ bottle)
bp <- variance(bp)
```

```
Variance components
 sample component     VC %Total     SD  CV[%]
-----------------------------------------------
    all    bottle 1.3236   55.1 1.1505 1.1485
    all     error 1.0768   44.9 1.0377 1.0359
```

The between-bottle SD is ≈ 1.15, CV ≈ 1.15%, and the within-bottle SD is ≈ 1.04, CV ≈ 1.04%. Assuming the protocol specifies a total precision limit of 2% (CV), the between-bottle difference is within the limit. **Note:** The "limit" here is only an example; formal studies must use the limits specified in the protocol or standard, and should compare the total CV (including between- and within-bottle) and each component separately.

```
bp <- ci(bp)
```

```
Confidence intervals (SD)
 sample component estimate  lower  upper
---------------------------------------
    all     total   1.5493 1.2449 2.0519
    all    bottle   1.1505 0.4254 1.5704
    all     error   1.0377 0.8807 1.2633

Confidence intervals (%CV)
 sample component estimate  lower  upper
---------------------------------------
    all     total   1.5467 1.2428 2.0485
    all    bottle   1.1485 0.4246 1.5678
    all     error   1.0359 0.8792 1.2612
```

## 8.4 Example 2: Different Reagent Lots Measuring the Same Quality-Control Material

### 8.4.1 Reading Data

Three reagent lots (L1, L2, L3) each measure the same quality-control material 15 times:

```
lot_data <- read_csv("./data/reagent-lot.csv", show_col_types = FALSE)
lot_data <- as.data.frame(lot_data)
lot_data$lot <- factor(lot_data$lot)
str(lot_data)
```

```
'data.frame':   45 obs. of  3 variables:
 $ lot      : Factor w/ 3 levels "L1","L2","L3": 1 1 1 1 1 1 1 1 1 1 ...
 $ replicate: num  1 2 3 4 5 6 7 8 9 10 ...
 $ value    : num  100 102 101 102 101 ...
```

```
lot_summary <- aggregate(value ~ lot, lot_data, function(x) c(n = length(x), mean =
mean(x), sd = sd(x)))
lot_summary <- data.frame(lot = lot_summary$lot, lot_summary$value)
lot_summary
```

```
  lot  n     mean        sd
1  L1 15 100.8173 1.1626353
2  L2 15 101.2760 1.4271040
3  L3 15 100.5673 0.7829201
```

### 8.4.2 Testing the Between-Lot Difference

```
ba_lot <- bottle_anova(lot_data, value ~ lot)
print(ba_lot)
```

```
Bottle ANOVA Analysis
  Response:        value
  Group variable:  lot (3 groups)
  Complete cases:  45 / 45
  Confidence level: 0.95

  Descriptive Statistics:
 ------------------------------------------------------------
  lot                 n      Mean        SD
 ------------------------------------------------------------
  L1                 15  100.8173    1.1626
  L2                 15  101.2760    1.4271
  L3                 15  100.5673    0.7829

  ANOVA Table:
 ------------------------------------------------------------
  Term              Df    Sum Sq    Mean Sq   F-value    p-value  Sig
 ------------------------------------------------------------
  lot                2    3.8754     1.9377    1.4528     0.2454
  Residuals         42   56.0183     1.3338
 ------------------------------------------------------------
  Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

  Assumption Checks:
    Bartlett test: K-squared = 4.6245, df = 2, p-value = 0.0990 (OK)
    Shapiro-Wilk:  W = 0.9817, p-value = 0.6870 (OK)

  Post-hoc: use tukey() for pairwise comparisons
```

```
tukey(ba_lot)
```

```
Tukey HSD Post-hoc Test
  Response:        value
  Complete cases:  45 / 45
  Confidence level: 0.95

  Term: lot
 ------------------------------------------------------------
  Comparison               Diff       lwr       upr   p-value  Sig
 ------------------------------------------------------------
  L2-L1                  0.4587   -0.5659    1.4832    0.5269
  L3-L1                 -0.2500   -1.2745    0.7745    0.8247
  L3-L2                 -0.7087   -1.7332    0.3159    0.2245
 ------------------------------------------------------------
  Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

  Compact Letter Display:
 ------------------------------------------------------------
    L2: a
    L1: a
    L3: a
```

The compact letter display shows between which lots the means differ significantly and in which direction. Even if the between-lot mean difference is statistically significant, it must still be judged against the pre-specified acceptable bias (for example, the allowable target-value range of the quality-control material) to determine whether it is within the acceptable range; do not draw conclusions from the p-value alone.

## 8.5 Key Points for Interpreting Results

1. **Set acceptance limits first**: Before analysis, specify the allowable difference magnitude between lots/bottles (SD/CV or mean-bias limit).
2. **Significance ≠ acceptability**: ANOVA/Tukey test whether there is a difference; whether the magnitude is acceptable is determined by the limits.
3. **Variance decomposition**: Use variance components to separate between-bottle/lot and within-bottle variability, and evaluate the contribution of each component separately.
4. **Pairwise-comparison scale**: With many bottles, the number of pairwise comparisons is large; use the compact letter display to focus on the difference structure.
5. **Consistent conditions**: Between-lot comparisons should keep conditions such as method, instrument, operator, and time constant, with only the lot varying.
6. **Complete records**: Report the mean, SD of each bottle/lot, the ANOVA table, Tukey results, variance components, and limit judgments.

# 9. Reference Intervals

## 9.1 Analysis Objectives

A reference interval (RI) is defined by the measurement results of a reference sample population and is used to interpret the normal range of an individual's results, corresponding to CLSI EP28-A3c. `ivdtools` provides three methods: the percentile method (nonparametric), the parametric method (normal or log-normal), and the robust method. The three methods differ in their sensitivity to sample size, distribution shape, and outliers; it is recommended to apply multiple methods to the same data simultaneously for comparison, and to perform outlier detection and normality assessment before establishing reference intervals.

This document demonstrates pre-analyses with `outliers_test()` and `normal_test()`, and then calls `reference_interval(method = "all")` to output the reference intervals of the three methods at once. The example data are deterministic teaching data (n=120, mildly right-skewed and containing outliers).

```
library(ivdtools)
library(readr)
```

## 9.2 Overview of Functions

| Function | Main purpose | Key input or output |
|---|---|---|
| `outliers_test()` | Outlier detection | Grubbs / ESD / Dixon / IQR |
| `normal_test()` | Normality test and graphics | `method="auto"` selects the test by sample size |
| `reference_interval()` | Reference intervals by three methods | `method="all"`, `interval`, `ci`, `id` |
| `plot.reference_interval()` | Plot intervals by method | jitter + upper/lower limits |

## 9.3 Example: Reference Intervals Using Three Methods Together

### 9.3.1 Reading and Validating Data

```
ri_data <- read_csv("./data/reference-interval.csv", show_col_types = FALSE)
ri_data <- as.data.frame(ri_data)
str(ri_data)
```

```
'data.frame':   120 obs. of  2 variables:
 $ id   : chr  "S001" "S002" "S003" "S004" ...
 $ value: num  131.1 100.8 106 74.3 116.9 ...
```

### 9.3.2 Pre-Analysis 1: Outlier Detection

EP28 recommends reviewing outliers before establishing reference intervals. With sample size n=120, Dixon is not suitable (Dixon applies only to n≤30), so use Grubbs and the generalized ESD:

```
outliers_test(ri_data, col = "value", method = "grubbs")
```

```
Grubbs Outliers Test
  Data:   ri_data
  Column: value (n = 120, missing = 0)
  Parameters: alpha = 0.05
  Outliers: 1

  Index            Value      G stat    Critical
  ------------------------------------------
  88            183.0000      4.0104      3.4451
```

```
outliers_test(ri_data, col = "value", method = "esd", r = 3)
```

```
Generalized ESD Outliers Test
  Data:   ri_data
  Column: value (n = 120, missing = 0)
  Parameters: alpha = 0.05, r = 3
  Outliers: 2

  Index            Value      R stat    Critical
  ------------------------------------------
  88            183.0000      4.0104      3.4451
  57            178.0000      4.0576      3.4424
```

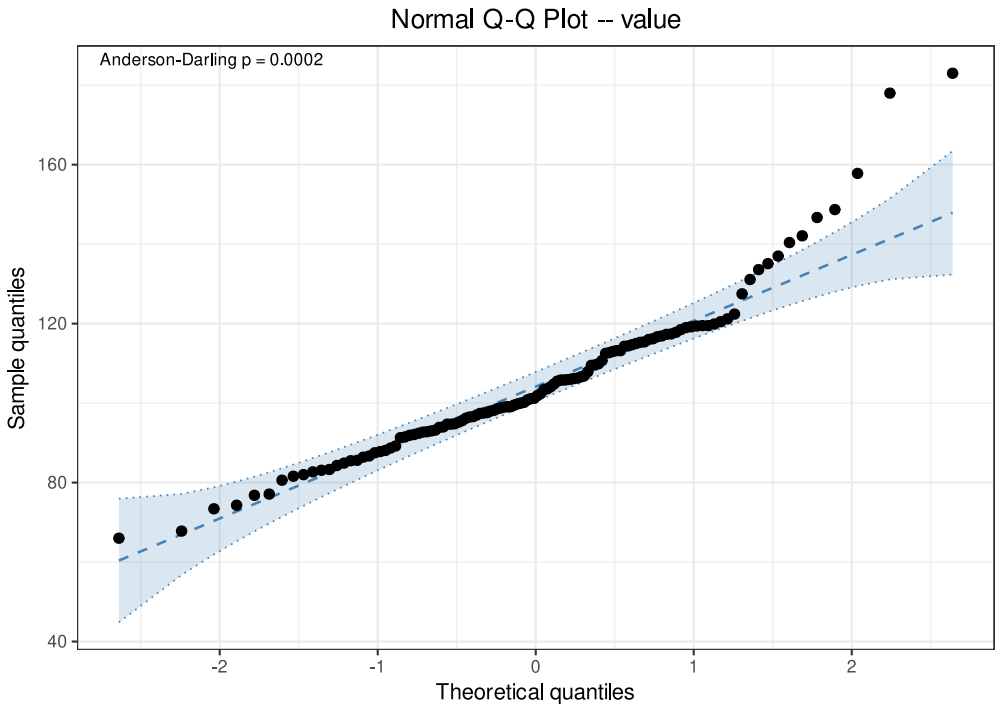
**Parameter notes:** The `method` of `outliers_test()` can be `"grubbs"`, `"esd"`, `"dixon"`, or `"iqr"`; ESD specifies the maximum number of detections via `r`, Dixon specifies the tail via `type` (`"both"`/`"min"`/`"max"`), and IQR specifies the multiplier via `coef` (default 1.5). **Detecting an outlier does not mean it should be deleted** — go back to the original records, experimental procedure, and protocol requirements, and perform an include/exclude sensitivity analysis when there is sufficient justification.

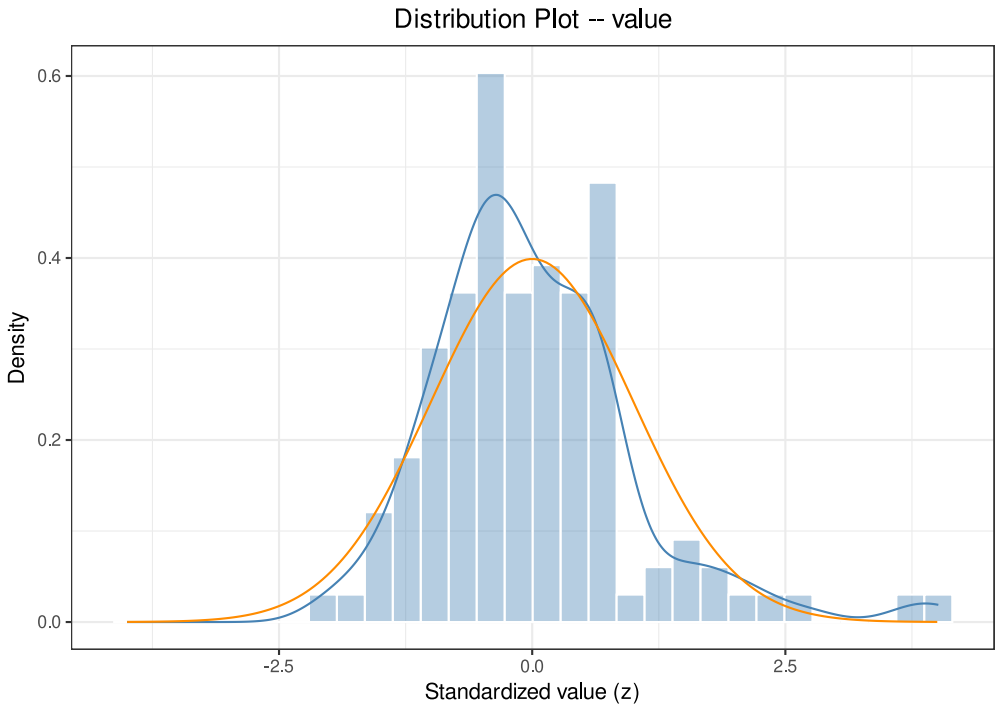
### 9.3.3 Pre-Analysis 2: Normality Test

```
nt <- normal_test(ri_data, col = "value", method = "auto")
nt
```

```

Normality Test
  Data:   ri_data
  Column: value (n = 120, missing = 0)
  Method:     Anderson-Darling

  A = 1.7577, p = 0.0002
```

```
plot(nt, type = c("qq", "his"))
```




`method="auto"` automatically selects the Anderson-Darling test for n=120; the p-value is very small, indicating that the data do not follow a normal distribution (the data are right-skewed and contain outliers). The QQ plot and histogram can visually confirm the skewness.

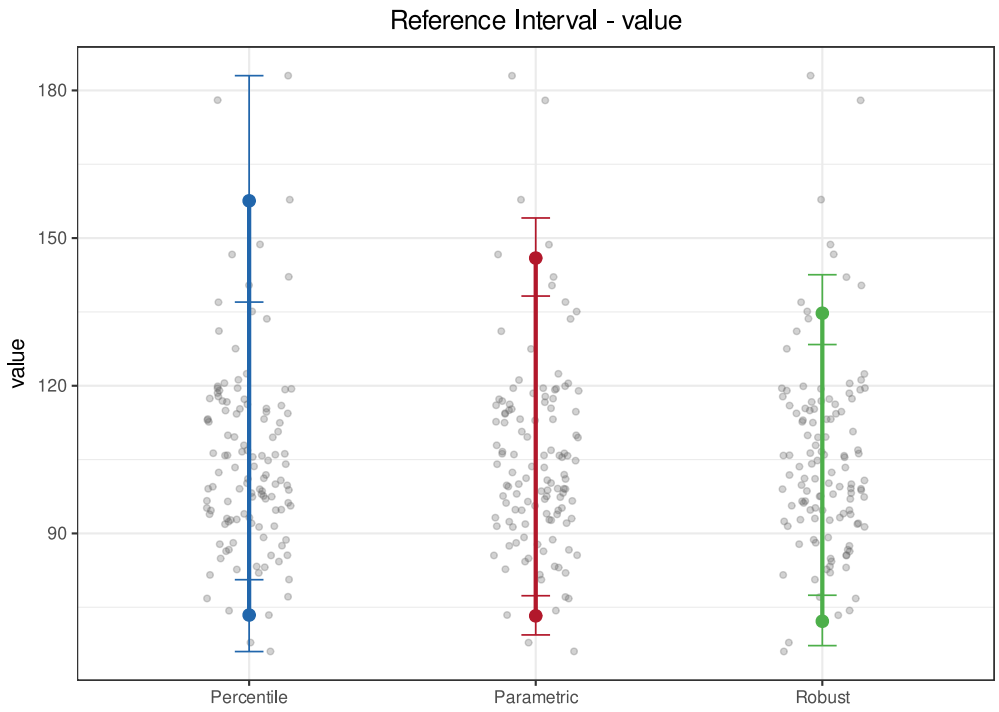
### 9.3.4 Reference Intervals Using Three Methods Together

```
ri <- reference_interval(
  ri_data,
  col = "value",
  interval = 0.95,
  ci = 0.95,
```

```
  method = "all",
  id = "id"
)
print(ri)
```

```
Reference Interval Analysis
  Column:              value
  Reference interval:  95% (alpha = 0.025)
  Confidence level:    0.95
  ID variable:         id
  Complete cases:      120 / 120

  No missing values.
  No duplicate IDs found.

  Percentile (quantile type 6, CLSI EP28-A3c)
    Lower = 73.4225  [66.0000, 80.6000]
    Upper = 157.5725  [137.0000, 183.0000]

  Parametric (log-transformed, Shapiro-Wilk p = 0.0661)
    Lower = 73.2539  [69.3807, 77.3434]
    Upper = 145.9407  [138.2241, 154.0880]

  Robust (Huber M-estimation (k=1.5, 16 iter) + MAD)
    Lower = 72.1573  [67.2065, 77.4407]
    Upper = 134.7365  [128.3734, 142.5561]
```

```
plot(ri)
```




**Parameter notes:** In `reference_interval()`, `interval` is the coverage of the reference interval (default 0.95, corresponding to 2.5%–97.5%), while `ci` is the confidence level of the interval endpoints; the two have different meanings. `method` can be `"percentile"`, `"parametric"`, `"robust"`, `"all"`, or any combination of these. `id` is used for duplicate-sample detection.

Comparison of the three methods in this example:

- **Percentile method** (73.4–157.6): nonparametric and makes no distributional assumption, but the upper limit is raised by two large outliers;
- **Parametric method** (73.3–145.9): automatically detects non-normality and applies a log transformation (`transformed=TRUE`) before back-transforming;
- **Robust method** (72.2–134.7): based on the Huber M-estimator and MAD, most robust to outliers.

The upper limit of the percentile method is clearly higher than that of the robust method, which is precisely the effect of the outliers. **The reporting method should be selected under the guidance of the sample source, sample size, distribution characteristics, and study protocol**, rather than simply picking the one whose result "looks best".

## 9.4 Key Points for Interpreting Results

1. **Protocol first**: Specify the reference population, inclusion/exclusion criteria, sample size (EP28 recommends ≥120 for the percentile method), and partitioning strategy.
2. **Check before compute**: Perform outlier detection and normality assessment first, and record the outlier-handling decision (whether to exclude, and the basis).
3. **Methods comparable**: Outputting the three methods together helps reveal the effect of outliers or distributional assumptions on the interval; state the chosen method and its rationale in the report.
4. **Distinguish parameters**: Distinguish `interval` (coverage) from `ci` (endpoint confidence); do not mix them up.
5. **Transparent transformation**: When the parametric method automatically applies a log transformation, report the transformation type and the back-transformation method.
6. **Non-unique conclusion**: A reference interval is not the same as a clinical decision limit, nor does it automatically constitute an acceptability determination.

# 10. Quality Control

## 10.1 Analysis Objectives

Quality control (QC) determines whether a measurement system is in control by periodically measuring quality-control materials and monitoring whether the results deviate from target values. `ivdtools` provides Levey–Jennings charts with Westgard multi-rule evaluation (`qc_chart()`), grouped monitoring in lot-change/group-change scenarios, and the Youden plot for two-level QC (`youden_plot()`).

This document demonstrates: obtaining and understanding the Westgard rules, whether daily QC results exceed the limits, how to evaluate grouping correctly when a reagent lot changes, and the Youden plot for two-level QC. The example data are deterministic teaching data; **the Westgard rule combination and allowable limits should be pre-determined according to the laboratory's risk-management strategy and protocol**.

```
library(ivdtools)
library(readr)
```

## 10.2 Overview of Functions

| Function | Main purpose | Key input or output |
|---|---|---|
| `list_westgard()` | View Westgard rule names and meanings | `brief=TRUE` returns only the rule vector |
| `qc_chart()` | Levey–Jennings chart + rule judgment | `rules`, `mean/sd`, `group`, `run` |
| `youden_plot()` | Youden plot for two-level QC | two sample columns, target means/SDs |

## 10.3 Example 1: Obtaining Rules and Daily QC

### 10.3.1 Viewing the Westgard Rules

```
list_westgard()
```

```
Westgard QC Rules
  ------------------------------------------------
  1-2s    1 point exceeds +/-2s (warning)
  1-3s    1 point exceeds +/-3s (out of control)
  2-2s    2 consecutive points exceed +/-2s (same side)
  R-4s    2 consecutive points differ by at least 4s
  4-1s    4 consecutive points exceed +/-1s (same side)
  8x      8 consecutive points on same side of mean
  10x     10 consecutive points on same side of mean
  ------------------------------------------------
```

### 10.3.2 Reading Daily QC Data

```
qc_daily <- read_csv("./data/qc-daily.csv", show_col_types = FALSE)
qc_daily <- as.data.frame(qc_daily)
head(qc_daily)
```

```
  run  value
1   1 103.24
2   2 103.06
3   3  94.52
4   4  99.51
5   5  99.14
6   6  95.62
```

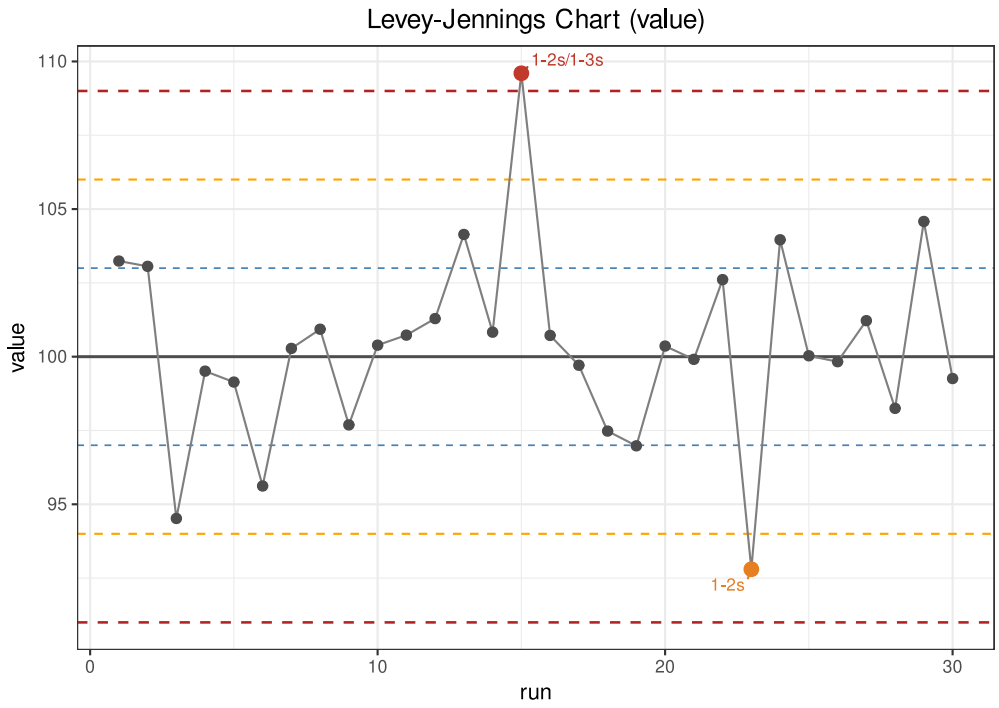
The data are daily QC results with target mean = 100, SD = 3, for a total of 30 run points.

### 10.3.3 Running Quality Judgments

```
qc <- qc_chart(
  qc_daily,
  value = "value",
  rules = "1-2s,1-3s,2-2s,R-4s,4-1s,10x",
  mean = 100,
  sd = 3,
  run = "run"
)
print(qc)
```

```

QC Control Chart
  Value column:    value
  Mean (target):   100.0000
  SD (target):     3.0000
  N (total):       30  |  Complete: 30  |  Missing: 0
  Rules applied:   1-2s, 1-3s, 2-2s, R-4s, 4-1s, 10x

  Violations:
  --------------------------------------------------
    1-2s  : points 15, 23
    1-3s  : points 15
  -> 2 violation(s): 1 warning(s), 1 out of control
```

```
plot(qc)
```




`qc_chart()` records the points violating each rule in `$violations`:

```
qc$violations
```

```
$`1-2s`
[1] 15 23

$`1-3s`
[1] 15

$`2-2s`
integer(0)

$`R-4s`
integer(0)

$`4-1s`
integer(0)

$`10x`
integer(0)
```

In this example, point 15 exceeds +3s (triggering `1-3s` and `1-2s`, judged out of control), and point 23 falls below −2s (triggering `1-2s`, judged a warning). `n_warn` and `n_oc` give the warning and out-of-control point counts, respectively.

**Parameter notes:** The `rules` of `qc_chart()` is a comma-separated rule string; `mean`/`sd` are the target values (when not given, they are estimated from the data, but monitored QC should use target values from the instructions). `run` specifies the run-order column, and `group` specifies the grouping column (see Example 2).

## 10.4 Example 2: Grouped Evaluation on Reagent Lot Change

### 10.4.1 Reading Data

In the data, the first 10 points are lot L1, the middle 10 are lot L2 (with an overall shift), and the last 10 are lot L3:

```
qc_lot <- read_csv("./data/qc-lot-change.csv", show_col_types = FALSE)
qc_lot <- as.data.frame(qc_lot)
table(qc_lot$lot)
```

```
L1 L2 L3
10 10 10
```

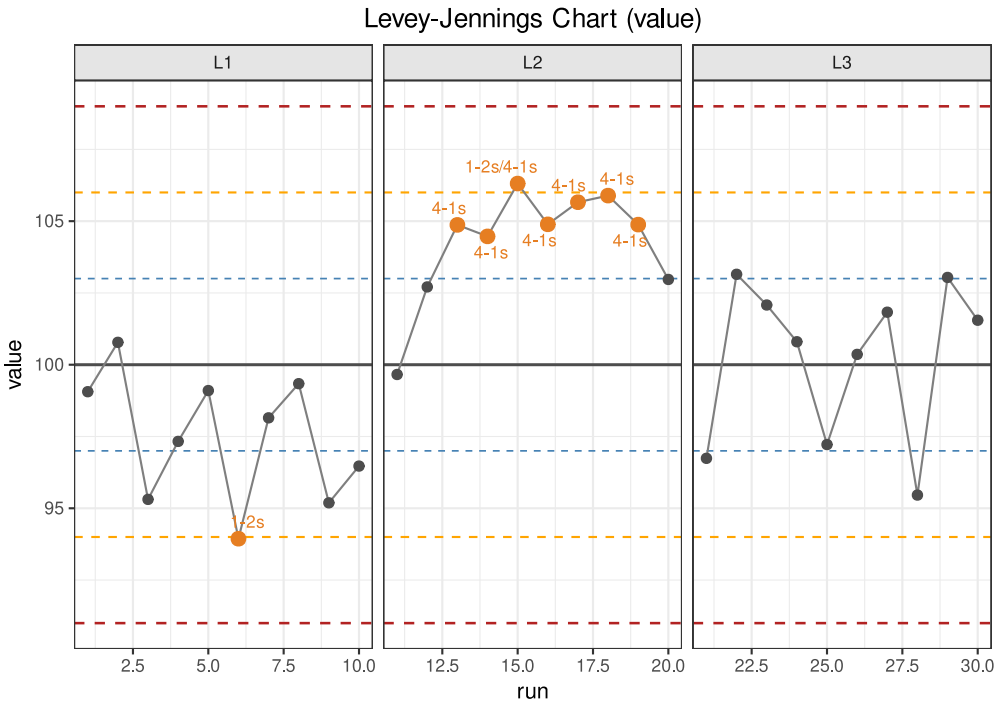
### 10.4.2 Grouping the Whole Sequence

Use `group = "lot"` for plotting; the Westgard rules are still judged on the **entire sequence**:

```
qc_lot_all <- qc_chart(
  qc_lot,
  value = "value",
  mean = 100,
  sd = 3,
  run = "run",
  group = "lot"
)
qc_lot_all$n_oc
```

```
[1] 0
```

```
plot(qc_lot_all)
```




Although the plot is faceted by lot, the rules are computed on the entire sequence, so observe the systematic shift caused by the lot change.

## 10.5 Example 3: Youden Plot for Two-Level QC

### 10.5.1 Reading Data

The Youden plot uses the results of two quality-control materials at different concentrations (levels) in the same run (wide-table format, one column per level):

```
youden_dat <- read_csv("./data/qc-youden.csv", show_col_types = FALSE)
youden_dat <- as.data.frame(youden_dat)
head(youden_dat)
```

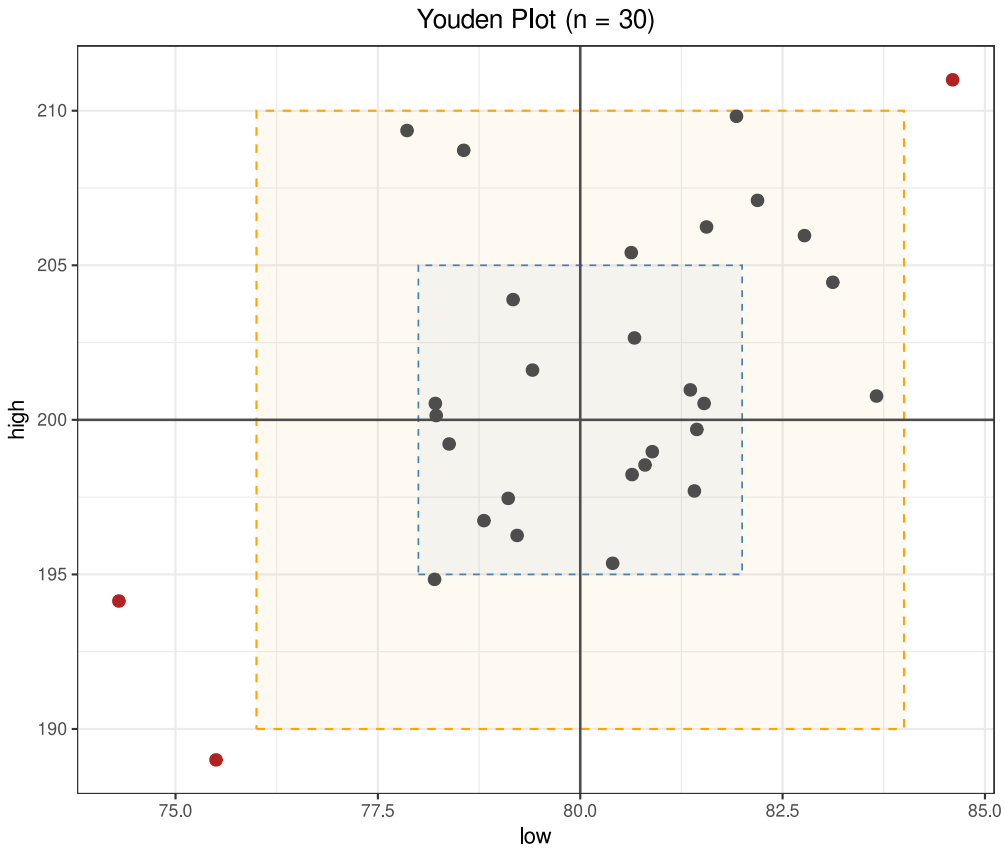
```
  run   low   high
1   1 77.86 209.36
2   2 82.19 207.10
3   3 75.50 189.00
4   4 80.40 195.36
5   5 80.64 198.23
6   6 78.22 200.14
```

### 10.5.2 Drawing the Youden Plot

```
yd <- youden_plot(
  youden_dat,
  sample1 = "low",
  sample2 = "high",
  mean1 = 80,
  mean2 = 200,
  sd1 = 2,
  sd2 = 5
)
print(yd)
```

```


Youden Plot
  Sample 1 (low):
    Mean:   80.0000
    SD:     2.0000
  Sample 2 (high):
    Mean:   200.0000
    SD:     5.0000
  Correlation: 0.5428
  N (total):   30  |  Complete: 30  |  Missing: 0
  Points outside +/-2SD: 3  (rows: 3, 17, 19)
```

```
plot(yd)
```




```
yd$outside_rows
```

```
[1]  3 17 19
```

`youden_plot()` uses a ±2s rectangle to mark points outside the range. In this example, 3 run points fall outside the rectangle (including an artificially set out-of-control point), suggesting that these runs may be affected by a systematic error at the same time. A shared shift at both levels (points moving outward along the diagonal) suggests a systematic error, while a large shift in a single direction is more likely to come from random error or a single-level problem.

**Parameter notes:** The `sample1`/`sample2` of `youden_plot()` are the numeric columns of the two levels; `mean1`/`mean2`/`sd1`/`sd2` are the target values (when not given, they are estimated from the data). `$n_outside` and `$outside_rows` give the points outside the rectangle.

## 10.6 Key Points for Interpreting Results

1. **Target-value source**: The target mean and SD of the QC chart should come from the instructions or validation data, not be estimated on the fly from the data.
2. **Rule combination**: The Westgard rule combination and the disposition process for judging out of control should be pre-specified in the protocol and re-evaluated with the risk.
3. **Two-level complementarity**: The Youden plot looks at the joint performance of both levels simultaneously, identifying signs of systematic and random error.
4. **Out-of-control disposition**: After detecting an out-of-control point, stop reporting, investigate the cause, re-validate and recall if necessary, and record the complete disposition chain.
5. **Statistics ≠ release**: QC judgment is one basis for the release decision; the final decision should comply with the laboratory's quality-system requirements.

# 11. Sample Size Calculation

## 11.1 Analysis Objectives

Sample size calculation determines the required sample size or the achievable test power during the research-design and performance-evaluation planning phase. `ivdtools` provides three standalone functions corresponding to three common problem types:

- `sample_size_bland_altman()`: sample size/power for Bland–Altman agreement evaluation (Lu et al. 2016);
- `sample_size_proportion_ci()`: sample size required for a single-proportion confidence interval to reach a specified half-width;
- `sample_size_proportion()`: sample size/power/significance for a one-arm target-value test (a single-sample proportion against a known target).

All three functions are parameter-driven and do not require raw data files. They return scalars with attributes; `print()` gives a friendly explanation, and `as.numeric()` extracts the numeric result. The parameters in the examples (power, confidence level, allowable half-width, etc.) are teaching settings; formal studies must use the parameters specified in the protocol.

```
library(ivdtools)
```

## 11.2 Overview of Functions

| Function | Main purpose | Solving direction |
|---|---|---|
| `sample_size_bland_altman()` | BA agreement evaluation | from `power` find n, or from `n` find power |
| `sample_size_proportion_ci()` | Single-proportion confidence interval | from `d` find n, or from `n` find `d` |
| `sample_size_proportion()` | One-arm target-value test | given any two of `n/alpha/power`, find the third |

## 11.3 Example 1: Sample Size for Bland–Altman Agreement

### 11.3.1 Finding Sample Size from Target Power

```
n_ba <- sample_size_bland_altman(
  power = 0.80,
  mu = 0.2,
  sd = 1,
  delta = 2.5
)
print(n_ba)
```

```
Sample size for Bland-Altman agreement assessment
  Mode             : Compute n from power
  Mean difference  : 0.2000
  SD of differences: 1.0000
  Clinical limit   : 2.5000
  Confidence level : 0.9500
```

```
  Agreement level  : 0.9500
  Sample size (n)  : 201
  Power            : 0.800300
```

```
as.numeric(n_ba)
```

```
[1] 201
```

Under the settings of mean difference `mu = 0.2`, difference SD `sd = 1`, and clinical agreement limit `delta = 2.5`, about 201 pairs are needed to achieve 80% power.

> **Parameter notes:** `mu` is the mean difference, `sd` is the difference standard deviation (>0), and `delta` is the clinically acceptable agreement limit (>0). `conf.level` is the confidence level of the agreement-limit confidence interval, while `agree.level` is the coverage of the LoA; the two are at different levels — the former affects the reliability of the estimate, the latter determines the width of the LoA itself.

### 11.3.2 Finding Power from Sample Size

Given an existing sample size, look up the achievable power:

```
sample_size_bland_altman(n = 201, mu = 0.2, sd = 1, delta = 2.5)
```

```
Sample size for Bland-Altman agreement assessment
  Mode             : Compute power from n
  Mean difference  : 0.2000
  SD of differences: 1.0000
  Clinical limit   : 2.5000
  Confidence level : 0.9500
  Agreement level  : 0.9500
  Sample size (n)  : 201
  Power            : 0.800300
```

This mutually verifies the above; n=201 exactly corresponds to power 0.80.

## 11.4 Example 2: Sample Size for a Single-Proportion Confidence Interval

### 11.4.1 Finding Sample Size from Target Half-Width

```
n_pci <- sample_size_proportion_ci(p = 0.3, d = 0.05)
print(n_pci)
```

```
Single proportion confidence interval
  Mode             : Compute n from p and d
  Method           : wilson
  Target proportion: 0.3000
  Confidence level : 0.9500
  Sample size (n)  : 320
  Half-width (d)   : 0.050000
```

For an expected proportion p = 0.3, to keep the half-width of the 95% confidence interval no larger than 0.05, about 320 cases are needed.

### 11.4.2 Finding Half-Width from Sample Size

```
sample_size_proportion_ci(p = 0.3, n = 320)
```

```
Single proportion confidence interval
  Mode             : Compute d from p and n
  Method           : wilson
  Target proportion: 0.3000
  Confidence level : 0.9500
  Sample size (n)  : 320
  Half-width (d)   : 0.049967
```

At n=320 the half-width is about 0.05, mutually verifying the above.

**Parameter notes:** `p` is the expected proportion (must be given, 0<p<1) and `d` is the desired half-width (0<d≤0.5). `method` accepts 7 confidence-interval methods (default `"wilson"`, also `"wald"`, `"wald-cc"`, `"agresti-coull"`, `"jeffreys"`, `"wilson-cc"`, `"clopper-pearson"`); the method choice affects the required sample size and should be stated in the protocol.

## 11.5 Example 3: Sample Size for a One-Arm Target-Value Test

### 11.5.1 Finding Sample Size

```
n_sp <- sample_size_proportion(
  p0 = 0.2,
  p1 = 0.35,
  alpha = 0.05,
  power = 0.8,
  alternative = "greater",
  method = "wilson"
)
print(n_sp)
```

```
Sample size for a single-arm target value test
  Mode             : Compute n from alpha and power
  Method           : wilson
  H0: p <= 0.2000
  H1: p > 0.2000
  Expected p       : 0.3500
  Sample size (n)  : 47
  Alpha            : 0.050000
  Power            : 0.815687
  Critical x       : 14
```

Under the settings of p0 = 0.2 (target/null-hypothesis proportion), p1 = 0.35 (expected proportion), one-sided $\alpha = 0.05$, and power 0.80, about 47 cases are needed.

**Parameter notes:** `alternative` can be `"greater"` (in which case p1 > p0 is required), `"less"`, or `"two.sided"`. This method is based on confidence-interval inversion: given any two of `n`/`alpha`/`power`, find the third; when all three are given, it serves as a consistency check.

### 11.5.2 Finding Power

```
sample_size_proportion(
  p0 = 0.2,
  p1 = 0.35,
  n = 47,
```

```
  alpha = 0.05,
  alternative = "greater"
)
```

```
Sample size for a single-arm target value test
  Mode             : Compute power from n and alpha
  Method           : wilson
  H0: p <= 0.2000
  H1: p > 0.2000
  Expected p       : 0.3500
  Sample size (n)  : 47
  Alpha            : 0.050000
  Power            : 0.815687
  Critical x       : 14
```

### 11.5.3 Finding the Significance Level (alpha)

```
sample_size_proportion(
  p0 = 0.2,
  p1 = 0.35,
  n = 47,
  power = 0.8,
  alternative = "greater"
)
```

```
Sample size for a single-arm target value test
  Mode             : Compute alpha from n and power
  Method           : wilson
  H0: p <= 0.2000
  H1: p > 0.2000
  Expected p       : 0.3500
  Sample size (n)  : 47
  Alpha            : 0.046728
  Power            : 0.815687
  Critical x       : 14
```

The three solving directions verify one another, and the results are consistent (n=47, power=0.8, alpha≈0.05).

## 11.6 Key Points for Interpreting Results

1. **Parameters first**: Before computing sample size, determine power, confidence level, allowable error/half-width, expected proportion, and the one- or two-sided direction.
2. **Results are conditional**: The sample size is valid only under the set parameters; inaccurate parameter estimates lead to inaccurate sample sizes.
3. **Consistent methods**: For proportion-type problems, determine the CI method (Wilson, etc.) in advance; do not choose after the fact.
4. **Direction constraint**: For the one-arm test, note the consistency between `alternative` and the relative magnitude of p0/p1.
5. **Record in protocol**: Write the parameters, function calls, and outputs together into the study protocol for easy review and update.

# 12. Method Comparison

## 12.1 Analysis Objectives

Method comparison evaluates the agreement between the results measured on the same samples by a candidate method and a reference method, corresponding to CLSI EP09. The `mcr()` function in `ivdtools` provides a progressive S3 workflow: descriptive statistics → correlation analysis → regression (OLS, WLS, Deming, weighted Deming, Passing–Bablok) → Bland–Altman → outliers → bias at medical decision levels, and provides plotting, prediction, and summarization. This document demonstrates the full workflow with a complete example containing different lots and sample types, and demonstrates subgroup comparison after splitting by lot/sample type.

The example data are deterministic teaching data, into which an outlier has been deliberately added to demonstrate outlier detection; statistical results do not automatically constitute a pass/fail determination; acceptance criteria should be defined before analysis.

```
library(ivdtools)
library(readr)
```

## 12.2 Overview of Functions

| Function | Main purpose | Key input or output |
|---|---|---|
| `mcr()` | Create a method-comparison object | `id`, `candidate`, `reference`, `weights` |
| `describe()` | Descriptive statistics | candidate/reference/difference and extra columns |
| `correlation()` | Correlation analysis | pearson/spearman/kendall |
| `regression()` | Regression fitting | ols/wls/deming/wdeming/pb, `lambda`, `weights` |
| `bland_altman()` | Agreement limits | difference/ratio/percent, `x_axis` |
| `outlier()` | Difference outlier | grubbs/esd/dixon/iqr, `type` |
| `bias()` | Bias at medical decision levels | `mdl`, `interval` |
| `predict()` | Forward/inverse prediction | requires regression first |
| `plot()` / `summary()` | Graphics and summary | `type` selects the graphic |

## 12.3 Example: A Complete Single-Sample Analysis

### 12.3.1 Reading and Validating Data

The data contain 40 cases, each with an ID `id`, candidate method result `candidate`, reference method result `reference`, as well as a lot `batch` (B1/B2) and a sample type `sample_type` (serum/plasma/urine):

```
mc_dat <- read_csv("./data/method-comparison.csv", show_col_types = FALSE)
mc_dat <- as.data.frame(mc_dat)
str(mc_dat)
```

```
'data.frame':   40 obs. of  5 variables:
 $ id         : num  1 2 3 4 5 6 7 8 9 10 ...
 $ batch      : chr  "B1" "B1" "B1" "B1" ...
 $ sample_type: chr  "serum" "plasma" "urine" "serum" ...
 $ reference  : num  10 14.9 19.7 24.6 29.5 34.4 39.2 44.1 49 53.8 ...
 $ candidate  : num  11.6 16.6 20.9 30.4 34 ...
```

```
dim(mc_dat)
```

```
[1] 40  5
```

```
knitr::kable(table(mc_dat$batch), caption = "Lot distribution")
```

| Var1 | Freq |
|---|---|
| B1 | 20 |
| B2 | 20 |

```
knitr::kable(table(mc_dat$sample_type), caption = "Sample type distribution")
```

| Var1 | Freq |
|---|---|
| plasma | 13 |
| serum | 14 |
| urine | 13 |

### 12.3.2 Creating the Object and Descriptive Statistics

```
mc <- mcr(mc_dat, id = "id", candidate = "candidate", reference = "reference")
```

```
mc <- describe(mc, cols = c("batch", "sample_type"))
```

```
Descriptive Statistics
  Variable                n      Mean        SD       Min        Q1    Median        Q3
Max  NA
  ---------------------------------------------------------------------------
   candidate (X)      40 112.4977  59.2513  11.5900  63.1075 112.5750 159.5925 209.9000
   reference (Y)      40 105.0000  56.9555  10.0000  57.4750 105.0000 152.5250 200.0000
   Diff (X-Y)         40   7.4977   3.9583   1.2000   5.2725   6.6100   9.7950  22.1000

  batch
  Level                Count  Percent
   ------------------------------------
  B1                      20    50.0%
  B2                      20    50.0%

  sample_type
  Level                Count  Percent
   ------------------------------------
  plasma                  13    32.5%
  serum                   14    35.0%
  urine                   13    32.5%
```

```
mc
```

```
Method Comparison
  Data class:        data.frame
  Total rows:        40
  Complete pairs:    40
  ID variable:       id
  Test method (X):   candidate
  Reference (Y):     reference
  Duplicate IDs:     none

Analysis slots:
    [x] Describe
    [ ] Correlation
    [ ] Regression
    [ ] Outlier
    [ ] Bland-Altman
    [ ] Bias
```

The `cols` of `describe()` specifies the categorical/numeric extra columns to include in the description, and supports exclusion with a `-` prefix.

### 12.3.3 Correlation Analysis

```
mc <- correlation(mc, method = "pearson")
```

```
Correlation
  Method: pearson
  r = 0.9985, 95% CI [0.9971, 0.9992]
  p-value: < 2.2e-16
  n = 40
```

**Parameter notes:** The `method` of `correlation()` can be `"pearson"`, `"spearman"`, or `"kendall"`. Correlation reflects the degree to which the two variables co-vary; it is **not the same as agreement** and cannot replace Bland–Altman or regression bias evaluation.

### 12.3.4 Regression Analysis

Compare OLS, Deming, Passing–Bablok, and WLS in turn. Because `regression()` updates the regression result within the object, the multiple methods are stored in separate objects for comparison:

```
mc_ols <- regression(mc, method = "ols")
```

```
Regression
  Method: OLS
  Intercept = -2.9723  (SE = 1.0971)
  Slope     = 0.9598  (SE = 0.0087)
  Intercept 95% CI: [-5.1933, -0.7513]
  Slope     95% CI: [0.9423, 0.9773]
    H0: slope = 1 -> t = -4.6494, p = 3.944e-05
    slope significantly different from 1 (CI excludes 1)
  R-squared = 0.9969
  Sigma     = 3.2015
  n = 40
```

```
mc_dem <- regression(mc, method = "deming")
```

```
Regression
  Method: Deming
  Intercept = -3.1323  (SE = 1.4133)
  Slope     = 0.9612  (SE = 0.0092)
  Intercept 95% CI: [-5.9934, -0.2712]
  Slope     95% CI: [0.9426, 0.9798]
    H0: slope = 1 -> t = -4.2313, p = 0.0001413
    slope significantly different from 1 (CI excludes 1)
  Sigma     = 3.2026
  n = 40
```

```
mc_pb  <- regression(mc, method = "pb")
```

```
Regression
  Method: Passing-Bablok
  Intercept = -2.1674  (SE =    NA)
  Slope     = 0.9556  (SE =    NA)
  Intercept 95% CI: [-3.1487, -0.9828]
  Slope     95% CI: [0.9444, 0.9669]
  Sigma     = 3.2299
  n = 40
```

```
mc_wls <- regression(mc, method = "wls", weights = "1/y")
```

```
Regression
  Method: WLS
  Weights: 1/y
  Intercept = -2.2924  (SE = 0.9153)
  Slope     = 0.9518  (SE = 0.0108)
  Intercept 95% CI: [-4.1452, -0.4396]
  Slope     95% CI: [0.9299, 0.9736]
    H0: slope = 1 -> t = -4.4712, p = 6.818e-05
    slope significantly different from 1 (CI excludes 1)
  R-squared = 0.9951
  Sigma     = 0.4819
  n = 40
```

```
knitr::kable(
  data.frame(
    Method = c("OLS", "Deming", "Passing-Bablok", "WLS(1/y)"),
    Intercept = c(mc_ols$regression$intercept, mc_dem$regression$intercept,
             mc_pb$regression$intercept, mc_wls$regression$intercept),
    Slope = c(mc_ols$regression$slope, mc_dem$regression$slope,
             mc_pb$regression$slope, mc_wls$regression$slope)
  ),
  digits = 4,
  caption = "Intercept and slope comparison across the four regression methods"
)
```

| Method | Intercept | Slope |
|---|---|---|
| OLS | −2.9723 | 0.9598 |

| Method | Intercept | Slope |
|---|---|---|
| Deming | −3.1323 | 0.9612 |
| Passing-Bablok | −2.1674 | 0.9556 |
| WLS(1/y) | −2.2924 | 0.9518 |

```
print(mc_ols)
```

```
Method Comparison
  Data class:        data.frame
  Total rows:        40
  Complete pairs:    40
  ID variable:       id
  Test method (X):   candidate
  Reference (Y):     reference
  Duplicate IDs:     none

Analysis slots:
    [x] Describe
    [x] Correlation (pearson)
    [x] Regression (OLS)
    [ ] Outlier
    [ ] Bland-Altman
    [ ] Bias
```

In this example, the OLS slope is ≈ 0.96, and its 95% CI does not contain 1, indicating that the candidate method has about a 4% proportional bias (the true bias in the data is 5%).

**Parameter notes:** The `method` of `regression()` can be `"ols"`, `"wls"`, `"deming"`, `"wdeming"`, or `"pb"`. Deming uses `lambda` to specify the variance ratio (ref/cand, default 1); `wls`/`wdeming` must provide `weights` (a built-in scheme or a data column name); `conf.level` controls the parameter confidence interval.

### 12.3.5 Bias at Medical Decision Levels

Use `bias()` to evaluate the predicted bias at medical decision levels (MDL):

```
mc_ols <- bias(mc_ols, mdl = c(30, 80, 150), interval = "both")
```

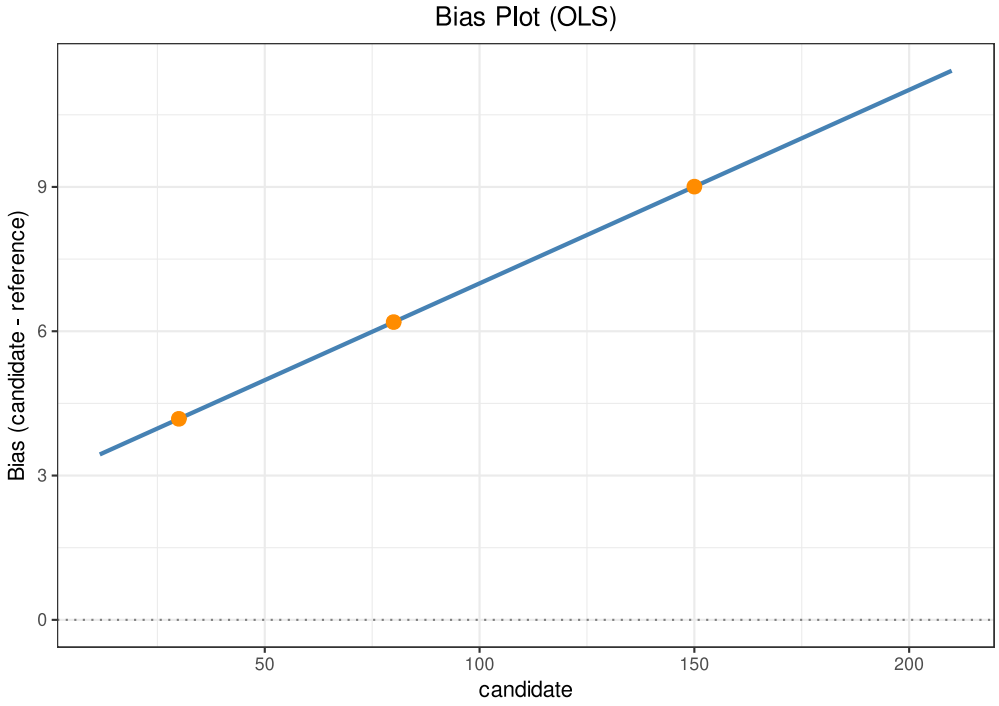
```
Bias
  Method: OLS
  Interval: both (95%)
  MDL points: 3
  Bias range: [4.1791, 9.0064]

 mdl     bias ci_lower  ci_upper  pi_lower pi_upper
-----------------------------------------------------
  30 4.179112 2.407658  5.950566 -2.539698 10.89792
  80 6.190463 5.018240  7.362685 -0.395770 12.77670
 150 9.006354 7.789153 10.223554  2.411967 15.60074
```

```
mc_ols$bias
```

```
Bias
```

```
  Method: OLS
  Interval: both (95%)
  MDL points: 3
  Bias range: [4.1791, 9.0064]

 mdl     bias ci_lower  ci_upper  pi_lower pi_upper
-------------------------------------------------
  30 4.179112 2.407658  5.950566 -2.539698 10.89792
  80 6.190463 5.018240  7.362685 -0.395770 12.77670
 150 9.006354 7.789153 10.223554  2.411967 15.60074
```

```
plot(mc_ols, type = "bias", mdl = c(30, 80, 150))
```




**Parameter notes:** In `bias(mdl, level, interval)`, `mdl` is a vector of medical decision levels; `interval` can be `""`, `"confidence"`, `"prediction"`, or `"both"`. Bias is defined as the difference between the fitted value of the candidate method at `mdl` and `mdl`; interpret it together with the regression direction and the allowable bias specified in the protocol.

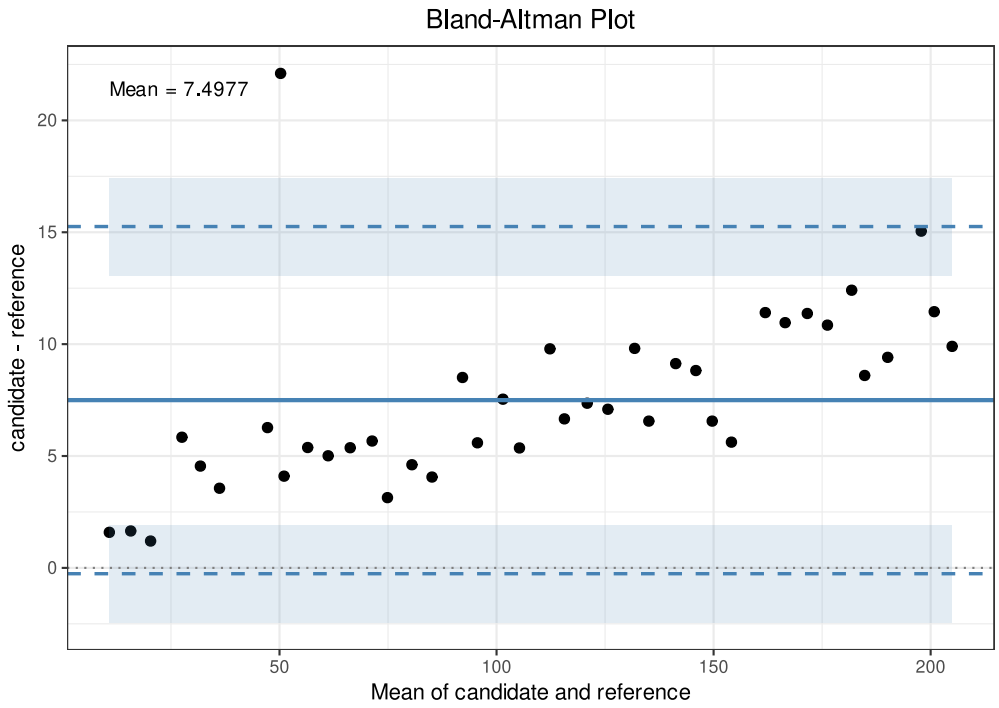
### 12.3.6 Bland–Altman Analysis

```
mc_ols <- bland_altman(mc_ols, type = "difference")
```

```
Bland-Altman
  Type:        difference
  Y-axis:      candidate - reference
  X-axis:      Mean of candidate and reference
  n = 40

  Mean diff:       7.4977
  SD diff:         3.9583
  95% LoA: [-0.2603, 15.2558]
```

```
  95% CI for LoA:
    Lower LoA: [-2.4419, 1.9213]
    Upper LoA: [13.0742, 17.4374]
```

```
plot(mc_ols, type = "bland_altman")
```




```
mc_ols$bland_altman[c("mean_diff", "sd_diff", "loa")]
```

```
$mean_diff
[1] 7.49775

$sd_diff
[1] 3.958259

$loa
     lower      upper
-0.2602947 15.2557947
```

**Parameter notes:** The `type` of `bland_altman()` can be `"difference"`, `"ratio"`, or `"percent"` (the Y-axis metric), and `x_axis` can be `"mean"`, `"candidate"`, or `"reference"` (the X-axis); `agree.level` controls the LoA width, and `conf.level` controls the LoA confidence interval.

### 12.3.7 Outlier Detection

Detect outliers in the paired differences:

```
mc_ols <- outlier(mc_ols, method = "grubbs", type = "difference")
```

```
Outlier
  Method: grubbs
```

```
  Data:   Difference (candidate - reference)
  Alpha:  0.05

  1 outlier(s) found:
  Row               Value   Statistic  Critical
   ------------------------------------------------
  7               22.1000      3.6891  3.0361
```

```
mc_ols$outlier
```

```

Outlier
  Method: grubbs
  Data:   Difference (candidate - reference)
  Alpha:  0.05

  1 outlier(s) found:
  Row               Value   Statistic  Critical
   ------------------------------------------------
  7               22.1000      3.6891  3.0361
```

```
mc_ols$outlier$indices
```

```
[1] 7
```

The difference for sample 7 is detected as clearly too large. **Detecting an outlier does not mean it should be deleted** — go back to the original records and experimental procedure, and perform an include/exclude sensitivity analysis when there is sufficient justification.

### 12.3.8 Prediction

```
predict(mc_ols, candidate = c(40, 100))
```

```
  candidate ref_pred
1        40 35.41862
2       100 93.00500
```

```
predict(mc_ols, reference = c(50, 120), inverse = TRUE)
```

```
  reference cand_pred
1        50  55.19253
2       120 128.12645
```

### 12.3.9 Subgroup Comparison

The method performance may differ across lots or sample types. First split by lot, then re-run regression and Bland–Altman on each subgroup:

```
for (b in c("B1", "B2")) {
  sub <- mc_dat[mc_dat$batch == b, ]
  mcs <- mcr(sub, id = "id", candidate = "candidate", reference = "reference")
  mcs <- regression(mcs, method = "ols")
  mcs <- bland_altman(mcs, type = "difference")
  cat(b, ": slope =", round(mcs$regression$slope, 4),
      ", mean diff =", round(mcs$bland_altman$mean_diff, 3), "\n")
}
```

```
Regression
  Method: OLS
  Intercept = -2.9704  (SE = 2.2333)
  Slope     = 0.9582  (SE = 0.0327)
  Intercept 95% CI: [-7.6623, 1.7215]
  Slope     95% CI: [0.8895, 1.0269]
    H0: slope = 1 -> t = -1.2784, p = 0.2173
    slope not significantly different from 1 (CI contains 1)
  R-squared = 0.9795
  Sigma     = 4.2427
  n = 20

Bland-Altman
  Type:        difference
  Y-axis:      candidate - reference
  X-axis:      Mean of candidate and reference
  n = 20

  Mean diff:       5.5550
  SD diff:         4.3129
  95% LoA: [-2.8981, 14.0081]

  95% CI for LoA:
    Lower LoA: [-6.4069, 0.6107]
    Upper LoA: [10.4993, 17.5169]
B1 : slope = 0.9582 , mean diff = 5.555

Regression
  Method: OLS
  Intercept = -1.3159  (SE = 2.3598)
  Slope     = 0.9502  (SE = 0.0142)
  Intercept 95% CI: [-6.2736, 3.6419]
  Slope     95% CI: [0.9203, 0.9801]
    H0: slope = 1 -> t = -3.4988, p = 0.002564
    slope significantly different from 1 (CI excludes 1)
  R-squared = 0.9960
  Sigma     = 1.8780
  n = 20

Bland-Altman
  Type:        difference
  Y-axis:      candidate - reference
  X-axis:      Mean of candidate and reference
  n = 20

  Mean diff:       9.4405
  SD diff:         2.3693
  95% LoA: [4.7968, 14.0842]

  95% CI for LoA:
    Lower LoA: [2.8693, 6.7244]
    Upper LoA: [12.1566, 16.0117]
B2 : slope = 0.9502 , mean diff = 9.44
```

Similarly, `split()` by sample type and analyze separately. Subgroup-comparison conclusions are used to judge whether method performance is affected by lot/sample type; only pre-defined acceptance criteria can support an acceptability conclusion.

### 12.3.10 Summary

```
summary(mc_ols)
```

```
Method Comparison Regression (mcr)
------------------------------------------------------

Method Comparison
  Data class:        data.frame
  Total rows:        40
  Complete pairs:    40
  ID variable:       id
  Test method (X):   candidate
  Reference (Y):     reference
  Duplicate IDs:     none

Analysis slots:
    [x] Describe
    [x] Correlation (pearson)
    [x] Regression (OLS)
    [x] Outlier (grubbs)
    [x] Bland-Altman (difference)
    [x] Bias

Descriptive Statistics
  Variable                n      Mean        SD       Min        Q1    Median        Q3
Max  NA
  ------------------------------------------------------------------------------
   candidate (X)     40 112.4977  59.2513  11.5900  63.1075 112.5750 159.5925 209.9000
   reference (Y)     40 105.0000  56.9555  10.0000  57.4750 105.0000 152.5250 200.0000
   Diff (X-Y)         40   7.4977   3.9583   1.2000   5.2725   6.6100   9.7950  22.1000

  batch
  Level                Count  Percent
   ------------------------------------
  B1                      20    50.0%
  B2                      20    50.0%

  sample_type
  Level                Count  Percent
   ------------------------------------
  plasma                  13    32.5%
  serum                   14    35.0%
  urine                   13    32.5%

Correlation
  Method: pearson
  r = 0.9985, 95% CI [0.9971, 0.9992]
  p-value: < 2.2e-16
  n = 40

Regression
  Method: OLS
  Intercept = -2.9723  (SE = 1.0971)
  Slope     = 0.9598  (SE = 0.0087)
  Intercept 95% CI: [-5.1933, -0.7513]
  Slope     95% CI: [0.9423, 0.9773]
    H0: slope = 1 -> t = -4.6494, p = 3.944e-05
    slope significantly different from 1 (CI excludes 1)
```

```
  R-squared = 0.9969
  Sigma     = 3.2015
  n = 40

Outlier
  Method: grubbs
  Data:   Difference (candidate - reference)
  Alpha:  0.05

  1 outlier(s) found:
  Row               Value   Statistic  Critical
   ---------------------------------------------------
  7               22.1000      3.6891  3.0361

Bland-Altman
  Type:        difference
  Y-axis:      candidate - reference
  X-axis:      Mean of candidate and reference
  n = 40

  Mean diff:       7.4977
  SD diff:         3.9583
  95% LoA: [-0.2603, 15.2558]

  95% CI for LoA:
    Lower LoA: [-2.4419, 1.9213]
    Upper LoA: [13.0742, 17.4374]

Bias
  Method: OLS
  Interval: both (95%)
  MDL points: 3
  Bias range: [4.1791, 9.0064]

 mdl     bias ci_lower  ci_upper  pi_lower pi_upper
-------------------------------------------------
  30 4.179112 2.407658  5.950566 -2.539698 10.89792
  80 6.190463 5.018240  7.362685 -0.395770 12.77670
 150 9.006354 7.789153 10.223554  2.411967 15.60074
```

## 12.4 Key Points for Interpreting Results

1. **Direction check**: For both regression and Bland–Altman, confirm the direction and bias sign of the candidate relative to the reference.
2. **Correlation ≠ agreement**: A high correlation coefficient does not mean the methods agree; focus on the regression slope and the Bland–Altman LoA.
3. **Method selection**: OLS assumes the reference has no error; use Deming/PB when the reference has error; consider WLS/WDeming when variances are unequal.
4. **Outlier handling**: After detecting an outlier, investigate and perform a sensitivity analysis; do not delete automatically.
5. **Subgroup interpretation**: Judge lot/sample-type splits separately to avoid the mean masking subgroup differences.
6. **Acceptance criteria**: Bias, LoA, and MDL bias must all be compared with the pre-specified allowable limits before drawing an acceptability conclusion.

# 13. Qualitative Analysis

## 13.1 Analysis Objectives

Qualitative analysis evaluates the degree of agreement between the candidate method and the reference method in positive/negative determinations, corresponding to CLSI EP12. `ivdtools` provides `raw_to_table()` to build a four-fold table from raw paired data, `counts_to_table()` to build one directly from existing TP/FP/TN/FN counts, and three accompanying analysis methods: `diagnostics()` (sensitivity, specificity, PPV, NPV, likelihood ratios, etc.), `kappa()` (Kappa and PABAK), and `mcnemar()` (paired-difference test). Quantitative results can be converted to qualitative with `continuous_to_binary()`.

This document uses three examples to demonstrate the complete analysis for the three data sources. The example data are deterministic teaching data; the positive direction and threshold should be determined according to the protocol.

```
library(ivdtools)
library(readr)
```

## 13.2 Overview of Functions

| Function | Main purpose | Key input or output |
|---|---|---|
| `raw_to_table()` | Build a four-fold table from raw paired data | `candidate`, `reference`, `positive`, `na.rm` |
| `counts_to_table()` | Build a four-fold table from TP/FP/TN/FN | `candidate`/`reference` naming |
| `continuous_to_binary()` | Convert quantitative to qualitative | `cutoff` (≥ cutoff is 1) |
| `describe()` | Description of raw data | only available for `raw_to_table()` sources |
| `diagnostics()` | Diagnostic accuracy metrics | `ci.method`, `prevalence` |
| `kappa()` | Kappa / PABAK | `prevalence` switch |
| `mcnemar()` | Paired-difference test | uses exact test when discordant pairs <25 |

## 13.3 Example 1: Complete Analysis of Raw Paired Data

### 13.3.1 Reading and Validating Data

```
qual <- read_csv("./data/qualitative-raw.csv", show_col_types = FALSE)
qual <- as.data.frame(qual)
str(qual)
```

```
'data.frame':   60 obs. of  5 variables:
 $ id    : chr  "S001" "S002" "S003" "S004" ...
 $ new   : chr  "positive" "negative" "negative" "negative" ...
 $ gold  : chr  "positive" "negative" "negative" "negative" ...
```

```
 $ age   : num  46.1 44.7 65.7 60 64.6 32.7 47 40.9 38.4 48.1 ...
 $ gender: chr  "M" "F" "M" "F" ...
```

```
table(qual$gold)
```

```

negative positive
      36       24
```

```
table(qual$new)
```

```

negative positive
      36       24
```

### 13.3.2 Building the Four-Fold Table

```
qa <- raw_to_table(
  qual,
  candidate = "new",
  reference = "gold",
  id = "id",
  positive = "positive"
)
qa
```

```

Fourfold Table
  Source            : raw data
  Total n           : 60
  Duplicate IDs     : none
  Candidate         : new
  Reference         : gold

  2x2 Contingency Table
  ----------------------------------------
              gold
                  positive  negative  Sum
  ----------------------------------------
  new   positive  23        1         24
        negative  1         35        36
        sum       24        36        60
  ----------------------------------------
```

**Parameter notes:** `raw_to_table()` requires that both the candidate and reference columns have exactly 2 levels; `positive` specifies the positive level (used for level mapping of factor/character columns); `na.rm` controls whether missing rows are deleted. `id` is optional and used for duplicate-sample detection.

### 13.3.3 Describing the Raw Data

```
qa <- describe(qa)
```

```
Data Summary
  Rows:    60
  Columns: 5
  ID:      60 unique, no duplicates

  new
    negative        36  ( 60.0%)
    positive        24  ( 40.0%)
  gold
    negative        36  ( 60.0%)
    positive        24  ( 40.0%)
  age
    Mean (SD):       51.05 (13.45)
    Median (Q1-Q3):  48.90 (42.15 - 57.28)
    Range:           24.30 - 90.10
  gender
    F               32  ( 53.3%)
    M               28  ( 46.7%)
```

`describe()` is only available for `raw_to_table()` sources; it outputs the row count, ID duplicates, per-column summaries, and missingness.

### 13.3.4 Diagnostic Accuracy Metrics

```
qa <- diagnostics(qa, ci.method = "wilson")
```

```
Diagnostic Performance Summary
  Sensitivity        0.958  (0.798, 0.993)
  Specificity        0.972  (0.858, 0.995)
  PPV                0.958  (0.798, 0.993)
  NPV                0.972  (0.858, 0.995)
  Accuracy           0.967  (0.886, 0.991)
  Prevalence         0.400  (0.286, 0.526)
  LR+                34.500  (4.986, 238.724)
  LR-                0.043  (0.006, 0.292)
  Odds Ratio         805.000  (47.921, 13522.837)

  # PPV & NPV are based on data prevalence of 0.400.
  # Confidence intervals for proportions use 'wilson' method.
```

```
qa$diagnostics[c("sensitivity", "specificity", "ppv", "npv", "accuracy")]
```

```
$sensitivity
      est     lower     upper
0.9583333 0.7975819 0.9926065

$specificity
      est     lower     upper
0.9722222 0.8583028 0.9950796

$ppv
      est     lower     upper
0.9583333 0.7975819 0.9926065

$npv
      est     lower     upper
0.9722222 0.8583028 0.9950796
```

```
$accuracy
      est     lower     upper
0.9666667 0.8863623 0.9908107
```

In this example, the sensitivity is 0.958 and the specificity is 0.972. If the prevalence in the target population is known, use `prevalence` to recompute PPV/NPV:

```
qa <- diagnostics(qa, prevalence = 0.4)
```

```
Diagnostic Performance Summary
  Sensitivity        0.958  (0.798, 0.993)
  Specificity        0.972  (0.858, 0.995)
  PPV                0.958  (0.790, 0.993)
  NPV                0.972  (0.864, 0.995)
  Accuracy           0.967  (0.886, 0.991)
  Prevalence         0.400
  LR+                34.500  (4.986, 238.724)
  LR-                0.043  (0.006, 0.292)
  Odds Ratio         805.000  (47.921, 13522.837)

  # PPV & NPV are based on user-specified prevalence of 0.400.
  # Confidence intervals for proportions use 'wilson' method.
```

```
qa$diagnostics[c("ppv", "npv")]
```

```
$ppv
      est     lower     upper
0.9583333 0.7895852 0.9926193

$npv
      est     lower     upper
0.9722222 0.8641373 0.9950711
```

**Parameter notes:** The `ci.method` of `diagnostics()` accepts 7 proportion confidence-interval methods (default `"wilson"`); when `prevalence` is given, PPV/NPV are recomputed by the Bayes formula; otherwise they are based on the observed prevalence.

### 13.3.5 Kappa Agreement

```
qa <- kappa(qa)
```

```
Cohen's Kappa
  new  vs  gold
  n = 60

  Kappa                0.9306
  SE                   0.0483
  95% CI               (0.8359, 1.0000)
  Observed agreement   0.9667
  Expected agreement   0.5200
```

```
qa <- kappa(qa, prevalence = 0.5)
```

```
PABAK
  new  vs  gold
  n = 60

  Kappa                0.9333
  SE                   0.0463
  95% CI               (0.8425, 1.0000)
  Observed agreement   0.9667
  Expected agreement   0.5000
  # PABAK (prevalence = 0.500): 2 * p_obs - 1, assumes expected agreement = 0.5
```

**Parameter notes:** `kappa()` gives Cohen's Kappa; when `prevalence` is given, it instead computes PABAK (`2·p_obs − 1`). The two handle prevalence bias differently; note the method in the report.

### 13.3.6 McNemar Test

```
qa <- mcnemar(qa)
```

```
McNemar Test
  new  vs  gold
  Discordant pairs:  b (pos, neg) = 1,  c (neg, pos) = 1
  n (b + c)        = 2

  Method: Exact binomial test
  P-value:          1.0000
  Conclusion: Not significant at 95% confidence level
```

The discordant pairs b+c = 2 < 25, so the function automatically uses the exact binomial test.

### 13.3.7 Summary

```
summary(qa)
```

```
Qualitative Analysis - summary
------------------------------------------------

Fourfold Table
  Source            : raw data
  Total n           : 60
  Duplicate IDs     : none
  Candidate         : new
  Reference         : gold

  2x2 Contingency Table
  ----------------------------------------
             gold
                 positive  negative  Sum
  ----------------------------------------
  new   positive  23        1         24
        negative  1         35        36
        sum       24        36        60
  ----------------------------------------

Analyses performed:
```

```
  [x] Describe
  [x] Diagnostics
  [x] Kappa
  [x] McNemar

Data Summary
  Rows:    60
  Columns: 5
  ID:      60 unique, no duplicates

  new
    negative        36  ( 60.0%)
    positive        24  ( 40.0%)
  gold
    negative        36  ( 60.0%)
    positive        24  ( 40.0%)
  age
    Mean (SD):       51.05 (13.45)
    Median (Q1-Q3):  48.90 (42.15 - 57.28)
    Range:           24.30 - 90.10
  gender
    F               32  ( 53.3%)
    M               28  ( 46.7%)

Diagnostic Performance Summary
  Sensitivity        0.958  (0.798, 0.993)
  Specificity        0.972  (0.858, 0.995)
  PPV                0.958  (0.790, 0.993)
  NPV                0.972  (0.864, 0.995)
  Accuracy           0.967  (0.886, 0.991)
  Prevalence         0.400
  LR+                34.500  (4.986, 238.724)
  LR-                0.043  (0.006, 0.292)
  Odds Ratio         805.000  (47.921, 13522.837)

  # PPV & NPV are based on user-specified prevalence of 0.400.
  # Confidence intervals for proportions use 'wilson' method.

PABAK
  new  vs  gold
  n = 60

  Kappa                0.9333
  SE                   0.0463
  95% CI               (0.8425, 1.0000)
  Observed agreement   0.9667
  Expected agreement   0.5000
  # PABAK (prevalence = 0.500): 2 * p_obs - 1, assumes expected agreement = 0.5

McNemar Test
  new  vs  gold
  Discordant pairs:  b (pos, neg) = 1,  c (neg, pos) = 1
  n (b + c)        = 2

  Method: Exact binomial test
  P-value:          1.0000
  Conclusion: Not significant at 95% confidence level
```

## 13.4 Example 2: Building Directly from TP/FP/TN/FN

When counts are already available, no raw data are needed:

```
tab <- counts_to_table(
  tp = 80, fp = 5, tn = 120, fn = 3,
  candidate = "new method",
  reference = "gold standard"
)
tab
```

```
Fourfold Table
  Source            : counts
  Total n           : 208
  Candidate         : new method
  Reference         : gold standard

  2x2 Contingency Table
  -----------------------------------------------
                     gold standard
                         positive  negative  Sum
  -----------------------------------------------
  new method   positive  80        5         85
               negative  3         120       123
               sum       83        125       208
  -----------------------------------------------
```

```
tab <- diagnostics(tab)
```

```
Diagnostic Performance Summary
  Sensitivity        0.964  (0.899, 0.988)
  Specificity        0.960  (0.910, 0.983)
  PPV                0.941  (0.870, 0.975)
  NPV                0.976  (0.931, 0.992)
  Accuracy           0.962  (0.926, 0.980)
  Prevalence         0.399  (0.335, 0.467)
  LR+                24.096  (10.198, 56.934)
  LR-                0.038  (0.012, 0.114)
  Odds Ratio         640.000  (148.773, 2753.180)

  # PPV & NPV are based on data prevalence of 0.399.
  # Confidence intervals for proportions use 'wilson' method.
```

```
tab <- kappa(tab)
```

```
Cohen's Kappa
  new method  vs  gold standard
  n = 208

  Kappa                0.9201
  SE                   0.0277
  95% CI               (0.8659, 0.9744)
  Observed agreement   0.9615
  Expected agreement   0.5184
```

```
tab <- mcnemar(tab)
```

```
McNemar Test
  new method  vs  gold standard
  Discordant pairs:  b (pos, neg) = 5,  c (neg, pos) = 3
  n (b + c)        = 8

  Method: Exact binomial test
  P-value:          0.7266
  Conclusion: Not significant at 95% confidence level
```

```
summary(tab)
```

```
Qualitative Analysis - summary
------------------------------------------------

Fourfold Table
  Source            : counts
  Total n           : 208
  Candidate         : new method
  Reference         : gold standard

  2x2 Contingency Table
  ------------------------------------------------
                    gold standard
                        positive  negative  Sum
  ------------------------------------------------
  new method   positive  80        5         85
               negative  3         120       123
               sum       83        125       208
  ------------------------------------------------

Analyses performed:

  [ ] Describe
  [x] Diagnostics
  [x] Kappa
  [x] McNemar

Diagnostic Performance Summary
  Sensitivity        0.964  (0.899, 0.988)
  Specificity        0.960  (0.910, 0.983)
  PPV                0.941  (0.870, 0.975)
  NPV                0.976  (0.931, 0.992)
  Accuracy           0.962  (0.926, 0.980)
  Prevalence         0.399  (0.335, 0.467)
  LR+                24.096  (10.198, 56.934)
  LR-                0.038  (0.012, 0.114)
  Odds Ratio         640.000  (148.773, 2753.180)

  # PPV & NPV are based on data prevalence of 0.399.
  # Confidence intervals for proportions use 'wilson' method.

Cohen's Kappa
  new method  vs  gold standard
  n = 208
```

```
  Kappa                0.9201
  SE                   0.0277
  95% CI               (0.8659, 0.9744)
  Observed agreement   0.9615
  Expected agreement   0.5184

McNemar Test
  new method  vs  gold standard
  Discordant pairs:  b (pos, neg) = 5,  c (neg, pos) = 3
  n (b + c)        = 8

  Method: Exact binomial test
  P-value:          0.7266
  Conclusion: Not significant at 95% confidence level
```

```
describe(tab)   # counts sources have no raw data; describe() is unavailable
```

Objects built by `counts_to_table()` have no raw data, so `describe()` is unavailable; `diagnostics()`, `kappa()`, and `mcnemar()` all work normally.

## 13.5 Example 3: Converting Quantitative Data to Qualitative

### 13.5.1 Reading Data

The candidate method is a quantitative result `result` and the reference is qualitative `gold`:

```
cont <- read_csv("./data/qualitative-continuous.csv", show_col_types = FALSE)
cont <- as.data.frame(cont)
str(cont)
```

```
'data.frame':   50 obs. of  3 variables:
 $ id    : chr  "S001" "S002" "S003" "S004" ...
 $ result: num  10.11 13.06 1.53 7.78 4.98 ...
 $ gold  : chr  "positive" "positive" "negative" "negative" ...
```

### 13.5.2 Converting to Binary

Convert to binary at cutoff = 8:

```
cont_bin <- continuous_to_binary(cont, cols = "result", cutoff = 8)
head(cont_bin)
```

```
    id    result     gold result_binary
1 S001 10.111087 positive             1
2 S002 13.057209 positive             1
3 S003  1.534680 negative             0
4 S004  7.782787 negative             0
5 S005  4.982424 negative             0
6 S006  9.443399 negative             1
```

`continuous_to_binary()` generates a `result_binary` column (≥ cutoff is 1, otherwise 0). To satisfy the `positive` mapping requirement of `raw_to_table()`, first relabel it with labels consistent with the reference:

```
cont_bin$result_binary_lab  <-  ifelse(cont_bin$result_binary  ==  1,  "positive",
"negative")
```

### 13.5.3 Building the Four-Fold Table and Analyzing

```
qa3 <- raw_to_table(
  cont_bin,
  candidate = "result_binary_lab",
  reference = "gold",
  id = "id",
  positive = "positive"
)
qa3 <- diagnostics(qa3)
```

```
Diagnostic Performance Summary
  Sensitivity        0.952  (0.773, 0.992)
  Specificity        0.897  (0.736, 0.964)
  PPV                0.870  (0.679, 0.955)
  NPV                0.963  (0.817, 0.993)
  Accuracy           0.920  (0.812, 0.968)
  Prevalence         0.420  (0.294, 0.558)
  LR+                9.206  (3.140, 26.994)
  LR-                0.053  (0.008, 0.361)
  Odds Ratio         173.333  (16.746, 1794.100)

  # PPV & NPV are based on data prevalence of 0.420.
  # Confidence intervals for proportions use 'wilson' method.
```

```
qa3$diagnostics[c("sensitivity", "specificity", "accuracy")]
```

```
$sensitivity
      est     lower     upper
0.9523810 0.7733064 0.9915440

$specificity
      est     lower     upper
0.8965517 0.7361492 0.9641851

$accuracy
      est     lower     upper
0.9200000 0.8116175 0.9684505
```

**Note:** The level specified by `positive` must exist in both columns, so first relabel the binary column as `positive/negative`. If `positive` is not specified, the function may reverse the positive/negative direction according to the sorted levels, flipping TP/FP; be sure to check.

## 13.6 Key Points for Interpreting Results

1. **Direction consistent**: Confirm that the positive-direction definitions of the candidate and reference are consistent, to avoid flipping the whole four-fold table.
2. **Levels unique**: Both the candidate and reference columns must have exactly 2 levels; multi-level or continuous columns must first be converted to binary.
3. **Prevalence**: PPV/NPV depend on prevalence; report the observed prevalence or specify the target-population prevalence.
4. **Method labeling**: Kappa and PABAK have different meanings; McNemar uses an exact test for small numbers of discordant pairs; report the method.
5. **Threshold recorded**: Converting quantitative to qualitative must record the cutoff and positive direction; the results hold only under that threshold.

# 14. ROC Curve and Cutoff Determination

## 14.1 Analysis Objectives

The receiver operating characteristic (ROC) curve evaluates the ability of a quantitative marker to distinguish positive from negative outcomes, and is commonly used to establish a diagnostic cutoff. The `roc()` function in `ivdtools` provides a progressive workflow: `roc()` creates the object → `describe()` → `auc()` (AUC and confidence interval) → `cutoff()` (Youden and closest-corner optimal cutoffs) → `plot()` and `summary()`.

This document demonstrates the complete ROC analysis workflow with a complete single-marker example. The reference variable must be binary; if the original reference is a continuous variable, first convert it to binary (see `continuous_to_binary()` in 《Qualitative Analysis》) and record the threshold and positive direction.

```
library(ivdtools)
library(readr)
```

## 14.2 Overview of Functions

| Function | Main purpose | Key input or output |
|---|---|---|
| `roc()` | Create a ROC object | `cols`, `reference`, `id`, `positive` |
| `describe()` | Marker descriptive statistics | reference forced to be treated as categorical |
| `auc()` | AUC and confidence interval | per marker |
| `cutoff()` | Optimal cutoff | Youden and closest corner; requires `auc()` first |
| `plot()` | ROC curve | `cutpoint` marks the optimal point |
| `summary()` | Summary | |

## 14.3 Example: Single-Marker ROC Analysis

### 14.3.1 Reading and Validating Data

```
roc_dat <- read_csv("./data/roc-example.csv", show_col_types = FALSE)
roc_dat <- as.data.frame(roc_dat)
str(roc_dat)
```

```
'data.frame':   100 obs. of  3 variables:
 $ sid: num  1 2 3 4 5 6 7 8 9 10 ...
 $ ref: num  0 0 0 0 0 0 0 0 0 0 ...
 $ x1 : num  7.59 8.31 5.89 5.59 9.57 ...
```

```
table(roc_dat$ref)
```

```
 0  1
50 50
```

The reference column `ref` is binary 0/1, with 50 positive and 50 negative cases.

### 14.3.2 Creating the Object and Description

```
ro <- roc(roc_dat, cols = "x1", reference = "ref", id = "sid")
ro
```

```
ROC Analysis
  Data class:        data.frame
  Total rows:        100
  Complete cases:    100
  Reference:         ref (binary)
  Positive / Neg:    50 / 50
  Evaluation cols:   1  (x1)
  Duplicate IDs:     none

Analysis slots:
    [ ] Describe
    [ ] AUC
    [ ] Cutoff
    [ ] MLR
```

```
ro <- describe(ro)
```

```
Descriptive Statistics
  Variable                n      Mean        SD       Min        Q1    Median        Q3
Max  NA
  ------------------------------------------------------------------------------
  x1                100   9.2576   4.4744  -0.3894   5.8224   8.5472  12.9591  18.7295

  ref
  Level                Count  Percent
   ------------------------------------
  0                       50    50.0%
  1                       50    50.0%
```

**Parameter notes:** The `cols` of `roc()` are the quantitative marker columns; `reference` is the binary reference column (either numeric 0/1, a two-level factor, or a two-value character is accepted); `positive` specifies the positive level (when not given, the last sorted level is taken by default). The direction of each marker (`geq`/`leq`) is determined automatically from the mean of the positive group.

### 14.3.3 AUC

```
ro <- auc(ro)
```

```
AUC Table
  Column               AUC          95% CI  Direction
  ------------------------------------------------------------
  x1                0.9420  [0.8940, 0.9900]       geq
```

```
ro$auc_table
```

```
AUC Table
  Column               AUC          95% CI  Direction
  -------------------------------------------------------------
  x1                0.9420  [0.8940, 0.9900]       geq
```

The AUC of marker x1 is ≈ 0.942 (95% CI [0.894, 0.990]), indicating strong discrimination.

### 14.3.4 Optimal Cutoff

```
ro <- cutoff(ro)
```

```
Optimal Cutoff Analysis
  Column: x1
    Youden / Corner: cutoff = 8.7617  (sens = 0.9000, spec = 0.9200, Youden = 0.8200)

  Full cutoff table:
       Cutoff      Sens      Spec     TP    FP    TN    FN    Youden
  ------------------------------------------------------------------------
         -Inf    1.0000    0.0000     50    50     0     0    0.0000
      18.7295    0.0200    1.0000      1     0    50    49    0.0200
      18.5675    0.0400    1.0000      2     0    50    48    0.0400
      18.2248    0.0600    1.0000      3     0    50    47    0.0600
      17.7837    0.0800    1.0000      4     0    50    46    0.0800
      16.4453    0.1000    1.0000      5     0    50    45    0.1000
      15.8893    0.1200    1.0000      6     0    50    44    0.1200
      15.8087    0.1400    1.0000      7     0    50    43    0.1400
      15.6058    0.1600    1.0000      8     0    50    42    0.1600
      15.5003    0.1800    1.0000      9     0    50    41    0.1800
      15.2855    0.2000    1.0000     10     0    50    40    0.2000
      15.0168    0.2200    1.0000     11     0    50    39    0.2200
      14.8639    0.2400    1.0000     12     0    50    38    0.2400
      14.6980    0.2600    1.0000     13     0    50    37    0.2600
      14.4757    0.2800    1.0000     14     0    50    36    0.2800
      14.4469    0.3000    1.0000     15     0    50    35    0.3000
      14.4042    0.3200    1.0000     16     0    50    34    0.3200
      14.1952    0.3400    1.0000     17     0    50    33    0.3400
      14.1938    0.3600    1.0000     18     0    50    32    0.3600
      13.9382    0.3800    1.0000     19     0    50    31    0.3800
      13.7845    0.4000    1.0000     20     0    50    30    0.4000
      13.3718    0.4200    1.0000     21     0    50    29    0.4200
      13.0699    0.4400    1.0000     22     0    50    28    0.4400
      12.9970    0.4600    1.0000     23     0    50    27    0.4600
      12.9854    0.4800    1.0000     24     0    50    26    0.4800
      12.9659    0.5000    1.0000     25     0    50    25    0.5000
      12.9569    0.5200    1.0000     26     0    50    24    0.5200
      12.7812    0.5400    1.0000     27     0    50    23    0.5400
      12.7534    0.5600    1.0000     28     0    50    22    0.5600
      12.5227    0.5800    1.0000     29     0    50    21    0.5800
      12.3440    0.6000    1.0000     30     0    50    20    0.6000
      12.2962    0.6200    1.0000     31     0    50    19    0.6200
      12.1394    0.6400    1.0000     32     0    50    18    0.6400
      12.1354    0.6600    1.0000     33     0    50    17    0.6600
      12.0393    0.6800    1.0000     34     0    50    16    0.6800
      11.7418    0.6800    0.9800     34     1    49    16    0.6600
      11.6642    0.7000    0.9800     35     1    49    15    0.6800
```

```
11.5082    0.7000    0.9600     35     2    48    15    0.6600
11.4038    0.7200    0.9600     36     2    48    14    0.6800
11.2133    0.7400    0.9600     37     2    48    13    0.7000
11.0696    0.7600    0.9600     38     2    48    12    0.7200
10.3902    0.7800    0.9600     39     2    48    11    0.7400
10.1719    0.8000    0.9600     40     2    48    10    0.7600
 9.8836    0.8200    0.9600     41     2    48     9    0.7800
 9.5685    0.8200    0.9400     41     3    47     9    0.7600
 9.3526    0.8400    0.9400     42     3    47     8    0.7800
 9.1944    0.8600    0.9400     43     3    47     7    0.8000
 9.1032    0.8800    0.9400     44     3    47     6    0.8200
 8.8558    0.8800    0.9200     44     4    46     6    0.8000
 8.7617    0.9000    0.9200     45     4    46     5    0.8200
 8.5538    0.9000    0.9000     45     5    45     5    0.8000
 8.5406    0.9000    0.8800     45     6    44     5    0.7800
 8.5296    0.9200    0.8800     46     6    44     4    0.8000
 8.5259    0.9200    0.8600     46     7    43     4    0.7800
 8.3125    0.9200    0.8400     46     8    42     4    0.7600
 8.2985    0.9200    0.8200     46     9    41     4    0.7400
 7.9078    0.9200    0.8000     46    10    40     4    0.7200
 7.8493    0.9200    0.7800     46    11    39     4    0.7000
 7.6187    0.9200    0.7600     46    12    38     4    0.6800
 7.5941    0.9200    0.7400     46    13    37     4    0.6600
 7.5556    0.9200    0.7200     46    14    36     4    0.6400
 7.5150    0.9200    0.7000     46    15    35     4    0.6200
 7.4355    0.9200    0.6800     46    16    34     4    0.6000
 7.1837    0.9200    0.6600     46    17    33     4    0.5800
 6.8235    0.9400    0.6600     47    17    33     3    0.6000
 6.8029    0.9400    0.6400     47    18    32     3    0.5800
 6.7765    0.9400    0.6200     47    19    31     3    0.5600
 6.6770    0.9400    0.6000     47    20    30     3    0.5400
 6.5446    0.9400    0.5800     47    21    29     3    0.5200
 6.5440    0.9600    0.5800     48    21    29     2    0.5400
 6.1603    0.9600    0.5600     48    22    28     2    0.5200
 6.0665    0.9600    0.5400     48    23    27     2    0.5000
 6.0554    0.9600    0.5200     48    24    26     2    0.4800
 6.0361    0.9600    0.5000     48    25    25     2    0.4600
 5.9000    0.9600    0.4800     48    26    24     2    0.4400
 5.8857    0.9600    0.4600     48    27    23     2    0.4200
 5.6326    0.9600    0.4400     48    28    22     2    0.4000
 5.6281    0.9600    0.4200     48    29    21     2    0.3800
 5.5851    0.9600    0.4000     48    30    20     2    0.3600
 5.5802    0.9600    0.3800     48    31    19     2    0.3400
 5.4514    0.9600    0.3600     48    32    18     2    0.3200
 5.3207    0.9600    0.3400     48    33    17     2    0.3000
 5.1582    0.9600    0.3200     48    34    16     2    0.2800
 5.0599    0.9600    0.3000     48    35    15     2    0.2600
 5.0167    0.9600    0.2800     48    36    14     2    0.2400
 4.5654    0.9600    0.2600     48    37    13     2    0.2200
 3.9706    0.9800    0.2600     49    37    13     1    0.2400
 3.9552    0.9800    0.2400     49    38    12     1    0.2200
 3.9103    1.0000    0.2400     50    38    12     0    0.2400
 3.7719    1.0000    0.2200     50    39    11     0    0.2200
 3.5881    1.0000    0.2000     50    40    10     0    0.2000
 3.4583    1.0000    0.1800     50    41     9     0    0.1800
 3.4422    1.0000    0.1600     50    42     8     0    0.1600
 2.9673    1.0000    0.1400     50    43     7     0    0.1400
 2.9111    1.0000    0.1200     50    44     6     0    0.1200
 2.7698    1.0000    0.1000     50    45     5     0    0.1000
 2.5203    1.0000    0.0800     50    46     4     0    0.0800
```

```
        2.3646    1.0000    0.0600     50    47     3     0    0.0600
        2.2596    1.0000    0.0400     50    48     2     0    0.0400
        0.4733    1.0000    0.0200     50    49     1     0    0.0200
       -0.3894    1.0000    0.0000     50    50     0     0    0.0000
           Inf    0.0000    1.0000      0     0    50    50    0.0000
```

```
print(ro$cutoff_table)
```

```

Optimal Cutoff Analysis
  Column: x1
    Youden / Corner: cutoff = 8.7617  (sens = 0.9000, spec = 0.9200, Youden = 0.8200)

  Full cutoff table:
        Cutoff      Sens      Spec     TP    FP    TN    FN    Youden
  ---------------------------------------------------------------------
          -Inf    1.0000    0.0000     50    50     0     0    0.0000
       18.7295    0.0200    1.0000      1     0    50    49    0.0200
       18.5675    0.0400    1.0000      2     0    50    48    0.0400
       18.2248    0.0600    1.0000      3     0    50    47    0.0600
       17.7837    0.0800    1.0000      4     0    50    46    0.0800
       16.4453    0.1000    1.0000      5     0    50    45    0.1000
       15.8893    0.1200    1.0000      6     0    50    44    0.1200
       15.8087    0.1400    1.0000      7     0    50    43    0.1400
       15.6058    0.1600    1.0000      8     0    50    42    0.1600
       15.5003    0.1800    1.0000      9     0    50    41    0.1800
       15.2855    0.2000    1.0000     10     0    50    40    0.2000
       15.0168    0.2200    1.0000     11     0    50    39    0.2200
       14.8639    0.2400    1.0000     12     0    50    38    0.2400
       14.6980    0.2600    1.0000     13     0    50    37    0.2600
       14.4757    0.2800    1.0000     14     0    50    36    0.2800
       14.4469    0.3000    1.0000     15     0    50    35    0.3000
       14.4042    0.3200    1.0000     16     0    50    34    0.3200
       14.1952    0.3400    1.0000     17     0    50    33    0.3400
       14.1938    0.3600    1.0000     18     0    50    32    0.3600
       13.9382    0.3800    1.0000     19     0    50    31    0.3800
       13.7845    0.4000    1.0000     20     0    50    30    0.4000
       13.3718    0.4200    1.0000     21     0    50    29    0.4200
       13.0699    0.4400    1.0000     22     0    50    28    0.4400
       12.9970    0.4600    1.0000     23     0    50    27    0.4600
       12.9854    0.4800    1.0000     24     0    50    26    0.4800
       12.9659    0.5000    1.0000     25     0    50    25    0.5000
       12.9569    0.5200    1.0000     26     0    50    24    0.5200
       12.7812    0.5400    1.0000     27     0    50    23    0.5400
       12.7534    0.5600    1.0000     28     0    50    22    0.5600
       12.5227    0.5800    1.0000     29     0    50    21    0.5800
       12.3440    0.6000    1.0000     30     0    50    20    0.6000
       12.2962    0.6200    1.0000     31     0    50    19    0.6200
       12.1394    0.6400    1.0000     32     0    50    18    0.6400
       12.1354    0.6600    1.0000     33     0    50    17    0.6600
       12.0393    0.6800    1.0000     34     0    50    16    0.6800
       11.7418    0.6800    0.9800     34     1    49    16    0.6600
       11.6642    0.7000    0.9800     35     1    49    15    0.6800
       11.5082    0.7000    0.9600     35     2    48    15    0.6600
       11.4038    0.7200    0.9600     36     2    48    14    0.6800
       11.2133    0.7400    0.9600     37     2    48    13    0.7000
       11.0696    0.7600    0.9600     38     2    48    12    0.7200
       10.3902    0.7800    0.9600     39     2    48    11    0.7400
       10.1719    0.8000    0.9600     40     2    48    10    0.7600
        9.8836    0.8200    0.9600     41     2    48     9    0.7800
```

```
     9.5685    0.8200    0.9400     41     3    47     9    0.7600
     9.3526    0.8400    0.9400     42     3    47     8    0.7800
     9.1944    0.8600    0.9400     43     3    47     7    0.8000
     9.1032    0.8800    0.9400     44     3    47     6    0.8200
     8.8558    0.8800    0.9200     44     4    46     6    0.8000
     8.7617    0.9000    0.9200     45     4    46     5    0.8200
     8.5538    0.9000    0.9000     45     5    45     5    0.8000
     8.5406    0.9000    0.8800     45     6    44     5    0.7800
     8.5296    0.9200    0.8800     46     6    44     4    0.8000
     8.5259    0.9200    0.8600     46     7    43     4    0.7800
     8.3125    0.9200    0.8400     46     8    42     4    0.7600
     8.2985    0.9200    0.8200     46     9    41     4    0.7400
     7.9078    0.9200    0.8000     46    10    40     4    0.7200
     7.8493    0.9200    0.7800     46    11    39     4    0.7000
     7.6187    0.9200    0.7600     46    12    38     4    0.6800
     7.5941    0.9200    0.7400     46    13    37     4    0.6600
     7.5556    0.9200    0.7200     46    14    36     4    0.6400
     7.5150    0.9200    0.7000     46    15    35     4    0.6200
     7.4355    0.9200    0.6800     46    16    34     4    0.6000
     7.1837    0.9200    0.6600     46    17    33     4    0.5800
     6.8235    0.9400    0.6600     47    17    33     3    0.6000
     6.8029    0.9400    0.6400     47    18    32     3    0.5800
     6.7765    0.9400    0.6200     47    19    31     3    0.5600
     6.6770    0.9400    0.6000     47    20    30     3    0.5400
     6.5446    0.9400    0.5800     47    21    29     3    0.5200
     6.5440    0.9600    0.5800     48    21    29     2    0.5400
     6.1603    0.9600    0.5600     48    22    28     2    0.5200
     6.0665    0.9600    0.5400     48    23    27     2    0.5000
     6.0554    0.9600    0.5200     48    24    26     2    0.4800
     6.0361    0.9600    0.5000     48    25    25     2    0.4600
     5.9000    0.9600    0.4800     48    26    24     2    0.4400
     5.8857    0.9600    0.4600     48    27    23     2    0.4200
     5.6326    0.9600    0.4400     48    28    22     2    0.4000
     5.6281    0.9600    0.4200     48    29    21     2    0.3800
     5.5851    0.9600    0.4000     48    30    20     2    0.3600
     5.5802    0.9600    0.3800     48    31    19     2    0.3400
     5.4514    0.9600    0.3600     48    32    18     2    0.3200
     5.3207    0.9600    0.3400     48    33    17     2    0.3000
     5.1582    0.9600    0.3200     48    34    16     2    0.2800
     5.0599    0.9600    0.3000     48    35    15     2    0.2600
     5.0167    0.9600    0.2800     48    36    14     2    0.2400
     4.5654    0.9600    0.2600     48    37    13     2    0.2200
     3.9706    0.9800    0.2600     49    37    13     1    0.2400
     3.9552    0.9800    0.2400     49    38    12     1    0.2200
     3.9103    1.0000    0.2400     50    38    12     0    0.2400
     3.7719    1.0000    0.2200     50    39    11     0    0.2200
     3.5881    1.0000    0.2000     50    40    10     0    0.2000
     3.4583    1.0000    0.1800     50    41     9     0    0.1800
     3.4422    1.0000    0.1600     50    42     8     0    0.1600
     2.9673    1.0000    0.1400     50    43     7     0    0.1400
     2.9111    1.0000    0.1200     50    44     6     0    0.1200
     2.7698    1.0000    0.1000     50    45     5     0    0.1000
     2.5203    1.0000    0.0800     50    46     4     0    0.0800
     2.3646    1.0000    0.0600     50    47     3     0    0.0600
     2.2596    1.0000    0.0400     50    48     2     0    0.0400
     0.4733    1.0000    0.0200     50    49     1     0    0.0200
    -0.3894    1.0000    0.0000     50    50     0     0    0.0000
        Inf    0.0000    1.0000      0     0    50    50    0.0000
```

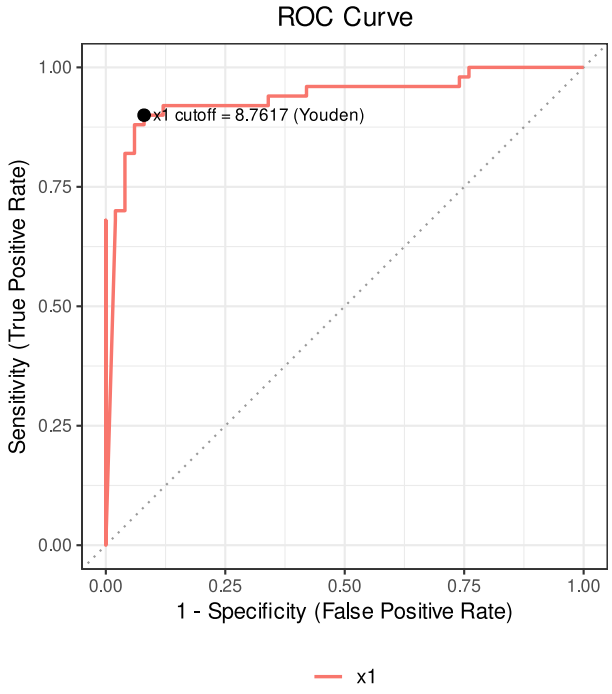
`cutoff()` gives all candidate cutoffs with their corresponding sensitivity, specificity, and Youden index, and marks the optimal one with two criteria: **Youden** (maximizing `Se + Sp − 1`) and **closest corner**

(minimizing the distance to (0,1)). In this example, the Youden/Corner optimal cutoff is 8.76 (sensitivity 0.90, specificity 0.92, Youden 0.82).

**Parameter notes:** `cutoff()` must be executed after `auc()`; `plot(cutpoint=...)` requires `cutoff()` to have been run. `cutpoint` can be `"Youden"`, `"Corner"`, `"all"`, or `NULL`.

### 14.3.5 Graphics and Summary

```
plot(ro, cutpoint = "Youden")
```




```
summary(ro)
```

```
ROC Analysis -- Summary
--------------------------------------------------

ROC Analysis
  Data class:        data.frame
  Total rows:        100
  Complete cases:    100
  Reference:         ref (binary)
  Positive / Neg:    50 / 50
  Evaluation cols:   1  (x1)
  Duplicate IDs:     none

Analysis slots:
    [x] Describe
    [x] AUC
    [ ] Cutoff
    [ ] MLR


Descriptive Statistics
  Variable               n      Mean        SD       Min        Q1    Median        Q3
```

```
Max  NA
  ------------------------------------------------------------------------------
  x1                100   9.2576   4.4744  -0.3894   5.8224   8.5472  12.9591  18.7295

  ref
  Level                Count  Percent
   ---------------------------------------
  0                       50    50.0%
  1                       50    50.0%


AUC Table
  Column               AUC          95% CI  Direction
  ---------------------------------------------------------------
  x1                0.9420  [0.8940, 0.9900]       geq


Optimal Cutoff Analysis
  Column: x1
    Youden / Corner: cutoff = 8.7617  (sens = 0.9000, spec = 0.9200, Youden = 0.8200)

  Full cutoff table:
        Cutoff      Sens      Spec     TP    FP    TN    FN    Youden
  ------------------------------------------------------------------------
          -Inf    1.0000    0.0000     50    50     0     0    0.0000
       18.7295    0.0200    1.0000      1     0    50    49    0.0200
       18.5675    0.0400    1.0000      2     0    50    48    0.0400
       18.2248    0.0600    1.0000      3     0    50    47    0.0600
       17.7837    0.0800    1.0000      4     0    50    46    0.0800
       16.4453    0.1000    1.0000      5     0    50    45    0.1000
       15.8893    0.1200    1.0000      6     0    50    44    0.1200
       15.8087    0.1400    1.0000      7     0    50    43    0.1400
       15.6058    0.1600    1.0000      8     0    50    42    0.1600
       15.5003    0.1800    1.0000      9     0    50    41    0.1800
       15.2855    0.2000    1.0000     10     0    50    40    0.2000
       15.0168    0.2200    1.0000     11     0    50    39    0.2200
       14.8639    0.2400    1.0000     12     0    50    38    0.2400
       14.6980    0.2600    1.0000     13     0    50    37    0.2600
       14.4757    0.2800    1.0000     14     0    50    36    0.2800
       14.4469    0.3000    1.0000     15     0    50    35    0.3000
       14.4042    0.3200    1.0000     16     0    50    34    0.3200
       14.1952    0.3400    1.0000     17     0    50    33    0.3400
       14.1938    0.3600    1.0000     18     0    50    32    0.3600
       13.9382    0.3800    1.0000     19     0    50    31    0.3800
       13.7845    0.4000    1.0000     20     0    50    30    0.4000
       13.3718    0.4200    1.0000     21     0    50    29    0.4200
       13.0699    0.4400    1.0000     22     0    50    28    0.4400
       12.9970    0.4600    1.0000     23     0    50    27    0.4600
       12.9854    0.4800    1.0000     24     0    50    26    0.4800
       12.9659    0.5000    1.0000     25     0    50    25    0.5000
       12.9569    0.5200    1.0000     26     0    50    24    0.5200
       12.7812    0.5400    1.0000     27     0    50    23    0.5400
       12.7534    0.5600    1.0000     28     0    50    22    0.5600
       12.5227    0.5800    1.0000     29     0    50    21    0.5800
       12.3440    0.6000    1.0000     30     0    50    20    0.6000
       12.2962    0.6200    1.0000     31     0    50    19    0.6200
       12.1394    0.6400    1.0000     32     0    50    18    0.6400
       12.1354    0.6600    1.0000     33     0    50    17    0.6600
       12.0393    0.6800    1.0000     34     0    50    16    0.6800
       11.7418    0.6800    0.9800     34     1    49    16    0.6600
```

```
      11.6642    0.7000    0.9800     35     1    49    15    0.6800
      11.5082    0.7000    0.9600     35     2    48    15    0.6600
      11.4038    0.7200    0.9600     36     2    48    14    0.6800
      11.2133    0.7400    0.9600     37     2    48    13    0.7000
      11.0696    0.7600    0.9600     38     2    48    12    0.7200
      10.3902    0.7800    0.9600     39     2    48    11    0.7400
      10.1719    0.8000    0.9600     40     2    48    10    0.7600
       9.8836    0.8200    0.9600     41     2    48     9    0.7800
       9.5685    0.8200    0.9400     41     3    47     9    0.7600
       9.3526    0.8400    0.9400     42     3    47     8    0.7800
       9.1944    0.8600    0.9400     43     3    47     7    0.8000
       9.1032    0.8800    0.9400     44     3    47     6    0.8200
       8.8558    0.8800    0.9200     44     4    46     6    0.8000
       8.7617    0.9000    0.9200     45     4    46     5    0.8200
       8.5538    0.9000    0.9000     45     5    45     5    0.8000
       8.5406    0.9000    0.8800     45     6    44     5    0.7800
       8.5296    0.9200    0.8800     46     6    44     4    0.8000
       8.5259    0.9200    0.8600     46     7    43     4    0.7800
       8.3125    0.9200    0.8400     46     8    42     4    0.7600
       8.2985    0.9200    0.8200     46     9    41     4    0.7400
       7.9078    0.9200    0.8000     46    10    40     4    0.7200
       7.8493    0.9200    0.7800     46    11    39     4    0.7000
       7.6187    0.9200    0.7600     46    12    38     4    0.6800
       7.5941    0.9200    0.7400     46    13    37     4    0.6600
       7.5556    0.9200    0.7200     46    14    36     4    0.6400
       7.5150    0.9200    0.7000     46    15    35     4    0.6200
       7.4355    0.9200    0.6800     46    16    34     4    0.6000
       7.1837    0.9200    0.6600     46    17    33     4    0.5800
       6.8235    0.9400    0.6600     47    17    33     3    0.6000
       6.8029    0.9400    0.6400     47    18    32     3    0.5800
       6.7765    0.9400    0.6200     47    19    31     3    0.5600
       6.6770    0.9400    0.6000     47    20    30     3    0.5400
       6.5446    0.9400    0.5800     47    21    29     3    0.5200
       6.5440    0.9600    0.5800     48    21    29     2    0.5400
       6.1603    0.9600    0.5600     48    22    28     2    0.5200
       6.0665    0.9600    0.5400     48    23    27     2    0.5000
       6.0554    0.9600    0.5200     48    24    26     2    0.4800
       6.0361    0.9600    0.5000     48    25    25     2    0.4600
       5.9000    0.9600    0.4800     48    26    24     2    0.4400
       5.8857    0.9600    0.4600     48    27    23     2    0.4200
       5.6326    0.9600    0.4400     48    28    22     2    0.4000
       5.6281    0.9600    0.4200     48    29    21     2    0.3800
       5.5851    0.9600    0.4000     48    30    20     2    0.3600
       5.5802    0.9600    0.3800     48    31    19     2    0.3400
       5.4514    0.9600    0.3600     48    32    18     2    0.3200
       5.3207    0.9600    0.3400     48    33    17     2    0.3000
       5.1582    0.9600    0.3200     48    34    16     2    0.2800
       5.0599    0.9600    0.3000     48    35    15     2    0.2600
       5.0167    0.9600    0.2800     48    36    14     2    0.2400
       4.5654    0.9600    0.2600     48    37    13     2    0.2200
       3.9706    0.9800    0.2600     49    37    13     1    0.2400
       3.9552    0.9800    0.2400     49    38    12     1    0.2200
       3.9103    1.0000    0.2400     50    38    12     0    0.2400
       3.7719    1.0000    0.2200     50    39    11     0    0.2200
       3.5881    1.0000    0.2000     50    40    10     0    0.2000
       3.4583    1.0000    0.1800     50    41     9     0    0.1800
       3.4422    1.0000    0.1600     50    42     8     0    0.1600
       2.9673    1.0000    0.1400     50    43     7     0    0.1400
       2.9111    1.0000    0.1200     50    44     6     0    0.1200
       2.7698    1.0000    0.1000     50    45     5     0    0.1000
```

```
    2.5203    1.0000    0.0800     50    46     4     0    0.0800
    2.3646    1.0000    0.0600     50    47     3     0    0.0600
    2.2596    1.0000    0.0400     50    48     2     0    0.0400
    0.4733    1.0000    0.0200     50    49     1     0    0.0200
   -0.3894    1.0000    0.0000     50    50     0     0    0.0000
       Inf    0.0000    1.0000      0     0    50    50    0.0000
```

## 14.4 Key Points for Interpreting Results

1. **Reference reliable**: ROC conclusions depend on the reference standard; errors in the reference propagate to the AUC and cutoff.
2. **Direction confirmed**: Confirm that the marker's judgment direction (geq/leq) is clinically meaningful.
3. **Cutoff selection**: Youden and closest corner are two criteria; the final cutoff should be determined in light of clinical costs (false-negative/false-positive weights) and the protocol, not only statistical optimality.
4. **Sample size**: The AUC confidence interval varies with sample size and event count; interpret cautiously with small samples or too few events.
5. **Method limitations**: This workflow is an empirical ROC/trapezoidal AUC; paired multi-curve comparison, partial AUC, etc., require separate designs.
6. **Independent validation**: The AUC on the training data is the apparent performance; it should be validated on independent data before generalization.

# 15. ROC Curve and Multivariate Regression

## 15.1 Analysis Objectives

The diagnostic ability of a single marker is often limited; combining multiple markers can improve the overall discrimination. Besides single-marker ROC, the `roc()` function in `ivdtools` also supports combining multiple markers into a joint score with multivariate logistic regression (`mlr()`), computing its AUC, and predicting the positive probability for new samples.

This document demonstrates the joint analysis of two markers (x1, x2): first compare the single-marker AUCs, then fit a multivariate logistic regression model, draw the multi-curve ROC, and predict new samples. The AUC of the joint model on the training data is the **apparent performance** and should be validated on independent data before generalization; the reference variable must be binary.

```
library(ivdtools)
library(readr)
```

## 15.2 Overview of Functions

| Function | Main purpose | Key input or output |
|---|---|---|
| `roc()` | Create a ROC object | `cols` supports multiple markers |
| `auc()` | Single-marker AUC | per column |
| `mlr()` | Multivariate logistic regression | `cols`, `name`; requires `auc()` first |
| `plot()` | ROC curve | `mlr="all"` overlays the joint-model curve |
| `predict()` | Predict positive probability | `newdata`, `column_map` |
| `summary()` | Summary | |

## 15.3 Example: Joint Multi-Marker Analysis

### 15.3.1 Reading and Validating Data

```
mlr_dat <- read_csv("./data/roc-multimarker.csv", show_col_types = FALSE)
mlr_dat <- as.data.frame(mlr_dat)
str(mlr_dat)
```

```
'data.frame':   120 obs. of  4 variables:
 $ sid: num  1 2 3 4 5 6 7 8 9 10 ...
 $ ref: num  0 0 0 0 0 0 0 0 0 0 ...
 $ x1 : num  5.51 3.19 8.13 6.16 11.12 ...
 $ x2 : num  8.3 12.9 12.4 16.1 11.9 ...
```

```
table(mlr_dat$ref)
```

```
 0  1
60 60
```

The data contain 120 cases (60 positive and 60 negative) and two quantitative markers, x1 and x2.

### 15.3.2 Creating the Object and Single-Marker AUC

```
ro2 <- roc(mlr_dat, cols = c("x1", "x2"), reference = "ref", id = "sid")
ro2 <- auc(ro2)
```

```
AUC Table
  Column               AUC          95% CI  Direction
  ------------------------------------------------------------
  x1                0.9219  [0.8712, 0.9727]       geq
  x2                0.8322  [0.7587, 0.9057]       geq
```

```
ro2$auc_table
```

```
AUC Table
  Column               AUC          95% CI  Direction
  ------------------------------------------------------------
  x1                0.9219  [0.8712, 0.9727]       geq
  x2                0.8322  [0.7587, 0.9057]       geq
```

The AUC of x1 is ≈ 0.922 and that of x2 is ≈ 0.832; among the single markers, x1 discriminates better.

### 15.3.3 Multivariate Logistic Regression

Model jointly with both markers:

```
ro2 <- mlr(ro2, cols = c("x1", "x2"), name = "combined")
```

```
Multivariate Logistic Regression (combined)
  ------------------------------------------------
  Formula: reference_binary ~ x1 + x2
  n = 120  (positive = 60, negative = 60)
  AUC = 0.9642  [0.9298, 0.9985]

  Coefficients:
                  Estimate Std. Error   z value     Pr(>|z|)
    (Intercept) -13.6586717  2.7569949 -4.954188 7.263286e-07
    x1            0.7638502  0.1549103  4.930919 8.184356e-07
    x2            0.3949617  0.1064673  3.709700 2.075049e-04

  * Note: AUC is evaluated on the same data used for fitting;
    the estimate may be optimistic.
```

```
ro2$mlr_results$combined$fit_summary
```

```
Call:
glm(formula = formula, family = binomial, data = model_data)

Coefficients:
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -13.6587     2.7570  -4.954 7.26e-07 ***
x1            0.7639     0.1549   4.931 8.18e-07 ***
x2            0.3950     0.1065   3.710 0.000208 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 166.355  on 119  degrees of freedom
Residual deviance:  58.112  on 117  degrees of freedom
AIC: 64.112

Number of Fisher Scoring iterations: 7
```

```
ro2$mlr_results$combined$auc
```

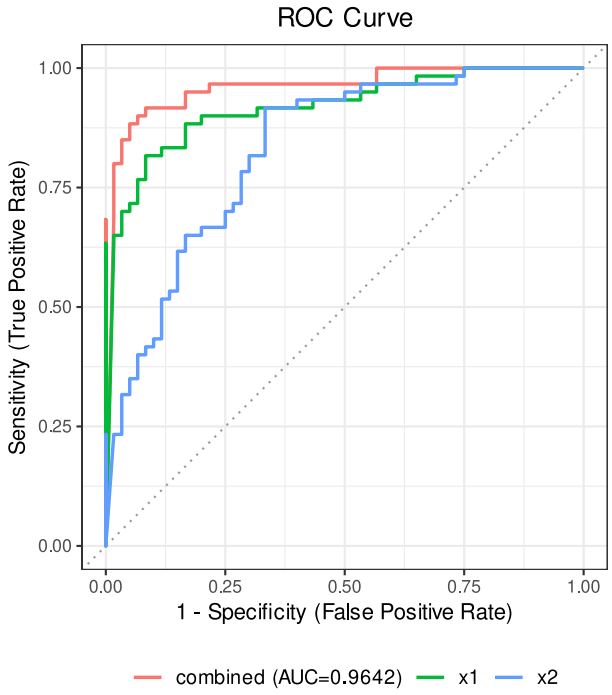
```
[1] 0.9641667
```

The joint model has an AUC of ≈ 0.964 (95% CI [0.930, 0.999]), higher than either single marker, indicating that combining the two markers does provide a gain.

**Parameter notes:** `mlr(cols, name, ...)` fits a binary logistic regression (`glm`) with the selected markers; `name` is the model name (auto-generates `MLR01`, `MLR02`, ...); `...` is passed through to `glm`. **`auc()` must be run first** before calling `mlr()`. The coefficients are interpreted as log odds ratios; the joint AUC is computed on the data used for fitting and is an apparent performance.

### 15.3.4 Multi-Curve ROC Plot

```
plot(ro2, mlr = "all")
```




`plot(ro2, mlr="all")` draws the ROC curves of each single marker and the joint model together for comparison.

### 15.3.5 Predicting New Samples

```
predict(ro2, newdata = data.frame(x1 = c(7, 12), x2 = c(16, 21)))
```

```
  x1 x2 pred_prob    pred_se   ci_lower  ci_upper pred_class
1  7 16 0.1200103 0.05313595 0.04840777 0.2677265          0
2 12 21 0.9781557 0.01751040 0.89984790 0.9955390          1
```

`predict()` returns each predictor variable, the positive probability (`pred_prob`), the standard error and confidence interval, and the predicted class (`pred_class`) at the default threshold of 0.5. The two cases obtain positive probabilities of 0.12 and 0.98, respectively, consistent with the expected direction.

**Parameter notes:** When the column names of `newdata` differ from those used in modeling, map them with the named vector `column_map` (for example `column_map = c(new_x1 = "x1")`). If the object contains multiple models, `predict()` must specify the model name with `mlr`.

### 15.3.6 Summary

```
summary(ro2)
```

```
ROC Analysis -- Summary
--------------------------------------------------

ROC Analysis
  Data class:        data.frame
  Total rows:        120
  Complete cases:    120
  Reference:         ref (binary)
  Positive / Neg:    60 / 60
  Evaluation cols:   2  (x1, x2)
  Duplicate IDs:     none

Analysis slots:
    [ ] Describe
    [x] AUC
    [ ] Cutoff
    [x] MLR


AUC Table
  Column               AUC          95% CI  Direction
  ------------------------------------------------------------
  x1                0.9219  [0.8712, 0.9727]       geq
  x2                0.8322  [0.7587, 0.9057]       geq

MLR Results

Multivariate Logistic Regression (combined)
  --------------------------------------------------
  Formula: reference_binary ~ x1 + x2
  n = 120  (positive = 60, negative = 60)
  AUC = 0.9642  [0.9298, 0.9985]

  Coefficients:
                  Estimate Std. Error   z value     Pr(>|z|)
    (Intercept) -13.6586717  2.7569949 -4.954188 7.263286e-07
    x1            0.7638502  0.1549103  4.930919 8.184356e-07
```

```
    x2           0.3949617  0.1064673  3.709700 2.075049e-04

  * Note: AUC is evaluated on the same data used for fitting;
    the estimate may be optimistic.
```

## 15.4 Key Points for Interpreting Results

1. **Joint gain**: A joint AUC higher than each single marker only indicates that the linear combination provides a gain; whether it is a significant improvement should be judged with the confidence interval and clinical decision needs.
2. **Apparent performance**: The AUC on the training data is an optimistic estimate; it must be re-estimated on an independent validation set to avoid overfitting misjudgment.
3. **Collinearity and separation**: When markers are highly correlated or perfectly separated, logistic regression may be unstable or fail to converge.
4. **Event count**: A multivariate model needs a sufficient number of events per parameter; use caution when events are too few.
5. **Cutoff and decision**: The final judgment threshold of the joint score should combine clinical costs and the protocol, not be limited to the default 0.5.
6. **Variable selection**: `mlr()` fits a given set of markers in one call; comparisons of multiple variable sets should be pre-specified in the protocol to avoid repeated trial and error.

# 16. Introduction to Algorithm Principles

## 16.1 Introduction

This document summarizes the algorithm principles underlying `ivdtools`, to support understanding of the statistical basis of each analysis module and the mapping with the package interface. It covers outliers/distribution, method comparison/qualitative evaluation, ROC, precision/ANOVA/reference intervals, and QC/stability/equation fitting. For ease of cross-reference, method comparison uniformly adopts the convention of candidate $X$, reference $Y$, and regression model $Y = a + bX$; the specific parameter names and directions in the package are governed by the registries such as `list_equation()`, `list_sadler()`, and `list_westgard()` in `ivdtools`.

The formulas are used to illustrate the principles; they do not replace textbook derivations, nor do they replace the judgment about method selection in the protocol.

## 16.2 General Conventions and Notation

Ordered samples $x_{(1)} \le \cdots \le x_{(n)}$, mean $\bar{x}$, standard deviation $s$, significance level $\alpha = 1 -$ level. Most tests require approximately independent samples; a detected "outlier" or "failure to follow some distribution" only indicates the strength of conflict with the assumed model, cannot quantify its actual effect on the final performance estimate, and does not automatically trigger deletion.

## 16.3 Outlier and Distribution Algorithms

### 16.3.1 Outliers

- **Grubbs**: $G = \max_i |x_i - \bar{x}|/s$, judged against a critical value derived from the t distribution for the most extreme point; approximately normal and applies to one point at a time. Mapping: `outliers_test(method="grubbs", alpha=...)`.
- **Generalized ESD**: successively removes the most extreme point, computes $R_i$ and compares it with $\lambda_i$ that changes with the remaining n; the maximum number of outliers is limited by an additional parameter.
- **Dixon**: uses the ratio of the endpoint gap to the sample range; the critical value depends on n and is suitable only for a small-sample range.
- **IQR**: marks values less than $Q_1 - k\,IQR$ or greater than $Q_3 + k\,IQR$; the default coefficient is determined by the interface implementation; this is a rule rather than a significance test.

`outliers_test()` first excludes missing values and retains the original row numbers, then returns the statistic, critical value, and outlier flag. A constant column makes statistics with s or the range as the denominator undefined; multiple testing increases false positives.

### 16.3.2 Normality

- **Shapiro–Wilk**: $W = (\sum a_i x_{(i)})^2 / \sum (x_i - \bar{x})^2$, with weights from the normal order statistics.
- **Anderson–Darling**: the weighted squared distance between the empirical distribution and the fitted normal CDF, with greater weight on the tails.
- **Lilliefors**: a Kolmogorov–Smirnov-type statistic when parameters are estimated from the sample.
- **Cramér–von Mises**: the overall integrated squared distance between the empirical CDF and the theoretical CDF.

`normal_test(method="auto")` selects an available test based on the sample conditions (Shapiro–Wilk for n≤50, Anderson–Darling for larger samples), and `level` maps to the rejection threshold. Graphics provide Q-Q and histogram; the p-value cannot quantify the actual effect of the deviation on the performance estimate.

## 16.4 Method Comparison and Qualitative Evaluation Algorithms

### 16.4.1 Regression, Correlation, and Bias

The unified direction is candidate $X$, reference $Y$, model $Y = a + bX$.

- **OLS** minimizes $\sum(y_i - a - bx_i)^2$; assumes the X error is negligible.
- **Deming** minimizes the weighted orthogonal distance to the line; the error variance ratio maps to `lambda`; the direction of $\lambda = \sigma_Y^2/\sigma_X^2$ must be checked against the source code or protocol.
- **Passing–Bablok** estimates b from the median of all pairwise slopes, then estimates a from the median of $y_i - bx_i$; it is robust to outliers but still requires linearity and independent samples.
- **Pearson** measures linear correlation; **Spearman/Kendall** measure rank association; `correlation(method=...)` answers "correlation" rather than "agreement" and cannot replace bias and agreement-limit evaluation.

The absolute Bland–Altman difference is recorded as $d_i = X_i - Y_i$ (the report must re-check the actual direction in the package object), with mean difference $\bar{d}$ and agreement limits $\bar{d} \pm z_{(1+p)/2} s_d$; relative BA uses percentage differences. `agree.level` controls the coverage proportion and `conf.level` controls the estimation interval. `bias()` maps the regression equation to the specified `level` and outputs the predicted bias and confidence/prediction intervals.

### 16.4.2 Four-Fold Table

From TP, FP, TN, FN:

$$Se = \frac{TP}{TP + FN}, \quad Sp = \frac{TN}{TN + FP}, \quad PPV = \frac{TP}{TP + FP}. \tag{16.1}$$

The Wilson interval is inverted from a score test; the exact interval is inverted from binomial tail probabilities. Cohen $\kappa = (P_o - P_e)/(1 - P_e)$, where $P_e$ is obtained from the marginal proportions. McNemar uses only the discordant cells b and c, with the asymptotic statistic approximately $(b - c)^2/(b + c)$ (with the continuity correction or exact version as recorded in the object). The package interfaces map to `diagnostics()`, `kappa()`, and `mcnemar()`; zero denominators, small cell counts, and extreme prevalence all require explicit explanation.

## 16.5 ROC Algorithms

For each threshold t, compute $TPR(t) = Se(t)$ and $FPR(t) = 1 - Sp(t)$ according to the marker direction. The empirical ROC treats all observations as candidate thresholds and connects them as a step/broken line; the AUC is summed by the trapezoidal rule over adjacent points:

$$AUC = \sum_i \frac{TPR_i + TPR_{i+1}}{2}(FPR_{i+1} - FPR_i). \tag{16.2}$$

`auc(roc_obj, cols=...)` saves the per-marker AUC table; the empirical AUC can be interpreted as the probability that a random positive score exceeds a random negative score (after direction adjustment). Sample-correlated multi-curve comparison, partial AUC, or complex sampling are not automatically resolved by this point estimate.

The Youden index $J(t) = Se(t) + Sp(t) - 1$; the closest-corner method minimizes $\sqrt{(1 - Se)^2 + (1 - Sp)^2}$. `cutoff()` outputs both types of optimal points. Tied thresholds, measurement resolution, misclassification costs, and clinical prevalence change the actual choice, so "mathematically optimal" is not an automatic clinical cutoff.

The multi-marker model is

$$\text{logit}\ \{P(Y = 1)\} = \beta_0 + \sum_j \beta_j x_j, \tag{16.3}$$

fitted by binomial logistic regression via `mlr(cols, name)`; `predict()` transforms the linear predictor into a probability through the logistic inverse, classifies at the default 0.5, and computes intervals using the model covariance. Perfect separation, collinearity, missingness, and small event counts lead to instability.

Re-drawing the ROC on the training set is an apparent performance; development and validation data must be distinguished.

## 16.6 Precision, ANOVA, and Reference Interval Algorithms

### 16.6.1 Variance Components

A nested design can be written as $y_{ijkl} = \mu + D_i + R_{j(i)} + \varepsilon_{k(ij)}$, with total variance $\sigma_T^2 = \sigma_D^2 + \sigma_R^2 + \sigma_e^2$. `precision(form = y ~ day/run)` passes the formula to `VCA::anovaVCA()`; `variance()`/`vc()` summarize the VC, $SD = \sqrt{VC}$, $CV = 100SD/\bar{y}$, and proportions. Satterthwaite uses

$$\nu \approx \frac{(\sum c_j MS_j)^2}{\sum (c_j MS_j)^2/\nu_j} \tag{16.4}$$

to approximate the effective degrees of freedom, then constructs variance/SD intervals from chi-square quantiles. Negative components, low degrees of freedom, and imbalance degrade the approximation.

The Sadler precision profile expresses the variance or SD as a function of concentration; `list_sadler()` lists 10 candidate forms, and `profile(model.no=...)` calls the VFP fit and selects by AIC. Different models extrapolate very differently at low concentrations; the concentration coverage and residuals must be checked.

### 16.6.2 ANOVA and Reference Intervals

`bottle_anova()` uses `stats::aov()` to decompose the between-group/within-group sums of squares; Bartlett tests variance homogeneity, the residual Shapiro–Wilk checks normality; Tukey HSD uses the studentized range to control the family-wise error rate of all pairwise comparisons at once.

The parametric reference interval is approximated as $\bar{x} \pm z_{1-\alpha/2} s$ under the normality assumption; the nonparametric percentile method uses R quantile type 6, with endpoints from the order statistics and endpoint intervals computed by binomial ordering or bootstrap paths. `interval` is the coverage and `ci` is the endpoint confidence; the two have different meanings. Sample selection bias cannot be repaired by any interval algorithm.

## 16.7 QC, Stability, and Equation Fitting Algorithms

### 16.7.1 QC and Stability

The Levey–Jennings standardization is $z_i = (x_i - \mu_0)/\sigma_0$; the Westgard rules combine these z values across single points, consecutive same-side/same-direction runs, or multiple levels in the same batch. `qc_chart()` evaluates in `run` order, and `youden_plot()` uses the target means/SDs of two control materials to form quadrant and ellipse-style judgments; the order, target values, and lot boundaries are inputs to the algorithm.

The node relative bias is $100(x_t - x_0)/x_0$ (relative) or $x_t - x_0$ (absolute). `stability_regression()` fits a time trend, and `stability_time()` finds the intersection of the confidence boundary/trend with the allowable limit. MKT is the Arrhenius-weighted temperature history:

$$T_k = -\frac{E_a}{R}\left[\log\left\{\sum_i w_i \exp(-E_a/RT_i)\right\}\right]^{-1}. \tag{16.5}$$

The Arrhenius relationship is $k = Ae^{-E_a/(RT)}$; zero-order is $C_t = C_0 - kt$, first-order is $\log C_t = \log C_0 - kt$. `mkt(ea=83.144)`, `arrhenius(order, target_temp, limit)` map these quantities. Temperature must be converted to Kelvin for the formulas, and extrapolation is limited by the assumption of an unchanged reaction mechanism.

### 16.7.2 Equation Fitting

Linear/polynomial are estimated by least squares; exponential and 4PL/5PL use `nlsLM`/constrained optimization. A typical 4PL is

$$y = d + \frac{a - d}{1 + (x/c)^b}, \tag{16.6}$$

and 5PL adds an asymmetry exponent g, but the specific parameter names and directions in the package are governed by `list_equation()`. Weighted fitting minimizes $\sum w_i(y_i - f(x_i, \theta))^2$. AIC approximates $-2 \log L + 2k$; residuals reveal heteroscedasticity, curvature, and anomalous points.

The mean confidence interval reflects the uncertainty of $E(Y|X)$, while the prediction interval also includes single-observation error. Nonlinear inverse prediction requires numerical root-finding; small signal errors near a flat plateau can translate into enormous concentration errors. Starting values, bounds, weights, equation versions, convergence codes, and failed models should all enter the audit record.

## 16.8 Module-to-Guideline Mapping

| Topic | Main functions | Corresponding CLSI guideline |
|---|---|---|
| Outliers and normality | `outliers_test()`, `normal_test()` | General (pre-analysis steps of the EP guidelines) |
| Method comparison | `mcr()` series | EP09 |
| Qualitative evaluation | `fourfold_table` series | EP12 |
| ROC | `roc()` series | EP24 (reference) |
| Precision | `precision()` series | EP05 |
| Bottle/lot ANOVA | `bottle_anova()` | EP15 |
| Reference intervals | `reference_interval()` | EP28-A3c |
| Stability | `stability_*()`, `arrhenius()`, `mkt()` | EP25 |
| QC | `qc_chart()`, `youden_plot()` | Westgard rules |
| Equation fitting | `fit_equation()` series | EP06 |
| Analytical specificity | `lob_lod_loq()` | EP17 |

## 16.9 Scope of Use

- Formulas and statistics are valid only under reasonable assumptions; when assumptions are not met, the results cannot be applied mechanically.
- "Statistically significant" and "mathematically optimal" are not the same as "clinically/product acceptable"; acceptance criteria must be defined before analysis.
- Performance on training data (such as ROC AUC, regression goodness of fit) is apparent performance; it should be validated on independent data before generalization.
- All computations should be conducted under the guidance of the protocol, standards, and professional judgment, and reviewed by appropriate personnel.

# 17. AI-Automated Analysis

## 17.1 Overview

`ivdtools-analysis` is an AI-powered analysis Skill (current version 0.1.0) for in-vitro diagnostic (IVD) evaluation data. It uses the R package `ivdtools` as its statistical engine and, starting from CSV, TSV, Excel, or RDS data, performs data quality checks and statistical analysis, then generates reproducible Chinese HTML reports, R code, result tables, and figures. It works in conversational agent environments such as Claude Code (`/ivdtools-analysis`) and OpenAI Codex (`$ivdtools-analysis`).

> **Scope statement:** This Skill is intended for statistical analysis and evidence compilation; it does not constitute clinical validation, product release, or regulatory approval. Decisions involving clinical, regulatory, or release matters should be independently reviewed by appropriate professionals.

The value of using the Skill in a conversation is not to "replace statistical judgment," but to standardize the repetitive technical steps: data precheck, parameter confirmation, statistical computation, result verification, and Chinese report generation are executed through a fixed workflow, while data sources, parameters, exclusion records, warnings, result tables, figures, and the software environment are all saved together, reducing the risk of omissions and manual copy errors.

## 17.2 Applicable Analysis Tasks

The Skill covers 10 categories of analysis tasks, all powered by the corresponding modules of `ivdtools` as their statistical engine:

| Analysis Category | Typical Tasks |
|---|---|
| Method comparison | Descriptive statistics, correlation, OLS/WLS/Deming/weighted Deming/Passing–Bablok, Bland–Altman, bias at medical decision levels |
| ROC analysis | Single/multi-marker ROC, AUC, cutoff values, sensitivity/specificity, multivariate Logistic |
| Qualitative agreement | 2×2 tables, sensitivity/specificity/predictive values, overall agreement, Kappa, McNemar |
| Precision | Variance components, SD/CV, confidence intervals, profile results |
| Analytical sensitivity | sadler fit, limit of blank/limit of detection/limit of quantitation establishment |
| Reference intervals | Percentile, parametric, and robust methods with confidence intervals |
| Stability | Condition trend, test/reference comparison, shelf life, MKT, Arrhenius, stability design |
| Quality control | Levey–Jennings plots, Westgard rule violation identification |
| Curve fitting | Curve search, nonlinear/weighted fitting, model comparison, residual diagnostics |

| Analysis Category | Typical Tasks |
|---|---|
| Data checks | Outlier candidates, normality tests, bottle-to-bottle ANOVA, Youden plots |
| Sample size | Bland–Altman, proportion tests, proportion confidence intervals, stability design sample size |

## 17.3 Standard Workflow

The Skill's analysis follows a fixed process, with a fixed output directory structure, to facilitate verification and traceability.

### 17.3.1 1. Create a Dedicated Output Directory

The default directory is named `<input-file-name>-ivdtools-analysis-YYYYMMDD-HHMMSS`, located outside the Skill directory, and does not overwrite existing directories. Source data is always kept read-only and is never modified.

### 17.3.2 2. Data Precheck

Generate a data inspection JSON with `inspect_data.R`:

```
Rscript scripts/inspect_data.R \
  --input /data/example.csv \
  --encoding UTF-8 \
  --output /output/data-inspection.json
```

The precheck reports the number of rows, duplicate rows/column names, per-column type and proportion of missing values, number of unique values, candidate binary fields and candidate ID fields, category levels, and numeric ranges, helping to uncover data quality issues before analysis.

### 17.3.3 3. Determine the Analysis Plan

Select the analysis type based on the research objective and confirm parameters that cannot be safely inferred from the data. The Skill **will not fabricate** clinical cutoff values, allowable bias, stability limits, or acceptance criteria on its own; if such parameters affect the conclusions and cannot be safely derived from the data or the study plan, the user will be asked to confirm them.

### 17.3.4 4. Execute the Analysis and Render the Report

The Skill copies `assets/report-template.Rmd` into the output directory, creates an `analysis.R` with explicit parameters and `ivdtools::` namespaced calls to perform the analysis, then renders the Chinese HTML report. The rendering process preserves warnings and messages rather than silently suppressing them.

### 17.3.5 5. Verify the Results

Before delivery, verify that the total input row count matches the rows excluded at each step, that missing/non-numeric/duplicate records have been reported, that the positive class and direction are correct, that models converge, that the HTML values match `results/*.csv`, and that `sessionInfo()` and the actual package versions are recorded.

## 17.4 Output Contents

A standard delivery directory contains:

```
analysis.R                  # Reusable analysis code
report.Rmd                  # Report source file
```

```
report.html                  # Chinese HTML report
analysis-manifest.json       # Manifest of data sources, parameters, and artifacts
session-info.txt             # R environment and package versions
results/*.csv                # Machine-readable result tables
figures/*.png                # Analysis figures
```

The Chinese report includes, at minimum, the analysis objective, data source, data quality findings, methods and parameters, result tables, figures, exclusions and warnings, interpretation of results, limitations, and reproducibility information.

## 17.5 Installation and Usage

### 17.5.1 Runtime Environment

R ≥ 4.1.0, a matching version of `ivdtools`, Pandoc, and dependent packages such as `ggplot2`, `VCA`, `VFP`, `rmarkdown`, and `knitr`. Run the environment check first:

```
Rscript scripts/check_environment.R
```

An output of `STATUS: READY` indicates readiness; otherwise, missing items are listed. Installing/supplementing dependencies writes to the R library and **may only be executed with explicit user authorization** (`Rscript scripts/install_ivdtools.R --yes`).

### 17.5.2 Use in Claude Code

The Skill is registered as an invocable skill; call it directly in the conversation:

```
/ivdtools-analysis analyze the Results worksheet of /data/method_comparison.xlsx.
The sample ID is sample_id, the candidate method is candidate, and the reference method
is reference.
Per the study plan, perform Deming regression (lambda=1) and difference-type Bland–
Altman analysis,
with a 95% confidence level, and output a reproducible Chinese HTML report.
```

### 17.5.3 Use in OpenAI Codex

Name `$ivdtools-analysis` in the conversation and attach the data and analysis requirements:

```
Please use $ivdtools-analysis to analyze /data/roc.csv.
sample_id is the sample ID, truth is the binary reference result where positive is the
positive class;
marker1 and marker2 are the markers under evaluation. Please output AUC, cutoff values,
sensitivity,
specificity, ROC plots, and a Chinese HTML report. The cutoff derived from the data is
exploratory only.
```

### 17.5.4 Recommended Details When Submitting a Task

1. Data file path (for Excel workbooks, also specify the worksheet);
2. Analysis objective and study design;
3. Mapping of fields such as sample ID, candidate/reference method, outcome, time, and lot number;
4. Positive class, direction, paired/repeated structure, and precision design formula;
5. Confidence level, power, statistical method, and plan-specified parameters;
6. Acceptance criteria such as medical decision levels, allowable bias, stability limits, and QC target mean/SD.

## 17.6 Data Handling Principles

- Preserve the original column names; do not automatically call `make.names()` to rename them;

- Do not default to imputing missing values, deleting outliers, averaging repeated measurements, or reversing the outcome direction;
- Outlier testing only produces candidate records for review and does not constitute a basis for automatic deletion;
- Keep a record of the rationale for excluded records and, where applicable, report sensitivity analyses with and without them;
- State the operations and rationale for any data transformation, exclusion, aggregation, and post-hoc analysis choices;
- Statistical significance does not equal clinical acceptability; acceptability conclusions can only be drawn against acceptance limits specified in the plan or provided by the user.

## 17.7 Frequently Asked Questions

- **Version mismatch**: When the environment check reports an `ivdtools` version mismatch, do not directly overwrite the newer installed version; choose a new writable R library, install the target version there after authorization, and verify it with `--library`.
- **Excel cannot be prechecked**: Confirm that `readxl` is installed; when there are multiple worksheets, use `--sheet` to select explicitly.
- **HTML cannot be rendered**: Confirm that `rmarkdown` and `knitr` are installed and that `rmarkdown::pandoc_available()` is `TRUE`; review the actual warnings or errors recorded in `report.Rmd` rather than silently ignoring them.
- **Binary direction is reversed**: Check the positive class, factor levels, marker direction, and test/reference mapping, and record the direction change and rationale before re-analyzing.
- **Outliers or repeated measurements present**: Do not automatically delete or average; first determine the handling approach based on the raw records, experimental design, and preset rules.

# 18. References

## 18.1 Introduction

This document summarizes the correspondence between the international guidelines (the CLSI EP series) for each analysis module of `ivdtools` and the main methodological references. The correspondences are used to assist understanding and analysis-plan design; they **do not constitute a regulatory determination**; for registration, review, or accreditation matters, implement according to the applicable standard in force at the relevant time.

> **About domestic (Chinese) standards:** Domestic standards fall into two categories — WS/T are health-industry standards (mainly aimed at clinical-laboratory practice), and YY are pharmaceutical/medical-industry standards (aimed at medical devices/in vitro diagnostic products; YY/T is recommended). For the same topic there may be both a laboratory-practice standard and a product-performance standard; when choosing, first clarify the applicable object (laboratory workflow or product evaluation).

## 18.2 Module-to-CLSI-Guideline Mapping

The correspondence between the modules of `ivdtools` and the guidelines of CLSI (Clinical and Laboratory Standards Institute) is as follows:

| Module | Main entry | CLSI guideline | Main functions |
|---|---|---|---|
| Method comparison | `mcr()` | EP09 | regression, correlation, Bland–Altman, bias, outliers |
| Precision | `precision()` | EP05 | variance components, confidence intervals, Sadler profile |
| Qualitative analysis | `raw_to_table()` etc. | EP12 | four-fold table, diagnostic metrics, Kappa, McNemar |
| Reference intervals | `reference_interval()` | EP28 | percentile/parametric/robust methods |
| Stability | `stability_*()`, `arrhenius()`, `mkt()` | EP25 | bias, regression, time-to-limit, MKT, Arrhenius |
| Linearity/curve fitting | `fit_equation()` series | EP06 | linear/polynomial/4PL/5PL fitting |
| Analytical sensitivity/detection limit | `lob_lod_loq()` | EP17 | LoB/LoQ, Sadler variance model |
| Bottle/lot ANOVA | `bottle_anova()` | EP15 | one-way/nested ANOVA, Tukey |
| ROC | `roc()` series | EP24 (reference) | AUC, cutoff, multivariate regression |

| Module | Main entry | CLSI guideline | Main functions |
|---|---|---|---|
| Quality control | `qc_chart()`, `youden_plot()` | Westgard rules | Levey–Jennings, Westgard, Youden |
| Sample size | `sample_size_*()` | General | agreement/proportion/one-arm test sample sizes |
| Outliers and normality | `outliers_test()`, `normal_test()` | General (pre-analysis steps) | Grubbs/ESD/Dixon/IQR, normality tests |

Before actual application, be sure to verify the officially published text and, if necessary, consult the competent authority or a qualified assessment body.

## 18.3 Methodological References

The main statistical methods of `ivdtools` refer to the following publications (included in the package DESCRIPTION):


- Bland J M, Altman D G. Statistical methods for assessing agreement between two methods of clinical measurement. *Lancet*, 1986, 327(8476): 307–310. doi:10.1016/S0140-6736(86)90837-8
- Passing H, Bablok W. A new biometrical procedure for testing the equality of measurements from two different analytical methods. *Journal of Clinical Chemistry and Clinical Biochemistry*, 1983, 21(11): 709–720. doi:10.1515/cclm.1983.21.11.709
- Linnet K. Evaluation of regression procedures for methods comparison studies. *Clinical Chemistry*, 1993, 39(3): 424–432. doi:10.1093/clinchem/39.3.424
- Hanley J A, McNeil B J. The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology*, 1982, 143(1): 29–36. doi:10.1148/radiology.143.1.7063747
- Horn P S, Pesce A J, Copeland B E. A robust approach to reference interval estimation and evaluation. *Clinical Chemistry*, 1998, 44(3): 622–631. doi:10.1093/clinchem/44.3.622
- Westgard J O, Barry P L, Hunt M R, et al. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clinical Chemistry*, 1981, 27(3): 493–501. doi:10.1093/clinchem/27.3.493
- Lu M J, Zhong W H, Liu Y X, et al. Sample size for assessing agreement between two methods of measurement by Bland–Altman method. *The International Journal of Biostatistics*, 2016, 12(2). doi:10.1515/ijb-2015-0039


## 18.4 Usage Suggestions

1. **Defer to the applicable standard**: Standards differ across countries/regions and applicable objects (laboratory vs product); first determine the applicable framework.
2. **Versions and supersession**: Standards are updated or superseded; cite the version and implementation date when referencing.
3. **Separate statistics from compliance**: This package provides statistical computation; whether all requirements of a given standard are met is determined by the complete validation/evaluation plan.
4. **Record and verify**: Write the standard list, versions, and DOI/number into the analysis plan for review and audit.